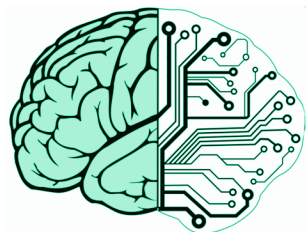


”Örület, de van benne rendszer.”
Shakespeare: Hamlet: 2. felvonás 2. szín

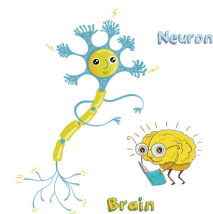
A Hodgkin-Huxley neuron modell V2.0

A biológiai neuronok valódi fizikája és fiziológiája



János Végh

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Contents

1	A neuronok termodinamikája	1
1.1	Termodinamikus tér	1
2	Speculations	3
2.1	Cell pressure	3
3	A neuronok elektromosságtana	9
3.1	Miért más	10
3.2	Concepts	12
3.2.1	Töltés	12
3.2.2	Erőtér	12
3.2.3	Feszültség	12
3.2.4	Áram	12
3.2.5	Condenser	18
3.2.6	Resistor	19
3.3	A membrán elektromosságtana	21
3.3.1	Elektrolitok	21
3.3.2	Membrán elektrolitban	23
3.3.3	Inhomogeneities	23
3.3.4	Force analysis	24
3.3.5	One segment	25
3.3.6	Two segments	26
3.3.7	Charge layers in segments	28
3.3.8	Bridging the micro- and macro views	34
3.4	Forming membrane's potential	35
3.4.1	States	37
3.4.2	Controlling states	46
3.4.3	Goldman-Hodgkin-Katz potential	51

4	Tévhitek a neuronokról	55
4.1	A K^+ áram az Action Potential (AP) folyamán	55
4.1.1	A folyamat formális leírása	55
4.1.2	A folyamat fizikája	59
4.2	A neuronális rezgőkör időállandói	61
4.2.1	GHK egyenlet	61
5	Nyugalmi potenciál	63
5.0.1	Thermodynamic field	63
5.0.2	Charge layers	67
5.0.3	Goldman-Hodgkin-Katz potential	74
5.0.4	Moving in the membrane's potential	75
6	Action Potential	79
6.1	The physical process	79
6.2	The oscillator model	81
6.3	Mathematics of the ActionPotential	84
	Appendices	87
	List of Tables	91
	List of Figures	95
	Index	96
	Glossary	99

Chapter 1

A neuronok termodinamikája

1.1 A "termodinamikai elektromos tér" származtatása

Szegmensekre osztott elektrolitban az elektromos töltések kiegyensúlyozottak; megfelelő fizikai körülmények között azonban kiegyensúlyozatlanná is válhatnak. Egy, a biológia számára fontos eset, amikor egy féligáteresztő hártya választ el két, nagyon eltérő koncentrációjú ionos oldatokat tartalmazó szegmenst. Az eltérő koncentrációk a jól ismert Nernst-Planck összefüggésnek megfelelően feszültséget generálnak, amint azt a **Physics panel 1.1.1** tárgyalja. A nyugalmi potenciál esetére vonatkozó számítási módot a **Physics panel 5.0.1**, annak eredményét pedig a **Numerics panel 5.0.1** mutatja.

Egyszerűen fogalmazva, a Nernst-Planck egyenlet azt állítja, hogy az ionok koncentrációjának változása változást okoz a elektromos térerőben (és megfordítva); továbbá, hogy stacionárius esetben azok változatlanok maradnak. Vegyük észre, hogy ionkeverék esetében közös, töltés-generált, elektromos erőter van, és az minden ionra azonos, míg a közös elektromos erőter az egyes ionoknál eltérő koncentrációkat generál. Megjegyezzük, hogy a neuronon belül a gradiensek élesen változnak, így ott csupán a differenciális alak alkalmazható; a statikus makroszkopikus integrális alak használata félrevezető.

**Physics panel 1.1.1:
The Nernst-Planck law**

The Nernst-Planck electrodiffusion equation in a balanced state

$$\frac{d}{dz} V_m(z) = -\frac{RT}{q * F} \frac{1}{C_k(z)} \frac{d}{dz} C_k(z) \quad (1.1)$$

describes in one dimension the mutual dependence of the *spatial gradients* of the electrical and thermodynamic fields on each other. In good text-books (see, for example, [1], Eq (11.28)), its derivation is exhaustively detailed. In the equation, z is the spatial variable across the direction of the changed invasion parameter, R is the gas constant, F is the Faraday's constant, T is the temperature, q the valence of the ion, $V_m(z)$ the potential, and $C_k(z)$ the concentration of the chemical ion. The two sides are the derivatives of the equation

$$V_m = \frac{RT}{q * F} \ln \left(\frac{C_k^{ext}}{C_k^{int}} \right) \quad (1.2)$$

known as *Nernst equation*. In other words, **Eqs.(1.1) and Eq.(1.2) are the differential and integral formulations of the same knowledge**. The limits of the integration are chosen arbitrarily (by choosing the reference concentration C_k^{int}).

Chapter 2

Speculations

2.1 Cell pressure

The electrolyte contains dissociated molecules (free charges), which, due to the repulsion between the ions, are located on the surfaces of the membrane and exert a pressure on the membrane surface. We assume that the relationship between the amount of the charge on the membrane and the pressure is linear, i.e., we conclude a pressure change from a charge change, and extrapolate it to the neuron in the resting state.

Numerics panel 2.1.1:
Constants from Johnston&Wu [23], page 12

Neuron's surface $8 * 10^{-9} [m^2]$

Numerics panel 2.1.2:
Other measured constants

Neuron's surface $8 * 10^{-9} [m^2]$

Physics panel 2.1.1:**Estimating the force exerted on ions in electric field**

By using that in an electric field $\vec{F} = q\vec{E}$, one can calculate the force exerted on a charge, that enables one to calculate also the particle's impulse, pressure, and energy. For example, in the case of Na^+ rush-in, the electric force field is 10^7 [V/m]. Since $([N/C] == [V/m])$, the value of the force exerted on a single ion, accelerating it across the ion channel, is

$$F_{Na^+} = 10^7 [N/C] \times 1.602 \times 10^{-19} [C] = 1.6 \times 10^{-12} [N] \quad (2.1)$$

(Here we used data from Johnston&Wu constants from **Numerics panel 2.1.1**)

Physics panel 2.1.2:**Estimating the acceleration of ions in electric field**

The force in the ion channel was calculated in **Physics Panel 2.1.1** and the mass of a Na^+ ion is $m_{Na} = 3.82 \times 10^{-26}$ kg. It causes an acceleration

$$a = \frac{F_{Na^+}}{m_{Na^+}} = \frac{1.6 \times 10^{-12} [N]}{3.82 \times 10^{-26} [kg]} = 0.42 \times 10^{14} [\frac{m}{s^2}] \quad (2.2)$$

The distance d to travel (without collision) is the thickness of the neuronal membrane 5×10^{-9} nm and we assume a continuously accelerating ion. Using $d = \frac{1}{2} \times a \times t^2$

$$t_{traverse} = \sqrt{\frac{2 \times 5 \times 10^{-9} [m]}{0.42 \times 10^{14} [\frac{m}{s^2}]}} = 1.54 \times 10^{-11} [s] \quad (2.3)$$

The acceleration of an ion is unbelievably large. It is sufficiently large to keep the potential at the same value over an extended object, provided that the ions must follow a small and slow change, such as due to the "leaking" current through the Axon Initial Segment (AIS). Similarly, the ions can 'instantly' follow quick changes such as a square wave gradient.

Physics panel 2.1.3:
Estimating the speed of ions in electric field

In **Physics Panel 2.1.2**, we estimated the acceleration and passage time of a Na^+ ion during a rush-in action. In this period the ion is accelerated to the speed $v = a * t$

$$v_{Na^+} = 0.42 \times 10^{14} \left[\frac{m}{s^2} \right] \times 1.54 \times 10^{-11} [s] = 600 \left[\frac{m}{s} \right] \quad (2.4)$$

(This speed is just a speed component along the channel. The thermal speed is about an order of magnitude higher, but not necessarily in the direction of the channel. The speed due to acceleration may convert in direction and magnitude due to a single collision, as soon as the diaphragm effect of the channel is over. The accelerated ion 'stops' as soon as the accelerating voltage is over.) See our other estimation in **Physics Panel 2.1.7**.

Physics panel 2.1.4:
Estimating the current of ion channels

Based on the time calculated in **Physics Panel 2.1.2**, we can attribute to an ion channel for this short period, provided that 10^3 ions pass through a single ion channel, a current

$$I_{channel} = \frac{10^3 \times 1.60 \times 10^{-19} [C]}{1.54 \times 10^{-11} [s]} = 10^{-5} A \quad (2.5)$$

It is in the range of several μAs , for a short period. It is distributed in the spatiotemporal surface of a neuron, and contributes to a current of dozen of pAs .)

Contents

2.1	Cell pressure	3
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Physics panel 2.1.5:
Estimating the pressure change due to rushin

Using the force from **Physics panel 2.1.1**, one can estimate how the pressure of the neural cell increases due to the rush-in changes at the beginning of the AP. As evidence shows, the local potential at the internal surface of the membrane is in the range of 100 mV in the resting state and increases by $\Delta U = 100$ mV in the transition state. This increase means a change in the force acting on an ion (see Eq. (2.1)) by $1.6 * 10^{-12}$ [N]. When we assume 10^7 rush-in ions and unchanged cell size, the total force acting on the membrane increases by $1.6 * 10^{-5}$ [N]. This change in force means on the neuron's $8 * 10^{-9}$ [m^2] surface a pressure change

$$\Delta P = \frac{1.6 * 10^{-5} \text{ [N]}}{8 * 10^{-9} \text{ [m}^2\text{]}} = 2 * 10^3 \left[\frac{\text{N}}{\text{m}^2} \right] \quad (2.6)$$

Again, the pressure can be calculated using electricity.
 (Here we used data from Johnston&Wu constants from **Numerics panel 2.1.1**)

Physics panel 2.1.6:
Estimating the force measurable due to rushin

The pressure change calculated in **Physics panel 2.1.5** enables to calculate the force measurable due to rushin. That pressure on a $10 \times 10 \text{ } \mu m$ tooltip of the measuring device is converted to a force $2 * 10^3 * 100 * 10^{-12} = 200 \text{ pN}$ force. The measured force value [43] is about 600 pN , so our estimation is in the correct range.

The measured pressure value is [?] $5 \frac{\text{dyn}}{\text{cm}^2} = 0.5 \frac{\text{N}}{\text{m}^2}$, which is an average value for some period. If we assume a 1 Hz frequency, and a 1 ms period for the duration of the AP peak, it means $1 * 10^3 \left[\frac{\text{N}}{\text{m}^2} \right]$. So, the measured mechanical change values [?] underpin our hypothesis that the increased pressure increases the neuron's size via its elasticity.

Physics panel 2.1.7:
Estimating the rush-in speed

A way to estimate the ions' rush-in speed is that one assumes that the Stokes-Einstein speed describes the ions' speed (Fig. 3 in [5], quantitatively underpins the hypothesis, see [?]) both in axons and ion channels. If so, the ratio of the potential gradients equals the ratio of the speeds of ions. One can estimate that the 100 mV AP on the 50 μm AIS generates a $2 * 10^4 \frac{V}{m}$ electrical gradient, and [5] measured 20 $\frac{m}{s}$ for the speed of the AP. By assuming that the electrical field across the membrane is $10^7 \frac{V}{m}$, one can estimate the speed at the exit of the ion channels as $10^4 \frac{m}{s}$. See our other estimation in **Physics Panel 2.1.3**

Physics panel 2.1.8:
Estimating the rush-in mechanical work

By using the result derived in **Physics panel 2.1.5**, one can also estimate the work done when the rush-in charge extends the neuron, as the $\Delta P \times \Delta V$. We consider that ΔP remains constant, and we calculate the change of the volume as neuron's surface $10^{-8} m^2$ multiplied by the change of the radius $\Delta r = 10^{-9} [m]$ multiplied by the pressure ΔP , that gives $E_{mech} = 2 * 10^{-14} [J]$. If one assumes that the pressure is of entirely mechanical origin [?], one must assume that the pressure is $10^6 \frac{N}{m^2}$, which is well above the measured value.

Physics panel 2.1.9:
Estimating the rush-in electrical work

We can also estimate the electrical energy from reset to threshold, see also [31]

$$E_{electr} = \frac{1}{2} C * (\Delta V)^2 = 1.4 * 10^{-12} [J]$$

see also [31].

Physics panel 2.1.10:
Estimating the rush-in time period

To estimate the time period required to change the pressure calculated in **Physics panel 2.1.5**, one can assume that the collision force of a Na^+ ion provides the pressure (the shock wave), and this can be calculated as $F = m \frac{\Delta v}{\Delta t}$. Given that Eq. (2.1) provides the force, and one must assume that the ion entirely loses its 10^4 to $10^5 \frac{m}{s}$ speed, one concludes that it generates a shock wave in $\Delta t = 10^{-8}$ to 10^{-7} seconds.

Physics panel 2.1.11:**Estimating the dissipated energy of thermodynamical cycle**

We estimate the work done when the rush-in charge extends the neuron, as the $\Delta P \times \Delta V$. We consider that ΔP remains constant, and we calculate the change of the volume as neuron's surface 10^{-8} m^2 multiplied by the change of the radius $\Delta r = 10^{-9} \text{ [m]}$ multiplied by the pressure ΔP , that gives $\mathbf{E}_{\text{mech}} = 2 * 10^{-14} \text{ [J]}$, as we calculated in **Physics panel 2.1.8**. [?] reported that an AP dissipates $Q_e = 0.25 \text{ } \mu\text{J m}^{-2}$, which on a $8 \times 10^{-9} \text{ m}^{-2}$ surface means $0.4 \times 10^{-14} \text{ J}$. As discussed in **Physics panel 2.1.9**, the electrical work is $\mathbf{E}_{\text{electr}} = 1.4 * 10^{-12} \text{ [J]}$.

The theoretically lowest possible energetic cost of information is $k_B T \ln(2)$, numerically ca. $3 \times 10^{-21} \text{ J}$ i.e., compared to the optimal limit set by physics, this efficiency value is 10^6 - 10^8 times less energy efficient, in line with [31]. Nature protests: the phenomenon is different from the one biology claim. In the neuron there is no single state which can be set or reset.

Chapter 3

A neuronok elektromosságtana

A kiváló tankönyv [2] (természetesen Hodgkin and Huxley (HH) (V1.0) szellemében) a neuron membrán állapotára a nyugalmi és az átmeneti állapotokat vezeti be. Az az állítása, hogy "the resting membrane potential results from the separation of charge across the cell membrane" csak az igazság egyik fele; mi elmondjuk a másik felét is a ?? szakaszban. Finomítjuk "by discussing how resting ion channels establish and maintain the resting potential" [2], 126. oldal, állításukat: az ioncsatornáknak csak a nyugalmi potenciál fenntartásában van szerepük, a létrehozásában nincs. Amint a 3.4 szakaszban tárgyaljuk, két különböző fizikai mechanizmus működik a két említett állapotban. A neurofiziológia tévútra kerülésének egyik oka az volt, hogy a nyugalmi állapotban (lényegében helyesen leírt) állapotot az átmeneti állapotra próbálták alkalmazni.

A [2] könyv felteszi ugyan a a központi kérdéseket "How do ionic [i.e., electrical and chemical] gradients contribute to the resting membrane potential? What prevents the ionic gradients from dissipating by diffusion of ions across the membrane through the resting channels?", de lényegében megválaszolatlanul hagyja azokat, mivel csak a membrán átjárhatóságát tárgyalja, a nyugalmi potenciál létrejöttének oka és szabályozásának mechanizmusa, a feltöltődés és polarizáció szerepének figyelembe vétele nélkül. A 3.4 szakaszban bemutatjuk, hogy a klasszikus elektromosságtan módszereivel ki tudjuk számítani a membrán potenciált, bár a teljes körű leíráshoz az elektrolitok speciális járulékát is figyelembe kell vennünk. Eredményeink alátámasztják, hogy "the crucial system in biology isn't a molecule or a molecular class whatsoever, but the interface created by biomolecules in water" [3]. Hozzáteesszük azonban, hogy főként fizikai folyamatok és sebesség gradiensek vezérlik a folyamatokat. Pontosabban, elválaszthatatlan termodinamikus és elektromos folyamatok tartják a potenciált azon az értéken, amelyet a sejt lipid/protein szerkezete által meghatározott elektromos és termodinamikus gerinc definiál. Ez a folyamat egyike azoknak a gyakori eseteknek, amikor az egyik hatás megvalósítja a

funkcionalitást (a gerinc szorosan meghatározza a működés kereteit) a másik pedig (ez esetben egy kombinált termodinamikai és elektromos hatás) korrigálja azt, miközben megvalósítja a komplex (dinamikus) funkcionalitást. "stimulated phase transitions enable the phase-dependent processes to replace each other ... one process to build and the other to correct" [4].

To avoid "reinventing the wheel", we heavily rely on the excellent site HyperPhysics, hosted by the Department of Physics and Astronomy of University of Georgia. Their material is excellent for discussing engineering electricity, where the charge carriers are electrons. However, they extended their material with bioelectric phenomena, tacitly agreeing with some fallacies created by biophysics. They do not scrutinize the differences caused in the electric phenomena by the structure of living matter, the effects of the limited volume of a cell, using slow positive ions in electrolytic solutions instead of fast free electron clouds in solids, the interaction with thermodynamics, among others. So, we must discuss the deviations in detail, which originate from the same reasons as discussed in section ??.

One of the famous dichotomies (see section ??) is that there exist laws, separately, for continuous and discrete electricity, without providing a transition between them. The fundamental concepts are the same, but we must rethink several approximations. Furthermore, sometimes we must consider the classical continuous electricity, sometimes particle-based electricity; sometimes the mixture of the two, or a transition between them. In biological cells, the usual "infinite" conditions (infinitely far, infinitely small, etc.) of the classical electronics are not fulfilled: it depends on the context what can be omitted with respect to some other quantity or size. Say, the charge in the intracellular volume is "infinitely smaller" than that in the extracellular volume, but "infinitely larger" than the charge handled in an ion-channel-related operations. The biological organization prevents discussing living matter as homogeneous and isotropic medium, the active "charge production" in some biological constructions needs a precise interpretation of the conservation laws, and the charge carriers' speed needs revisiting the interpretation of the fundamental physical laws.

3.1 Miért más az ionok elektromosságtana

This section recalls dichotomies discussed in section ?? from the viewpoint of electricity. Electricity is one of the fundamental disciplines of classical physics. It operates with concepts such as charge, current, voltage; furthermore, their spatial or temporal change described by their gradients (spatial or temporal derivatives). Physical and biological electricity are similar, but not identical. The former was already an established discipline when the latter started to discover that biological objects also show signs of electric processes occurring inside them. Of course, the already available concepts, laws, measurement devices and procedures were taken over, and based on the resemblance between those two kinds of electricities, their identity was and is assumed. The laws of physical electricity, in most cases, describe also biological electricity to some measure.

However, in some cases, there are drastic differences between them. Not considering those differences leads to misunderstandings and wrong conclusions.

Physical electricity operates with concepts of ideal (voltage or current) generators, ideal (that is single-substance) charge carriers, and discrete components, connected by ideal conducting wires. The wires are passive components, just connect the discrete components and the electric phenomena happen in the discrete elements, with one idealized feature. Physical electricity inherited the Newtonian picture that everything happens simultaneously and it applies a hybrid picture. It considers that no delay exists on the connecting wires, although charge passes in the current. The current is defined as temporal change of charge, but it is explicitly considered only in the slightly non-ordinary phenomena capacitance, inductance and impedance; it is considered as instant in the phenomenon called resistance. In biological electricity, neither of the above disciplinary concepts can be interpreted in unchanged form, they must be reinterpreted. The charges are moved not necessarily by an electric field, but their movement is interpreted as current (and, simultaneously, as mass transport) furthermore, that movement creates change in the potential (as well as in the concentration), due to the limited resources. Since the charge carriers are slow, local, instead of global, potentials must be used. The integral forms of laws of electricity are not valid in the form we used to in physical electricity. The charge (and mass) conservation are valid only in the global sense; while in classical physics, with its concept 'instant interaction', local and global coincide. Fortunately, physiology inherited the ambition from physics to measure its subject, together with the measurement methodologies and devices. Unfortunately, as we detail in section ??, measuring living matter (because of its "special construction") requires at least considerable care, sometimes different methodologies and appropriate interpretation; including the understanding of the special kind of electricity it applies.

The different charge transmission mechanism introduces another important difference, the relation to thermodynamics. In physical electricity, the medium is a solid, where the electrons heavily interact with the nodes in the solid (forming a kind of 'electron cloud'). In a good approximation, they are not comprising free and independent particles, so their phenomena are not related to thermodynamics. The ions, in contrast, are nearly independent, so they would be described by thermodynamics. However, they are simultaneously also charged particles, so they are governed also by the laws of electricity. These two circumstances together represent a case which is not belonging to either discipline. More precisely, neither of the two disciplines can describe the phenomena without the other. The case is somewhat similar to the case when an ion moves in simultaneously applied electric and magnetic fields, and it is considerably different in that the thermodynamic interaction also represents a mass transport, inseparably, so the driving force is the sum of the thermal and electric gradients. *The real complication and difference is, however, that the interactions have different speeds.* However, it is not a different physics ("or whatnot"). It is only using different laws for the different environment from the same first principles.

3.2 Fundamental concepts

The disciplinarily separated science defines electricity's concepts through forces and infinitely small test charge. In living matter, neither is valid in an unchanged form: constraints and physical/chemical additional forces are present, furthermore, the charges of the carriers are not negligible compared to the charges that generate the field acting on the charges. Furthermore, *since the carriers have mass, one cannot apply laws of disciplinary electricity since they are valid for "slow" currents in a different form.* One may have fundamental problems with the mathematical descriptions: the discrete nature of charge carriers question marks the differentiability of the corresponding functions, the charge-mass dichotomy of ions the interpretation of partial derivatives, furthermore, the simultaneity of different phenomena.

3.2.1 Töltés

Az ion tovább már nem egyszerűsíthető tulajdonsága, hogy töltést hordoz

3.2.2 Elektromos erőter

3.2.3 Feszültség

Az ion tovább már nem egyszerűsíthető tulajdonsága, hogy töltést hordoz

3.2.4 Feszültség

Az ion tovább már nem egyszerűsíthető tulajdonsága, hogy töltést hordoz

Ionic currents in a cell

On a microscopic scale, a charge generates a potential field that acts on other charges. It is a common fallacy in physiology that a "potential field travels": neither potential nor current changes without a physical movement of charges. In a conducting wire filled with ions (the solids have a different current transfer mechanism), there are free charges; their number per unit volume is given by n , and q is the amount of charge on each carrier. If the conductor has a cross-section of A , in the length dx of the wire, we have charge $dQ = q * n * A * dx$. If the charges move with a macroscopic speed $v = \frac{dz}{dt}$, at the macroscopic level, we define the current I as the charge moved per unit of time as

$$I = \frac{dQ}{dt} = q * n * A * v \quad (3.1)$$

The effect was hypothesized at the very beginning: "it seems difficult to escape the conclusion that the changes in ionic permeability depend on the movement of some component of the membrane which behaves as though it had a large charge or dipole moment." "it is necessary to suppose that there are more carriers and that they react or move more slowly" [5]. The charged ions, attracted by the

opposite charges on the other plate and repelled toward the single exit point AIS, really move slowly and behave as if they were a component of the membrane.

Notice that if any of the factors is zero, the macroscopic current I is zero. *Microscopic carriers must be present in the volume*, and have charge, the cross section must not be zero, and the charge carriers must move with a potential-assisted speed, which needs an electrical field generated by the system, see Eq.(3.18). When there is no constraint force, the thermodynamic and electrical forces will produce an effective force that moves the ions. That is, the ions can move without an external electrical field, as experienced in biological systems. One of the fundamental mistakes by HH was neglecting the Coulomb force in ion-electrical interactions and assuming that diffusion processes or ideal electrical batteries create the electrical potential.

In biological systems, where ions represent the "current", it is either a native current (without an external potential) or an artificially injected current or potential generating a current. This way, the current is consistently accompanied by a change in concentration gradient, as the moving ions represent both mass and charge transfer simultaneously (also, the repulsion leads to a "skin effect": in electrical wires and axons, the charge travels near the outer surface). The potential and current are connected through the features of the medium (material) that hosts our measurement.

Current's speed

According to Stokes' Law, to move a spherical object with radius a in a fluid having dynamic viscosity η , we need a force

$$F_d = 6 * \pi * \eta * a * v \quad (3.2)$$

(drag force) acting on it. A (microscopical) electrical force field $\frac{dV}{dz}$ inside the wire would accelerate the charge carriers continuously

$$F_{electric} = k * E(z) * q \quad (3.3)$$

with a constant speed v . It is not the *drift* speed because of the electrical repulsion, it is a *potential-assisted* or *potential-accelerated* speed that can be by orders of magnitude higher. The medium in which the charge moves shows a (macroscopic, speed-dependent) counterforce F_d , which in steady state equals $F_{electric}$, that is :

$$I = \frac{k * q^2 * A}{6 * \pi * \eta * a} * n * E(z) \quad (3.4)$$

We extend the original idea by using

$$E(z) = E_{electric} + E_{thermal} + E_{external} \quad \text{in electrolytes}$$

that is, for living matter

$$E(z) = \frac{\Delta V}{\Delta z} \quad \text{in linear potential} \quad (3.5)$$

$$= \frac{dV}{dz} \quad \text{in graded potential} \quad (3.6)$$

$$= E_{electric} + E_{(thermal)} (+E_{(external)}) \quad \text{in electrolytes} \quad (3.7)$$

The amount of current in a wire is not only influenced by the electric force field (specific resistance) but also by the number of charge carriers n . While the latter is commonly considered constant and part of the former, this is not necessarily the case for biological systems with electrically active structures inside. The medium's internal structure ("the construction) introduces significant modifications. Applying an electric field to a biological wire can generate varying amounts of current as the number of charge carriers changes due to biological effects. For axons, we use a single-degree-of-freedom system, a viscous damping model, so the ions will move with a field-dependent constant velocity in the electrical field; so it takes time while they appear on the membrane. The activity of potential-controlled ion channels in its wall may change n in various ways; furthermore, that change can result in 'delayed' currents during measurement, for example, in clamping, as physiological measurements witness.

If we have a concentration $C(z)$, in the volume $A * dz$, we have $dQ = C(z) * A * dx * q$ charge, resulting in another expression for the current

$$I = \frac{C_k(z) * A * dz * q}{dt} = C_k(z) * A * q * v \quad (3.8)$$

By combining equations (3.4) and (3.8):

$$v(z) = \frac{k * q}{C_k(z) * \eta * 6 * \pi * a} * E(z) \quad (3.9)$$

The higher the resulting space derivative (gradient) and the fewer ions that can share the task of providing a current, the higher the speed. We hypothesized (it needs a detailed simulation) that in the case of this charged fluid, the electrical repulsion plays the role of 'viscosity' [6]. The higher the charge density, the stronger the force equalizing the potential; so η is the lower, the higher the charge density (proportional to C_k). For the sake of simplicity, we assume that the speed is proportional to the space gradient of the local concentration and potential. Recall that *our equations refer to local gradients only. The electrical gradient can propagate only with the speed of the concentration gradient*, given that only the chemically moved ions can mediate the electrical field [6]. *The lower interaction speed limits the other interaction speed if the interactions generate each other.*

The dependence of the diffusion coefficient on the viscosity can be modeled by the Stokes-Einstein relation:

$$D = \frac{k * T}{6 * \pi * \eta * a} \quad (3.10)$$

so we can express the speed with the diffusion coefficient D and temperature T

$$v(z) = \frac{D}{T} * \frac{q}{C_k(z)} * E(z) \quad (3.11)$$

We introduce a *time-dependent speed gradient* $v(z, t)$ which plays a vital role in the processes that occur in living matter.

It is important to remember for alternating current experiments: the different ions will move at different speeds under the effect of the same $E(z, t)$, which is the function of the local concentrations and the frequency of the alternating current. The conduction speed sensitively depends on the temperature, and through it, the shape parameters [7] of the AP and especially the length of the "relative refractory" period [8].

Imitating slow currents

At the dawn of finding methods for describing neuronal operation, HH published high-precision measurements [5] enabling detailed testing of theories explaining the observed physiological behavior. *Their good physical model that "movement of any charged particle in the membrane should contribute to the total current" lacked considering the mutual repulsion and finite speed* of those particles (they are about 50,000 times heavier and a million times slower than the electrons, and they move in a closed space segment). Furthermore, they have started from the commonly used wrong assumption that conductance is a primary electrical entity. They could "find equations which describe the conductances with reasonable accuracy and are sufficiently simple for theoretical calculation of the AP and refractory period". The "reasonable accuracy" and the need for a simple calculation method (performed on a mechanical calculator) obscured the fact that the laws for electrons are not the same as the laws derived for ions.

As discussed in connection with Eq. (3.1), the current depends linearly on the number of charge carriers n . The physical processes changing the number of charge carriers also define the intensity of the currents, providing a way to imitate slow currents by fast currents (by using current generators using special function shapes), where the spatiotemporal time course of the physical process of producing ions in the system modulates the number of charge carriers.

As an example, we discuss the case of slow current traveling in an axon. The ions travel a finite distance with a finite speed, so there must be a delay between their entry and exit times. The case of switching a clamping voltage on, to some measure, is analogous to the arrival of a spike. Initially, the axon contains no ions. In the classical picture, the axonal current flows into the membrane with capacity C_m and increases the membrane's voltage V_m with a time constant

$$\frac{dV_m}{dt} = -\frac{1}{C_m} I_{axon}; \quad V_m(t) = \frac{I_{wall} * (1 - e^{-\alpha * t})}{C_m} \quad (3.12)$$

that generates a change in the membrane current

$$\frac{dI_m}{dt} = \frac{1}{R_m} \frac{dV_m}{dt}; \quad I_m^{on}(t) = g_m(V) V_m(t) \quad (3.13)$$

where $g_m = \frac{1}{R_m}$ is the conductance of the membrane. That is, the measurable current equals the product of the conductance and the clamping voltage. Eqs.(3)-(5) in [5] express this relation. *If we assume that the axonal current is “fast”, we arrive at the wrong conclusion that the conductance is voltage- or time-dependent.* In contrast, if we assume that the axonal current is “slow”, we naturally conclude that Ohm’s Law is correct and valid also for biology: *the conductance/resistance is constant, but the number of charge carriers changes,* see section ??.

There is no voltage-dependent conductance [9]. Instead, the finite speed of ions and the wrong assumption that conductance is a primary entity mislead physiological research. *With wording that “conductance changes”, one states that charge carriers appear/disappear/reappear; that is, the charge and mass conservation are not fulfilled.* Claims such as “the current cannot disappear, it has to go somewhere” [1], page 9, are wrong: *there is no conservation law for current, which is the derivative of charge with respect to time.* The correct physics background behind the early phenomenon [10] is that the number of charge carriers, at the cross section where the measurement is carried out, changes as a wave of ions with variable intensity moves slowly along the axon (they are apparently “created” for the static measurement arrangement). The changing conductance is a fallacy stemming from the idea of “fast current”, which is valid for electrons but not for slow ions.

In contrast, when the clamping voltage is switched off, the axon is still filled with charge carriers. The resting potential reaches the end at the membrane “instantly” (in this situation, the conduction mechanism is similar to that in metals). The driving force disappears, the ion stream stops, and the ions discharge into the membrane. The lack and the presence of ions in the axon when switching clamping on and off, respectively, produce the difference that “*conductances [actually, currents] increase with a delay when the axon is depolarized but fall with no appreciable inflexion when it is repolarized*” [5]. The potential is equalized by the AIS current, producing a net exponential decay:

$$I_m^{off} = I_{W_{all}} * e^{(-\frac{\alpha}{R_m C_m} * t)} \quad (3.14)$$

During the regular operation of a neuronal membrane, after opening the ion channels, a vast amount of ions flows into the intracellular space from the extracellular space, somewhat similar to the effect of switching a clamping voltage. The essential difference is that the ions arrive through the axon to the joining point in clamping. In contrast, through the membrane’s ion channels, they directly contribute to the current on the membrane’s entire surface. The membrane’s size is finite, so with a finite current speed, it takes time until the charges on the membrane’s surface arrive at the AIS, as discussed for the axonal current. Fig. 3.1 shows the measured rising and falling edges of the clamping experiment performed by HH [5]. The continuous lines show the diagram lines fitted to the experimental data (notice again that the fitted polynomial significantly differs from the correct exponential function around the time 0.5 msec); the dashed line shows the expected result of the rising edge, calculated with

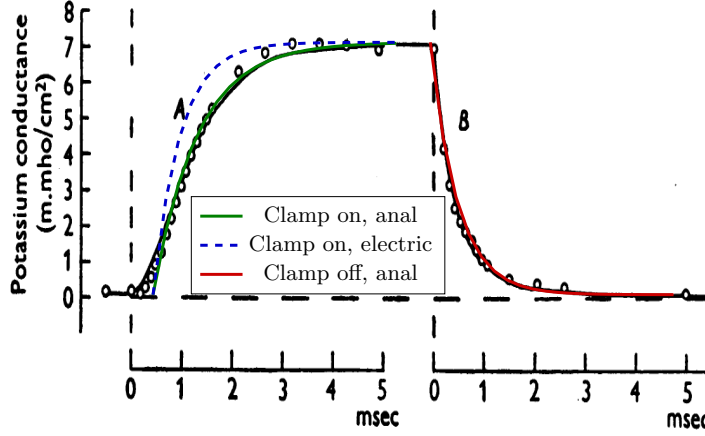


Figure 3.1: The measured current, instead of conductance, (Fig. 2 from [5]), demonstrating that two different physical effects result in the rising and falling edge of the measured current.

the time constant of the falling edge. The number of charge carriers follows a saturating curve, which decreases the current in the axon. HH measured that their time constants differ far outside their experimental error, which would not be possible if they were the result of the charge-up and discharge of the same oscillator circuit. The measured rising edge corresponds to a charge-up process where the charge-up current is a saturating curve due to the change in the number of charge carriers in the axon; see Eq.(3.4).

The *slow* input currents (although due to slightly differing physical reasons for the different current types) are described by the following analytic form valid for *fast* currents:

$$I_{in} = I_o * (1 - \exp(-\frac{1}{\alpha} * t)) * \exp(-\frac{1}{\beta} * t) \quad (3.15)$$

where I_o is the current amplitude per input, and α and β are timing constants for current in- and outflow, respectively. They are (due to geometrical reasons) approximately similar for the synaptic inputs and differ from the constants valid for the rush-in current. To implement such an analog circuit with a conventional electronic circuit, voltage-gated current injectors (with time constants around 1 msec) are needed. For example, see in Fig 3.2 the PSP measured function shape published in [11], fitted with our theoretical function Eq.(3.15). The corresponding voltage derivative is as follows:

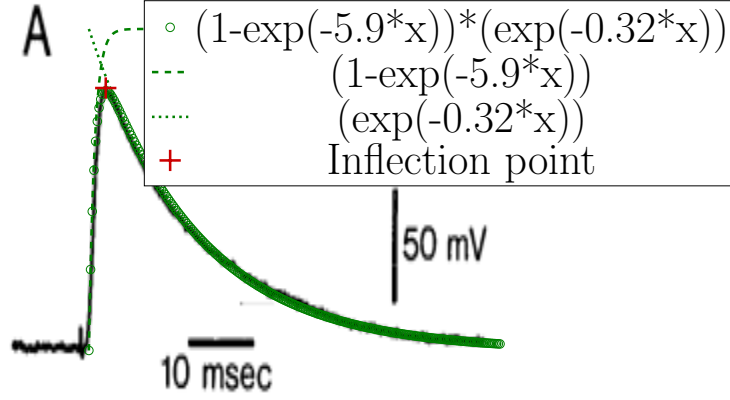


Figure 3.2: The Post-Synaptic Potential (PSP) shape measured by [11] and fitted by function forms given by Eq.(3.15).

$$\frac{dV_{in}}{dt} = \frac{I_o}{C} * \left(\frac{1}{\alpha} * \exp\left(-\frac{1}{\alpha} * t - \frac{1}{\beta} * t\right) - \frac{1}{\beta} * \exp\left(-\frac{1}{\beta} * t\right) * \exp\left(1 - \exp\left(-\frac{1}{\alpha} * t\right)\right) \right) \quad (3.16)$$

3.2.5 Condenser

Cell biochemistry sufficiently well describes the lipid bilayers that constitute the membrane [12, 13]. The negative and positive ends of the lipids that constitute the membrane, in an electrolyte solution, attract and bind ions, forming a charged layer on the membrane surfaces. In this way, the membrane in electrolytes acts as a condenser, comprising two charged sheets on its surfaces, with ill-defined parameters. Their boundaries, thicknesses, and compositions depend on the operating state (unlike conducting plates with well-defined sizes and boundaries). The condenser is not ideal: the charged sheet comprises ions with attached aqueous constituents from the electrolyte, so its thickness is far from the ideal zero thickness of classical electricity. Furthermore, in its wall, it inseparably comprises ion channels (from an electrical point of view: resistors), and the condenser plates are in polarizable electrolyte solutions.

Cell science does not establish the membrane's electrical phenomena; instead, it attempts to elucidate complex protein activities that explain charge transfer, resting and transient potentials, and explains the entire *biological operation using mathematical formulas without a realistic physical background*. Electrophysiology combines currents of electrons and ions, and *uses Ohm's Law* for

its admittedly non-Ohmic systems. It introduces foreign (clamping) currents into biological systems, attributing their effects to changes in membrane conductance without explaining how *voltage and current can be independent of charge carriers.*

3.2.6 Resistor

Ions differ from electrons we used to in physical electricity in many features. Their conduction mechanisms are entirely different, and they are not necessarily present in the discussed medium (say, when the medium is surrounded by a charged surface, the field moves free ions out of the volume). They may be produced by biological mechanisms, and they are slow (have a couple of m/s speed, creating the illusion of 'delayed current'). Both disciplines provide an atomic level description, but unfortunately they use the same words for their concepts, although their meaning can be entirely different.

In solids, as the Drude model describes, electrons constantly bounce among heavier, stationary crystal ions, that make up the structure of the material. With each collision, though, the electron is deflected in a random direction *with a speed that is much larger than the speed gained by the electric field.* The net result is that electrons take a zigzag path due to the collisions, but generally drift in a direction along the electric field. Important, that *the collisions are inelastic*, so the electrons lose (most of) their energy. The solid's gridpoints absorb the energy, which later is released in form of heat. The process is irreversible.

In biological matter, ions convey electricity. So called ion channels are used for the transfer. Here diffusion and electromigration takes place, by orders of magnitude different speeds. The process is mostly reversible. The resistance is interpreted as the interaction of charge carriers with the particles in the material that are transported by a driving force (that can comprise also a force of chemical origin). In solid state, the volume considered is "infinitely large" (that is, the field is homogeneous and the propagation is isotropic), and the charge carriers are uniform that is only one type of interaction is present. In biological objects, the volume is more or less limited (is finite), there may be charged objects present that make the propagation non-isotropic. Furthermore, the charge carriers are non-uniform that introduces thermodynamic interaction in addition to the electrical interaction. Ions, under the electrical/thermodynamic field (and maybe mechanical constraints) accelerate to the Stokes-Einstein speed (see Eq.(3.9)), move against a friction in the electrolyte. The ions suffer (essentially elastic) collisions with neutral molecules and transfer their momentum (and so: energy) to them. From the point of view of ions, the process is energy loss and momentum loss. From the point of neutral molecules, the process is energy gain and momentum gain. In a free space, the energy and momentum quickly distribute in the entire volume, so the process is finally irreversible.

If the acceleration occurs in a narrow tube such as an ion channel, the wall prevents gaining momentum in the direction perpendicular to the direction of acceleration. Given that the collisions are elastic, no energy dissipates; so, in principle, the process is reversible. Strangely enough, this way the electrical

force accelerates also the neutral molecules; furthermore, their momentum (using an elastic membrane) can fully contribute to the operation of neuron (forms the AP).

In a narrow tube (such as an ion channel) the momentum exchange is limited to one direction. If the ions are accelerated through the channel, due to the frequent collisions, they can entirely transfer their energy to a large number of neutral molecules. Those accelerated molecules having the same speed may represent a significant "impact force", sufficient to cause significant mechanical changes throughout biological systems. The momentum transfer by collisions is a two-way process, it is reversible also in the practice. If a small amount of ionized particles is present in the tube, under an external electrical field, the charged particles can accelerate the neutral ones. Vice versa, if the bulk of the particles are moved mechanically (say, when an ultrasound wave presses the neuron or a rush-in process transfers electric energy to the membrane), the one-directionally moving neutral molecules transfer momentum to the charged particles, this way producing measurable electrical waves (such as an AP).

The secondary electric quantities can be derived from the primary quantity: the bulk of the undissociated molecules 'resist' transferring ions under the effect of an electric field (the liquid must be pressed through the tube). One can measure potential and current along the ion channel (and so, to calculate resistance), but the behavior of the resistance/conductance is drastically different. One has a pneumatic system instead of a voltage generator and the resistance originates in viscosity. Most of the energy is converted to mechanical energy instead of heat, making the process (mostly) reversible.

The AP initially produces an impulse force that compresses the membrane. The compressed membrane, while performing a damped oscillation, pushes and pulls the incompressible electrolyte that contains a small amount of dissociated ions, this way (per definitionem) creating an observable current and voltage wave. In the positive and negative half periods of the elastic oscillation, due to opposite elastic/pneumatic forces, the current and voltage change their sign. Given that the charges move at their Stokes-Einstein speed, the voltage-, current- and pressure waves are synchronized, as observed. The different experimental observations measure the same effect by using their disciplinary measuring devices.

The energy of the ions cannot dissipate; instead, the neutral and charged particles share it, and convert between kinetic and elastic energies (the electric energy is much smaller). That is, the neuron can produce heat in the positive phase of the elastic oscillation (the electrolyte is compressed), and can reabsorb heat in the negative phase (decompression), in full accordance with the correctly interpreted laws of physics. This way, by introducing the correct interpretation of "resistance" for ion channels, the seven-decades-old contradiction [14] between the electrical and thermodynamic observations are finally resolved, furthermore, the conflicting theories can be naturally unified.

3.3 A membrán elektromosságtana

The importance of the subject was correctly estimated decades ago [15]: "For the past forty years, our understanding and our methods of studying the biophysics of excitable membranes have been significantly influenced by the landmark work of HH. Their work has had a far-reaching impact on many different life science sub-disciplines where concepts of cell biology have come to be important. The ionic currents and electrical signals generated by neuronal membranes are of obvious importance in the nervous system." To fix the flawed and incomplete hypotheses has a similar importance. Recall that if you have the HH model in mind, and measure with a voltmeter, you will surely measure voltage and current, and nothing more (no other quantity). This way the accompanied other changes can remain hidden.

3.3.1 Elektrolitok

Specific chemical substances can naturally hold either a positive or negative electrical charge and react to their micro-, macro-, and electrical/mechanical/thermodynamic environments. A molecule has internal electrical forces that keep its ions in place, so it has two charge centers (dipoles). When another dipole or a macroscopic external electrical field (which can be of electrical, magnetic, mechanical, or chemical origin) appears near the molecule, its perturbing effect can affect the relation of the ions to each other. The two charge centers initially increase their distance (the molecule polarizes). When that disturbance is strong enough, the ions can entirely separate (the molecule ionizes). The local electrical field fluctuates, so their state is dynamic: molecules dissociate, and free ions recombine. Furthermore, the process also plays an important role in the energy supply of neurons: the strong electric field near the membrane hydrolyzes Adenosine Triphosphate (ATP) molecules, and moves ions to the "condenser plates", thereby accumulating "potential energy". (The claim that the ions pass through ion channels by using the "downhill method" does not use energy, is wrong: they are accelerated by the condenser's field and consume energy in that way.) Depending on their environment, substances can exist in base, ionized, and polarized states (and exhibit corresponding behaviors).

(In physics, the term 'polarization' is reserved for the redistribution of charges in multiply charged systems without generating an external current. Changes in the membrane's potential difference are a consequence of complete charge separation (and the movement of slow charges), even though, unfortunately, physiology refers to it as polarization. As discussed below, true polarization is also present, which confuses the description. External currents contribute to the changes of potential.)

In biological cells, in a resting state, a small (approximately 10^{-3}) fraction of the molecules dissociates (i.e., ions separate from their counterparts with opposite charges) and moves freely within the volume. In other words, an electrolyte liquid can conduct electricity (under the effect of an electrical or thermodynamic

field) because the positive and negative ions are mobile. Notice, however, that, unlike the electrons in solids, the ions do not form a "free electron cloud": they must move "in person", that is they deliver charge with the speed of the carrier. The rest of the molecules can be more or less polarized, allowing for the possibility of producing an internal electrical field in the solution. In a transition state, all components (including ions, molecules, and ATP) have gradients, and they move towards their balanced state. Here comes into the picture that in the biological structures, the ions are in a closed space and counterforces may exert on them. When a mechanical counterforce does not keep back an ion, it moves out from a space where electrical or thermodynamic gradient is present. That is, near biological objects that are charged and inside them, in a balanced state, no freely moving ions can be present.

In the absence of external influences (that includes the lack of a separating membrane), both gradients are balanced and are at zero. When something perturbs this state, the system attempts to find a new balanced state, using temporal processes. The details of how it reserves and restores its balanced state are discussed in section ???. In this scenario, two interactions can influence the 'carrier' (the ion). We discussed in section ?? and in [16] that those phenomena in living matter need deriving and applying 'non-ordinary' laws of science, and the laws of motion of biology can describe the time course of those processes. Fundamentally, due to the mixing of interaction speeds, considering only one of the interactions leads to fundamentally incorrect conclusions.

The cellular thermoelectrical phenomena are very complex, and it is not a simple task to choose which physical/chemical effects can be omitted so that their omission does not prevent us from explaining physiological phenomena. When ions are contained in a closed volume, they exist in a state of thermal and electrical equilibrium. In the absence of external influences (that includes the lack of a separating membrane), both gradients are balanced and are at zero. When this state is perturbed, the system attempts to find a new balanced state, using temporal processes. In this scenario, the 'carrier' - the ion - can be influenced by two different types of interactions, each represented by a distinct abstraction in these processes. In this section we discover some conditions how electrical charges in electrolytes, furthermore, electrical and thermodynamic processes (excluding biochemical details) cooperate for concerting processes commonly called 'life'. We discuss that those phenomena are not against the laws of science, only against discussing them in terms of a single grasped science discipline. Simply, the structures in living matter shall be discussed at another abstraction level, and those structures need deriving and applying 'non-ordinary' laws of science. More precisely, we must discuss them in a non-disciplinary way, due to that the participating ions belong simultaneously to the disciplines electricity and thermodynamics. When measuring them, we must consider that "the construction is different from anything we have yet tested in the physical laboratory". Actually, ions' behavior is not different; the environment ("the construction") is different; see section ???. Fundamentally, due to mixing interaction speeds, considering only one of the interactions (the disciplinary discussion) leads to wrong conclusions.

We discover that in segmented electrolytes, the interplay of disciplines, especially if the segments have largely different concentrations, produce very thin but important layers, which, by using 'Maxwell's-demon'-like objects, produce self-contained phenomena known as 'signs of life' in biology. Furthermore, the features of living matter may change during measuring them (by their internal laws) in a way usual in physics laboratories. Given that, in many cases, inappropriate physical principles, notions, and methods are used in measuring and modeling neurons, we need to discuss the true physics (the correct approximations, valid in cross-disciplinary approach) behind biological phenomena.

3.3.2 Membrán elektrolitban

Az elektrolitokban működő membrán alapvető fontosságú a neuronális működés számára, így azt részletesebben tárgyaljuk. Az anyag nagy részének megértéséhez azonban még az egyetemi fizikai végzettség sem elegendő, így szokatlanul sok anyagot helyeztünk el a fizika és matematika panelekbe, és azokat a törzs-anyagban megpróbáljuk az említett tárgykörökben kevésbé járatos olvasónak is érthetővé tenni.

A membrán működése szorosan kötődne az élő anyaghoz. Azzal, hogy ezt az anyagot a fizikához kötődő részben tárgyaljuk, azt kívánjuk hangsúlyozni, hogy nem szükséges valamiféle 'életerő' a (avagy 'isten érintés') a biológiai anyag működéséhez, azt a (megfelelően alkalmazott) fizika le tudja írni; bár az lényegesen nagyobb körütekintést és diszciplínákon átívelő szemléletet igényel.

3.3.3 Inhomogeneities in the solution

In a segmented electrolyte we experience different forces, furthermore, the electrical fields have different sources. For a closed system, the electrical charges are balanced, but locally they may be unbalanced due to physical reasons. In a steady state, some other force must counterbalance the mentioned forces. That force may be a mechanical one: the ions sitting on the surface of the membrane press the surface due to the attractive force on ions and the membrane mechanically provides the needed counterforce. In addition, due to a concentration gradient, a thermodynamic force may act on the ions.

When any inhomogeneity is present (an ion is forcefully moved, by investing energy, to a place other than the one which is needed to be in a steady state), the ions may move due to different reasons, which, per definitionem, means current (and means potential energy). Notice the important aspect that, at a microscopic level, moving an ion simultaneously means redistributing charge and mass. At a macroscopic level, it results in simultaneous changes in the local macroscopic characteristics such as concentration and potential. These observations are expressed by Onsager's reciprocal relations [17].

That means when describing an ionic transfer process, *we must not separate the electrical current from the mass transfer*: they happen simultaneously, and mutually trigger each other. Notice that *the thermodynamic term is ion specific, while the electrical term is not*. To be entirely balanced, the system must be

balanced to all elements. In this way, changing one concentration implicitly changes all other concentrations and the electrical field.

It is a huge mistake to introduce equivalent electrical circuits with their fixed-voltage voltage generators. It forces one to assume that the conductances of the players (membrane, synapses, AIS) change without any reason, and prevents understanding how the competition of the thermodynamic and electrical processes govern neuronal operation. It leads, among others to attributing conductance change to membranes which are simple isolators with no charge carriers that could implement charge transfer, see section ???. This assumption neglects the driving force the mentioned forces provide. Instead, it attributes the magic ability to the ion channels that they can change their transmission ability as the actual situation requires, without having driving force. This approach obscures that biological electricity is of thermodynamic origin, and block understanding neuronal operation, that is the brain's understanding.

3.3.4 Force analysis

In the steady state, some other forces also contribute to the mentioned forces. To discuss how nature restores the steady state when a microscopic change happens in a balanced state of a biological solution, we write the well-known Nernst-Planck equation (see Eq.(1.1)) in a slightly extended form:

$$0 = \underbrace{e_{el} * E_{Gap}^{Total}(c, \Delta z)}_{\text{Electrical force}} + \underbrace{e_{el} * E_{thermal}^{C_k}(d)}_{\text{Thermodynamic force}} \underbrace{(+F_{ext}(z))}_{\text{Constraint}} \underbrace{(+F_{Transp}(z))}_{\text{Transport force}} \quad (3.17)$$

We multiplied the usual two terms by the elementary charge, so its terms are expressed as forces, plus we added a transport-related external force (its role is discussed below). Furthermore, changing one force triggers a corresponding counterforce and/or causes the ion to move in a viscous fluid. (An excellent summary of the possible biology-related forces is given in [18], with a lot of cited references.)

This equation, expressing Newton's first law, has no place for effects such as "Pumps that maintain ion gradients ... transport ions *against their electrical and chemical gradients*" [2], page 101. It implies some *new force* that moves ions, confronting Schrödinger's science-based opinion: "And that not on the ground that there is *any 'new force'* or whatnot, directing the behaviour of the single atoms within a living organism"[19]. Biophysics still did not provide either the origin of the 'new force' or the reason why Newton's laws are not valid for ions in living matter, which undermines the credibility of those protein-based mechanisms.

Expressed differently

$$-F_{constraint} = Eq.(3.29) + Eq.(5.2) + Eq.(3.11) + External \quad (3.18)$$

That means when describing an ionic transfer process, *we must not separate the electrical current from the mass transfer* (in other words, to discuss electrical

or thermodynamic models in a disciplinary way): they happen simultaneously and mutually trigger each other. Notice that *the thermodynamic term is ion-specific while the electrical term is not*. To be entirely balanced, the system must be balanced to all elements. This way, changing one concentration implicitly changes all other concentrations and the electrical field.

Notice the special role of the constraint force by the closed volume represented by the membrane's wall. The thermodynamic contribution also means a pressure from the colliding particles, and the electrical repulsion exerts a force on the membrane's wall. Those disciplinary forces act simultaneously and inseparably.

3.3.5 One segment

We prepare a tiny electrolyte volume filled with a solution containing ions (such as Na^+ , K^+ and Ca^+ ; furthermore, of course Cl^- or similar). The overwhelming majority of those ions is chemically bound, but a minority might exist separately from each other; especially under external macroscopic changes applied to the volume. For the discussion below, we assume that the segment has a two-dimensional surface boundary (see Fig. ??) and we discuss the gradients along a line, perpendicular to that plane surface. We compose the segments from such layers (sheets), (x, y) parallel plates, and describe the gradients in direction of z . Having the membrane's shape in mind, we introduce the idea of 'thin physical layer', that is parallel with the membrane and has a finite width. We compose the segments from such layers having different potentials.

We assume that when ions are present in an electrolyte having concentration c in a 'conducting layer' of thickness Δz on an isolating surface, they produce such an electrical field. Furthermore, we assume that the electrical field is formed (after integrating the contributions of the rings over the surface) as shown in Fig. 3.3 (for deriving that equation, see section 5.0.2). Notice that here we run into conflict between the 'infinitely thin' layer of physics and the biologically implemented (finitely) 'thin physical layer', so we use a thickness parameter Δz .

For the discussion below, we assume that the segment has a two-dimensional surface boundary, and we discuss the gradients along a line perpendicular to that plane's surface. We neglect the electrokinetic and surface-related electrical effects connected to membranes and nanopores [20] and focus on the balanced state of membranes in ionic solutions. We compose the segments from such layers (sheets), (x, y) parallel plates, and describe the gradients in the direction of z . We introduce the concept of a 'thin physical layer' parallel to the membrane, characterized by a finite width. The detailed calculations are shown in **Numeric panel 3.3.1**.

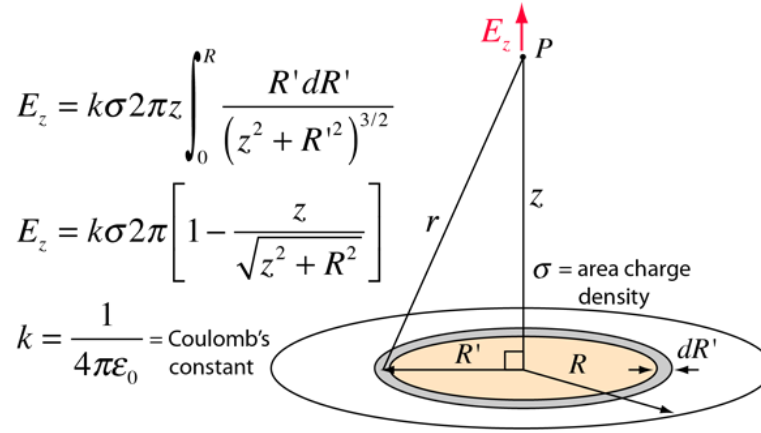


Figure 3.3: The electrical field of a disc of charge can be found by superposing the point charge fields of infinitesimal charge elements. This can be facilitated by summing the fields of charged rings.

3.3.6 Two segments

Now, we split the volume into two segments and compose compose the segments from 'thin physical layers' (sheets) having different concentrations (and such potentials) of ions. We assume the membrane is transparent for the electrical interaction (the electrical field affects the ions in the other segment on the other side of the membrane) but not for their masses (mechanically separates the segments). Separating a volume into two segments by a thin membrane has no initial effects: it actually does not affect the electrical and thermal distributions; the *bulk* concentration and potential remain the same on the two sides of the membrane (even the double layers are electrically neutral).

We consider the immediate environment of the neuron's membrane as adjacent parallel (x, y) plates and find the electrical field's z component of those plates in the points as shown in Fig. 3.3. **Physics panel 3.3.1** discusses the difference between the classical and biological condenser. Furthermore, **Physics panel 3.3.2** discusses the issues with the ill-defined biological boundaries.

We consider the immediate environment of the neuron's membrane as adjacent parallel (x, y) plates and find the electrical field's z component of those plates in the points as shown in Fig. 3.3. In classical electricity, the charge remains on the surface of the conducting layer and the electrical field is zero inside the conducting layer; the electrical field contributions cancel each other. In electrolyte segments, (a tiny fraction of) the molecules decompose into an ionized state (dissociate), see Fig. ??, and the created ions interact with a bounding membrane using a not entirely understood mechanism [21]. Although not explicitly, we consider electrical double layers to be present proximate to the membrane and ions of the dispersion medium are adsorbed on the particle's surface; depending on the chemical features of the medium.

Numerics panel 3.3.1:
The electric field of a physical charged sheet

In the calculations below, we require the 'surface charge density' (interpreted as an 'infinitely thin layer' in physics, expressed in $C * m^{-2}$). We can derive (assuming singly-charged ions) the *volume charge density*

$$\sigma_V(c) = c * N_A * e_{el} = c * 6.023 * 10^{20} * 1.602 * 10^{-19} = 96.5 * c \left[\frac{C}{mM * m^3} \right] \quad (3.19)$$

where N_A is the number of ions in a *millimol* and e_{el} is the elementary charge. Notice that all ions contribute to the charge density, so simply $c = \sum_k C_k$.

Here, we run into a conflict between the 'infinitely thin' layer of physics and the biologically implemented (finitely) 'thin physical layer', so we use a thickness parameter Δz . By assuming an arbitrary 'physical layer thickness' Δz , we can calculate the *surface charge density*

$$\sigma_A(c, \Delta z) = 96.5 * c * \Delta z \left[\frac{C}{m^2 mM * nm} \right] \quad (3.20)$$

We interpreted classical physics' notions from the theory of 'continuous' electricity, interpreted for 'infinite' cases, for the finite world of biological objects and the quantized atomic world. Our idea is similar to Boltzmann's for connecting the continuous medium to particles. When ions are present in an electrolyte having concentration c in a 'conducting layer' of thickness Δz on an isolating surface, they produce a field

$$E_z(c, \Delta z) = \frac{\sigma_A(c, \Delta z)}{\epsilon_o} = 10.89 * 10^3 * c * \Delta z \left[\frac{V}{m nm * mM} \right] \quad (3.21)$$

Although it is not discussed in the classical textbooks, the rest of the dipole molecules get polarized, but do not dissociate; in this way forming virtual charges. Unlike in the conductors, the electrolyte segment next to the surface layer contains dipole molecules, that have more or less balanced charges (the polarization depends also on the external electrical field), so they have much less mobility than the dissociated ions: their size and mass is by orders of magnitude bigger and their electrical force is a fragment of that of the ions. Their charge is virtual: it comes to light only in the appropriate environment, but creates an electrical field in the same way as real charge does. The final reason of creating virtual charges are real charges and/or external fields, so an electrical field due to virtual charges can exist also inside dielectrical matters such as biological cells.

Physics panel 3.3.1: The classical condenser

In classical electricity, the charge remains on the surface of the conducting layer, and the distribution is homogeneous (although the charge is concentrated on the surface and the electrical field is zero inside the conducting layer). Hence, the electrical field contributions cancel each other. In electrolyte segments, (a tiny fraction of) the molecules decompose into an ionized state (dissociate), and the created ions interact with a bounding membrane using a not entirely understood mechanism [21]. The rest of the dipole molecules become polarized but do not dissociate, forming virtual charges that allow the electrolytes to contain electrical fields. Although not explicitly stated, we assume that electrical double layers are present near the membrane, and ions from the dispersion medium are adsorbed onto the particle's surface, depending on the medium's chemical properties.

Correspondingly, the theory non-equivocally explains the potential near the membrane surface; see the discussion and review in [22], mainly their Figure 1. That model considers a symmetrical condenser, that is, both plates contain charges of the same sign. The Donnan function is a (discontinuous) step-like potential. Its continuous version introduces a mathematically smooth, non-physical function at the boundary of the charged layer. One must describe a physically discontinuous layer (electrolyte, ion layer on the membrane's surface, and isolating membrane) with a single continuous function and gradients (potential and concentration) with physical meaning.

3.3.7 Charge layers in segments

As we show, the finite thickness will result in a lack of balance (introducing inhomogeneity and creating a voltage and concentration gradient) proximal to the surfaces of the membrane, even if the concentrations on the two sides are the same. Changing the bulk concentration or potential in one of the segments creates a corresponding gradient across the separating membrane, increasing the inhomogeneity proximal to the membrane. The ions will experience an extra force due to the gradient; however, the mechanical counterforce of the membrane will keep them back in the segment. The *concentration and potential, inseparably and having the same time course*, will change across the two sides of the membrane, just because of the gap's physical features the membrane represents.

We consider that the segment is composed of electrically conducting discs (the ions are free to move on the surface), and the charged discs' contribution to the electrical field at point P (see Fig. 3.3) can be calculated as known from the theory of electricity. Due to the symmetry in the z direction, in a homogeneous solution, the resulting electrical field in the plane perpendicular

Physics panel 3.3.2:
The boundary problem of a physics condenser

The behavior near the boundary layers is problematic, primarily because the layers themselves are ill-defined. Physically, they are not ideally plain and well-defined surfaces; the layer thicknesses are limited by the size of the ions/molecules and their associated ions; the layers do not necessarily satisfy the ideal requirements of the respective physical laws. Mathematically, a single 'universal' function that spans all regions surely cannot accurately describe the behavior of the physical model. So, we chose a piecewise description where the pieces seamlessly fit at the boundaries. We assume that between the surfaces of the membrane, classical electricity describes the electrical field; the charged layer and the electrolyte jointly produce the electrical field; we have a uniformly charged layer from separated charges on the surface and a polarized electrolyte outside them.

Unlike in the classical theory, the electrolyte segment next to the surface layer contains dipole molecules that have more or less balanced charges (the polarization also depends on the external electrical field), so they have much less mobility than the dissociated ions: their size and mass are bigger, and their charge is a fraction of that of the ions. Their virtual charge becomes apparent only in the appropriate environment, yet it generates an electrical field similar to that of a real charge. The final reasons for creating virtual charges are real charges and/or external fields, so an electrical field due to virtual charges can also exist inside dielectric matter. We assume the membrane is transparent for the electrical interaction (the electrical field affects the ions in the other segment on the other side of the membrane) but not for their masses (mechanically separate the segments). The membrane does not need to generate any counterforce (except for collisions).

to the z direction is zero, as we have contributions of equal magnitude with opposite signs.

Notice that here we used the abstraction of an infinitely thin "charged sheet", and there is a step-like gradient in the electrical field in the gap. The physical reality is that the charge is carried by ions, which simultaneously carry mass. In equilibrium, the forces due to the voltage and concentration gradients must be counterbalanced by an external force.

Simple conducting sheet (a physical condenser)

First, we consider a condenser that obeys the laws of classical electricity and has the geometrical size of a biological neuron. We apply the physics terms to that pseudo-biological object and test whether the values we derive are consistent

with physiological phenomena. A tiny (according to [23], page 12, about $2 \cdot 10^{-3}$) portion of the positive ions leaves their negative counterions behind and forms a thin positively charged ion layer on the surface of the membrane (of course, the same happens on the opposite side with negative ions). Experience shows that – also in biology – two skinny layers of ions are formed on the two sides of the membrane of finite width, see the caption of figure (Fig. 11.22 in [24]): "The ions that give rise to the membrane potential lie in a thin ($< 1 \text{ nm}$) surface layer close to the membrane" (As shown below, these ions give rise only to part of the potential, and the dipoles in the electrolyte contribute the rest). A sub-nanometer-thick conducting layer covers the two surfaces at the top of a 5 nm -thick insulator. So, we model the cell with two finite-width "conductive sheet" layers. These layers represent an electrical condenser (so we can calculate the internal electrical field between the plates) and counterbalance each other's electrical field outside the condenser. We may assume that the solid surface provides the needed counterforce to keep the charges at rest.

We cannot expect geometrically flat surfaces, as some biological objects of up to 1 nm in size are present on the surface. So, we must assume a physical 'uniformly charged sheet' with an average thickness $\Delta z = 0.5 \text{ nm}$ on the surface. For the sake of simplicity, we assume a step-like function for the electrical field. It is zero in the segment outside the "conducting sheets" (the 'bulk' portions) and inside the sheets. Between the sheets, it jumps to the value of the electrical field of a charged sheet with surface density σ_A (see Eq.(3.20)). (In this picture, we see a jump in the value of the electrical field on the two sides of the physical layer. The physical layer emulates the well-defined boundary on the side opposite to that proximal to the membrane. Here, we still assume that a mechanical counterforce due to the surface's roughness keeps the charges inside the charged layer.)

Based on the calculations using the classical model, the numerical figures **Numerical panel 3.3.2** suggest our simple model to be a strong oversimplification. We are in the correct order of magnitude, but we arbitrarily assumed a layer thickness, a non-dielectric medium, and a physically unreasonable step-like electrical field. The deviation from the expected values suggests that dipoles in the electrolyte bulk significantly alter the achievable electrical field and potential across the plates.

A usual parallel plate condenser (shown in Fig. 5.2) can be derived with such a model. We have two infinite conducting sheets and an insulator layer between them. The magnitude of the opposite charges on the two plates must be the same as in the classical picture. *Given that the amount of surface charge densities is concluded from the ionic concentration in the segment (in a balanced state), the sum of ionic concentrations of the neuron must be the same in the segments.* A constant electrical field is present inside the membrane (across the plates of the condenser), and there is zero electrical field inside the parallel conducting plates and outside the condenser. In this ideal picture, the charges are aligned along the boundary between two (infinitely thin) conducting layers and cannot

Numerics panel 3.3.2:
The voltage in a classical neuronal condenser

We assume that all unbalanced (dissociated) ions are in the layer, and the rest contains no dissociated ions; furthermore, the ion concentration is unchanged in the segment. In the case of a 400 *mM* solution and a 0.5 *nm* 'thin physical layer' Eq.(3.21) evaluates to an electrical field

$$E_{Classic}^{Gap}(400, 0.5) = 0.22 * 10^7 \left[\frac{V}{m} \right]$$

and at a 5 *nm* membrane thickness, the voltage across the plates becomes

$$U_{Classic}^{Gap}(400, 0.5, 5) = E_{Classic}^{Gap}(400, 0.5) * 10^{-3} * d = 10.84 [mV] \quad (3.22)$$

The number of uncompensated ions needed for the cell, using the method in [23], is $0.88 * 10^7$. The calculated values are about a factor of 5 lower than the experimentally derived values. "An electrical potential difference about 50 – 100 *mV* ... exists across a plasma membrane only about 5 *nm* thick, so that the resulting voltage gradient is about 100,000 *V/cm*" [24]. "The number of uncompensated ions needed for the cell is $4.7 * 10^7$ " [23].

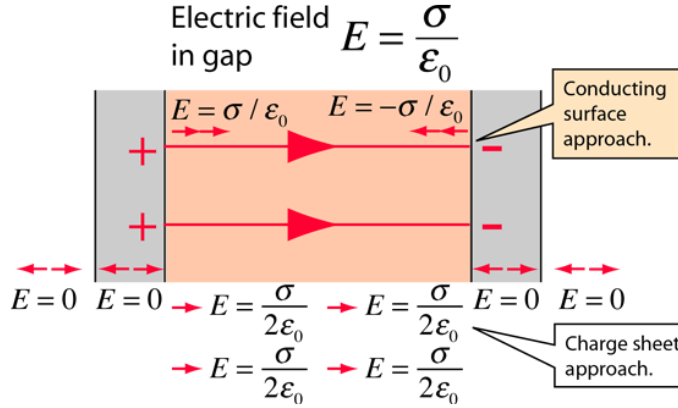


Figure 3.4: The oppositely charged parallel surfaces of the membrane are treated like conducting plates (infinite planes, neglecting fringing), then one can use Gauss's law to calculate the electrical field between the plates. Presuming the plates are at equilibrium with zero electrical fields inside the conductors, the result from a charged conducting surface can be used. "from Hyperphysics by Rod Nave, Georgia State University". ©<http://hyperphysics.phy-astr.gsu.edu/hbase/>

move. The attractive force between them and the opposite charges on the other plate keeps them fixed in the direction of z , and the repulsive force between the charges with the same sign, the surface density in the (x, y) plane remains uniform: the infinitely thin plates are equipotential. This picture illustrates the equilibrium state of charges, consisting of infinitely small non-dissociating charge carriers and perfectly smooth surfaces.

Condenser in dielectric material

In its integral form, Gauss's flux theorem states that the flux of the electrical field out of an arbitrary closed surface is proportional to the electrical charge enclosed by the surface, irrespective of how that charge is distributed. The surface layer represents a steep potential and concentration gradient. Above, we assumed that the counterforce that keeps the charges in their place against their electrical attraction on the membrane's surface is a mechanical force: the ions cannot pass through the membrane. However, such a counterforce does not exist on the side toward the 'bulk' part of the segment. The charges on the plates do not generate an electrical field toward the bulk, but the concentration they represent does, according to the Nernst-Planck equation, provided there are charge carriers in the bulk. We hypothesize that virtual electrical charges exist in the electrolytes, and their field provides the missing electrical field.

When explaining the effect of dielectricity, we must explicitly consider the duality of ions that they obey laws of electricity and thermodynamics *simultaneously*; furthermore, the complexity of the electrical structure of the solution and that the charge carriers have finite size (see "electron size" vs. "dipole size"). According to the theory of electricity, the free charge carriers (the dissociated ions) are located on the surfaces where they form powerfully charged thin layers (condenser plates) in the segments separated by a membrane on the two surfaces of the membrane; two proximal charge layers on the surfaces of the membrane. We assume that those ions behave as point charges, i.e., due to the attraction from the opposite charges on the opposite plate, they do not produce an electrical field toward the side of the bulk of the electrolyte layers. In the classical picture, those layers represent step-like gradients in the electrical field along the z axis, and the constraint that ions must not enter the membrane provides the necessary counterforce.

From a thermodynamic point of view, a driving force acts as long as a concentration gradient between neighboring layers exists. From an electrical point of view, the layers are simple parallel-plate condensers that produce no electrical field outside their closed volume and are connected serially; plus, the top layer has free ions. The electrical field is proportional to the bulk concentration in the layer. Changes in the local electrical gradient may also alter the degree of dissociation and polarization, producing a graded local electrical field. These two driving forces have opposite directions and in a balanced state, the same magnitude. The counterforce acts in a way that constrains the molecules to stay in their layers; it adapts (also compensates for the different interaction speeds) while the gradients are changing. It presses the proximal layer against

the membrane and the neighboring layers against each other.

Due to the membrane's finite width and surface roughness, the conducting layer is also finite. The electrical field is homogeneous within the layers. The rest of that electrical field toward the bulk layer directs the dipoles proximal to the neighboring dipole layer, and their polarization increases; in other words, they "produce" an electrical field. The final effect is that a dipole layer is attracted to the charged layer, and it forms another layer that shows a charged layer toward the bulk. The process repeats and results in a decaying electrical field based on the distance from the membrane's surface.

In our model, we build the volume of the electrolyte from thin layers of directed dipole molecules (the size of molecules limits the thinness Δz) in the volume, separated into two segments with electrolytes by a membrane with a finite thickness d (that is, a point's distance in the electrolyte from the two disks will be z and $z + d$, respectively). The contributing potential of an infinitesimal volume on the axis at point z due to the charged sheet is

$$dU(c, \Delta z, z) = \underbrace{E_{Classic}^{Gap}(c, \Delta z) * z}_{\text{Potential from sheet}} * \frac{1}{z^2} = E_{Classic}^{Gap}(c, \Delta z) \frac{1}{z}$$

The total potential due to all elements dz from the left side is (in the mathematical formulas below, we express the distance in units of $\frac{z}{d}$, so here z is dimensionless).

$$U_{left}(c, \Delta z, d) = E_{Classic}^{Gap}(c, \Delta z) * d \int_{-\infty}^0 \frac{1}{z} dz = E_{Classic}^{Gap}(c, \Delta z) * d \left[\ln |z| \right]_0^{\infty} \quad (3.23)$$

In the case of a single-segment volume, within the segment, a similar potential with an opposite sign is generated by the charges on the neighboring sides of the considered charged sheet, counterbalances the potential described by Eq.(3.23). However, when a membrane with thickness d separates the segments (with no charges in the gap), the potential on the right side will be

$$U_{right}(c, \Delta z, d) = E_{Classic}^{Gap}(c, \Delta z) * d * \left[\ln |z + 1| \right]_1^{\infty} \quad (3.24)$$

We can write the electrical field in the form

$$E(c, \Delta z, d) = E_{Classic}^{Gap}(c, \Delta z) * \ln \left(\frac{z}{z + 1} \right) \quad (3.25)$$

That is, the gap sets the potential difference across the membrane to

$$U(c, \Delta z, d) = E_{Classic}^{Gap}(c, \Delta z) * d * \left(\left[\ln |z| \right]_{-\infty}^0 - \left[\ln |z + 1| \right]_1^{\infty} \right) \quad (3.26)$$

We use the approximation that $\ln(\infty) \approx \ln(1 + \infty)$, and we arrive at that

$$U(c, \Delta z, d) = E_{Classic}^{Gap}(c, \Delta z) * d * \left[\ln \left(\frac{z}{z + 1} \right) \right]_0^1 \quad (3.27)$$

describes the potential across the plates; that is, the classical potential shall be multiplied by the result of the integral. It was experimentally confirmed [25] that the potential of the membrane is proportional to the logarithm of the distance from the membrane.

Given that the same number of charged ions must be present on the two sides of the membrane, their resultant surface density σ_A must be the same. Notice that the effect is purely electrostatic, resulting in an asymmetric ion distribution; in that balanced state, no permeability is needed. If the membrane is (slightly) permeable, ions will move across the membrane until equilibrium is reached. The resulting potential difference is directly proportional to the concentration difference.

The classical condenser has an electrical field, as the blue dashed line shows in Fig. 5.3. No field is outside the condenser or inside the condenser plates (see Fig. 5.2). There is a sudden jump in the field on the conductor's surface (the armatures) and a constant field between the plates. (The charged layer can be ideally thin if electrons are sitting on the surface, but is of finite thickness on the opposite side where ions are on the surface. This latter effect is not discussed in classical electricity.)

The biological condenser behaves differently; see the red diagram line. An electrolyte (instead of an insulator) outside the condenser significantly alters the field's structure. The electrical field on the surface and between the plates changes by a factor of 3-4. Outside the plates, the polarization creates a field that changes logarithmically. An ion layer with finite thickness is built near the membrane's surface. It is uniformly charged, so the field inside it is linear. At the internal surface of the condenser, it takes the field value calculated for the gap; on the other side of the layer, it takes the logarithmic field value at that position. The dashed line represents the gap field, the continuous line represents the charged layer's field, and the dotted line represents the electrolyte's field. We observe the "correspondence principle": discrete and continuous fields join seamlessly; furthermore, as the electrolyte's polarizability decreases toward zero, the field approaches the classical value.

3.3.8 Bridging the micro- and macro views

We assumed that the function can be interpreted for regions $(-\infty, 0)$ and $(1, +\infty)$. However, we have arrived at the boundary between particles and continuous electricity. We use a trick similar to the one Boltzmann used in his famous equation, except that we calculate the number of charge carriers from geometrical instead of statistical assumptions. The minimal thickness of layers is limited, at least by the size of ions. Furthermore, the membrane's surface is not flat; there are structures (mainly proteins/lipids) with sizes of up to 1 nm, so it is likely realistic to consider a layer thickness of 0.5 nm as we did above (on our mathematical scale $0.1 = \frac{0.5 \text{ nm}}{5 \text{ nm}}$), for which a multiplier of 1.7 between the classical and the electrolytic fields exists. A similar calculation using a layer thickness of 0.1 nm (on our mathematical scale, 0.02) results in a multiplier of 3.2; so we use a multiplier of value 3 (that corresponds to a layer thickness of

0.125 nm (the radius of a Na^+ ion is 0.116 nm)). Correspondingly, we assume an electrical field

$$E_{Dielectric}^{Gap}(c, \Delta z) = 3 * E_{Classic}^{Gap}(c, \Delta z) \left[\frac{V}{m \text{ nm} * mM} \right] \quad (3.28)$$

due to the dielectric properties of the segment. This contribution arises from the dielectricity in the segment (for simplicity, we have neglected the relative permittivity). It is to be added to the value of the classical contribution, the field generated by the conducting plates, so the final multiplier is 4. (The experience is known in technical electricity: the electrolytic capacitors *achieve several times higher charge storage capacity by using (pseudocapacitance)*. Using "roughened anode foil" increases the electrolyte thickness, and the roughening provides the necessary mechanical support. A thicker electrolyte layer wraps the condenser plates, and the dipoles in the thicker electrolyte provide an additional charge storage facility.)

Correspondingly,

$$E_{Gap}^{Total}(400, 0.5) = 4 * E_{Classic}^{Gap}(400, 0.5) = 8.8 * 10^6 \left[\frac{V}{m} \right] \quad (3.29)$$

and at a 5 nm membrane thickness, the voltage across the plates becomes

$$U_{Gap}^{Total}(400, 0.5, 5) = E_{Gap}^{Total}(400, 0.5) * (5 * 10^{-9}) = 45 \text{ [mV]} \quad (3.30)$$

"The plasma membrane of all cells, including nerve cells, is approximately 6 to 8 nm thick and consists of a mosaic of lipids and proteins." [2], page 71. Using such a value for the membrane's thickness may yield a thickness up to 60% higher. Furthermore, the usual concentration is also up to 10% higher. Hodgkin, in 1964, measured molarity values in squid axons for ions K^+ , Na^+ , and Cl^- , (400,50,40-150) inside and (20,440,560) outside, and they provided potential values 55–75 mV [23]. Given that we used a plausible but ad hoc "charged layer thickness" and gap distance, we cannot expect a better agreement. However, our qualitative conclusions remain valid and can be adjusted quantitatively to the results of dedicated measurements.

We have the boundary conditions that we know the electrical field at the membrane's surface through the volume charge density, and the Nernst-Planck equation delivers the change of the electrical field due to the change in concentration. The physical constraint is that the layer thickness cannot be infinitely small. If we use a parameter for layer thickness equal to that of the surface layer, the calculated electrical field distribution will be scalable.

3.4 Forming membrane's potential

In this section, we model the neuron as an electrical circuit that comprises an electrolyte condenser implemented by its membranes, surrounded by two electrolyte segments. The membrane comprises low-transmission-capacity ion

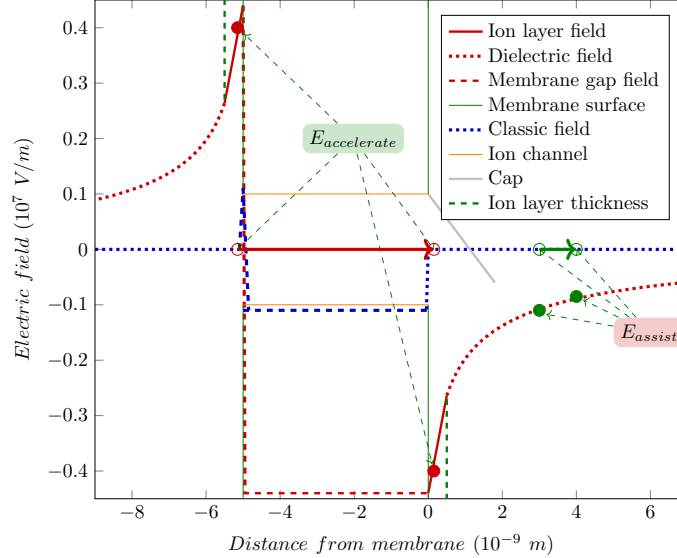


Figure 3.5: The neuronal membrane's electrical field in the function of the distance from the membrane's surface. The thickness of the 'atomic layers' proximal to the membrane's surfaces are also shown. The electrical field is linear inside the uniformly charged atomic ion layer on the surface and is described by Eq.(3.25) in the electrolyte. For visibility, a layer thickness 0.5 nm is assumed. For illustration, a simple capped ion channel in the membrane's wall is also displayed.

channels in its wall and high-transmission-capacity channels toward its axon; it also uses slow ions rather than electrons. The condenser produces a potential (that depends on the membrane's thickness and the concentrations in the two segments, as discussed above). Furthermore, the two electrolyte segments produce Nernst voltages that are combined to yield a single voltage. According to Kirchhoff's Voltage Law, in a balanced state, the sum of those voltages between the intracellular and extracellular segments must be zero. However, one must take into account the effects of the "slow current" (emulated by "fast current" with a plausible time course in the simulator). We omit biological regenerative processes (such as ATP production via hydrolysis, ion pumps and channels functioning, and so on) and discuss only the most important physical primary processes. In light of our findings, we must reconsider several key concepts in neurophysiology.

The electrical processes are inseparable from the corresponding thermodynamic processes [6], which we discuss by deriving an "equivalent thermodynamic electrical field" in section 3.3.3. The other minor effect is a low-intensity "passive flux" produced by the "resting channels". They are different from the ion

channels in the AIS [26, 27, 28, 29, 30]. The "resting current" is about two orders of magnitude lower [31, 32] than HH assumed at the time of writing their papers; correspondingly, its role is much smaller in the transient state, and it even drastically changes the electrical model used in producing an AP which we discuss elsewhere [33].

3.4.1 States

Figure 5.1 shows, in the function of the membrane's thickness, the electrical field across the membrane due to electrical charges (it does not depend on the thickness of the membrane), for different physically plausible assumptions; furthermore, the thermal electrical field. The two electrical fields show significantly different dependencies on the neuron's parameters. Recall that although the parameters we used are biologically plausible, they are only estimations from non-dedicated, incomplete measurements. The qualitative conclusions, however, remain valid.

The balanced state is set where the thermodynamic and electrical "electrical field" diagram lines cross each other; around 5 nm ; the value we assumed in evaluating our equations. On the other hand, we can check the electrical field's dependence on the summed-up concentration of the ions in the segment (see Fig. 3.6) (dehydration, for example, changes the setpoint through changing concentrations). Here, the thermodynamic electrical field is constant but different at different width parameters. The field of the classical condenser does not provide a reasonably balanced state (a matching line). The dielectric diagram line seems to provide a realistic estimation when using biologically reasonable parameters. One can describe the neuron as a balanced system at the intersection of the sloping and the horizontal lines. In the resting state, only minor deviations (fluctuations) from this state exist in both the electrical field and the chemical concentration.

In the transient state, if we assume that a charge transfer causes a significant potential change around 100 mV in the layer of width 10 nm , it means a $10^7 [\frac{V}{m}]$ jump in the electrical field that is caused by a $100 [mM]$ jump in the concentration (for only a short time and only in the mentioned layer, but not in the bulk; see also Fig. 3.8). One can infer that a neuron can minimize its offset voltage (and so: its energy) by either decreasing its membrane's width, the sum concentration of the electrolyte, the electrical field, or by a process with a temporal course that changes their combination, thereby implementing a true multi-physics phenomenon. During issuing an AP, the system finds its way back to the crossing point (the path in the phase space is not simple: concentrations on both sides of the membrane and the electrical field in the mentioned layers change, all having different speeds; electrical, thermodynamic, mechanical and so on, changes happen; the process is a relaxation (a waving), as it is seen in the shape of an AP).

We can imagine how the electrical process occurs in this controlled system (for visibility, the positions of the marks are not proportional to the mentioned numbers). The system is in a balanced state at a crossing point. The two

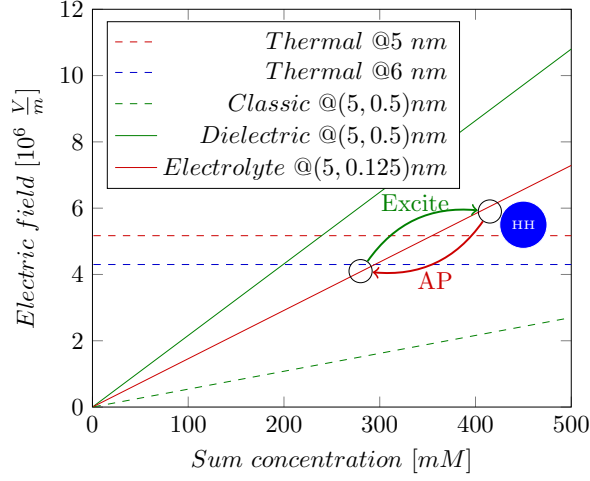


Figure 3.6: The electrical and the thermal "electrical fields" at different concentrations, see Eq.(3.29). The thermal electrical field varies with the thickness parameter and depends on the concentration; see Eq.(5.2). Balanced states are where the sloping lines cross the horizontal ones. The dot marks where HH provided measured data.

circles connected by bent arrows represent that when an excitation begins at the point with lower concentration and higher potential (the cytoplasm side and negative resting potential), ions rush into the surface layer, so the ion concentration increases, which increases the electric field; this is seen that the membrane potential suddenly increases. (We note here that, due to ions' repulsion, the pressure also increases proportionally with the potential.) The increased potential starts a current, but the current is slow, and the intensity of the current through the channels in the membrane's wall is much lower than the current of the drain (AIS) so the system issues an AP as it returns to its (balanced) starting point. Given that it receives a large amount of Na^+ ions, its primary effort is directed to get rid of them (the electrical gradient cannot distinguish the electrical charges by their chemical nature), so the output current through the AIS comprises mainly Na^+ ions. The system will move along a line, with the restriction that it is a movement in phase space (the change in concentration is not proportional to time), and the different points on the membrane's surface may have different potentials at different times (spatiotemporal behavior). Due to the capacitive current, the local potential may temporarily drop below the setpoint (as observed in physiology, this phenomenon is known as hyperpolarization). The time course of the processes is described by the laws of motion of biology [6] (the time derivatives of the Nernst-Planck equation). The simplified

model operates as described in section 4 of [34] and its computational details are provided in [35].

Based on the correct physical model, we can get closer to understanding and reducing the effects of neurological diseases [36, 37], which is not possible while having a wrong model in mind, with batteries and autonomously changing resistors inside the neuron. The change in parameter values potentially leads to neurological diseases as suspected: "Structural alterations of the AIS are likely to have an impact on normal neuronal activity" [36]. When lipid debris accumulates on the surface of the neuron membrane [36], the thickening of the layer Δz reduces the electric field strength (i.e., changes the neuron's 'setpoint'). In that case, the resting potential (and the threshold potential) may change, which weakens AP generation. The elongated shape of the neuron means that the resistance of the AIS increases that changes the parameters of the AP; see Eq.(6.3). "In brain tissue from Alzheimer's disease patients ... the most common changes were a proximal shift or a lengthening of the AIS" [36]. Also, as [38] observed, the setpoint changes with temperature (the "thermodynamic electrical field" changes with the temperature, while the "electrical electrical field" does not, so their crossing point changes), reflected in the dynamics of the AP.

Resting state

In the resting state, the goal is to provide stability with slow, less intense currents. As [2] discusses, there are 'holes' (non-controlled ion channels) in the membrane where the counterforce is missing, so the difference in the electrical and thermodynamic forces could accelerate ions until the Stokes-Einstein speed (see Eq.(3.11)) is reached. Given that the deviation from the setpoint is slight, the current's intensity and speed are low. The slow ion transport delivers both charge and mass, so the gradients permanently direct the process control variable toward the reference point. Given that the 'resting channels' are scattered on the surface of the membrane and it takes time for the gradients delivered by slow ions to reach the entrance of an ion channel, some slight variations in the actual value of the membrane's potential necessarily exist. Given that the AIS comprises non-gated ion channels with much higher density than those in the membrane, the variations and the control ("leaking") are marginal, and the current flows mainly through the ion channels in the membrane's walls, as observed in experimental physiology.

In Fig. 3.7, the electrical voltage contribution is unidirectional. In contrast, the two-component thermodynamic contributions, which depend on the concentrations, are bidirectional. If choosing the zero potential at the middle of the electrical potential, we can interpret that the concentration combinations maintain their ionic concentrations (and so also thermodynamic voltage contributions) independently on both sides, starting from a zero potential level. They generate potentials autonomously in the segments proximal to the opposite membrane surfaces. The electrical charge on the membrane generates an additional voltage independently. However, that voltage must fit between the

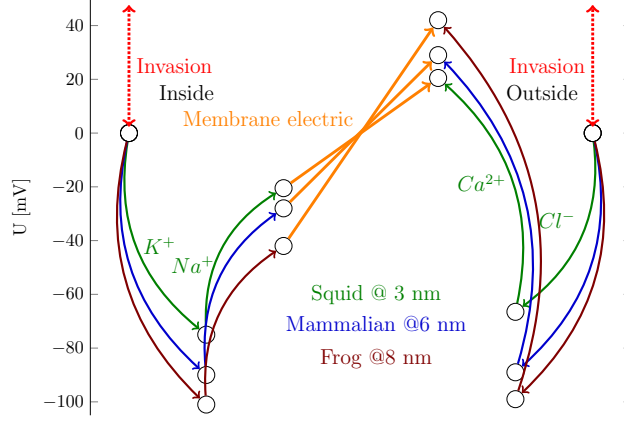


Figure 3.7: The mechanism for controlling neuronal operation is the same in all cases shown in Table 3.1. It appears that the thicker membrane produces a higher membrane electrical potential, requiring a higher thermal potential to compensate. However, it can solve the task using a lower salt concentration, since the electrical membrane potential accelerates ion flow through the membrane channels, thereby speeding up the operation. The electrical fields for the different species have similar values, as shown in Fig. 5.1. Furthermore, the total concentration decreases as the membrane thickness and potential increase.

two thermodynamic voltages. Any deviation in potential between the intracellular and extracellular sides initiates a current that drives the concentrations and voltages toward a balanced state. The process is complex: the sum of surface concentration of ions must be the same on both sides for the condenser, and the ratios on the intracellular and extracellular sides must be adjusted to satisfy the conditions

$$U_{internal}^{K^+, Na^+} = -U_{electric} + U_{internal}^{K^+} + U_{internal}^{Na^+} (+U_{internal}^{invasion}) \quad (3.31)$$

$$U_{external}^{Cl^-, Ca^{2+}} = U_{electric} - U_{external}^{Cl^-} - U_{external}^{Ca^{2+}} (+U_{external}^{invasion}) \quad (3.32)$$

In a balanced state, the left sides are zero: at the contact point between the electrolyte and the membrane, there are no potential differences, while in a perturbed state, they are non-zero and so they represent driving forces for restoring the balance. (Recall that the ions are slow and that the local electrical and concentration gradients control the local potentials.) From the derivation, it follows that

$$U_{electric} = -2 * (U_{internal}^{K^+} + U_{internal}^{Na^+}) \quad (3.33)$$

$$U_{electric} = -2 * (U_{external}^{Cl^-} + U_{external}^{Ca^{2+}}) \quad (3.34)$$

$$U_{electric} = U_{internal}^{K^+} + U_{internal}^{Na^+} = -(U_{external}^{Cl^-} + U_{external}^{Ca^{2+}}) \quad (3.35)$$

Our statement is the precise mathematical formulation of that "Thus ions are subject to two forces driving them across the membrane: (1) a *chemical driving force*, a function of the concentration gradient across the membrane, and (2) an *electrical driving force*, a function of the electrical potential difference across the membrane." [2], page 129. Fig. 3.7 quantitatively underpins that "action potential generation in nearly all types of neurons and muscle cells is accomplished through mechanisms similar to those first detailed in the squid giant axon in early research by HH" [39]. *In all species, the chemical and thermodynamic processes keep a balance in the cells of different construction in both resting and transient states.*

According to the above derivation, we consider three characteristic segments in the neuron. In the middle, we have an electrostatically charged isolator plate with positive and negative charge carriers on its two surfaces and none inside, so it has a homogeneous electrical field. The magnitude of the field depends only on the total concentration of the corresponding ions in the segments. On both sides are ionic solutions of different compositions. If the membrane is permeable for the ions, they can move from one segment to the other as the resultant field dictates, until the fields get balanced.

The membrane's electrical potential defines the resting potential, and the "electrical" and "thermodynamic" potentials work together to control neuronal operation. The anatomically defined "electrical" membrane potential, combined with the vast number of ions in the extracellular space, provides a stable concentration that serves as a solid reference point (a setpoint). In the resting state, the membrane potential is balanced by the presence of always-open "resting ion channels". When ions flow into the membrane, the "resting ion channels" can keep the balance: the internal K^+ and Na^+ concentrations are adjusted as the Na^+ flows in.

Figure 3.7 also shows that in its native state, the neuron is electrically balanced: in the internal and external segments, the resulting thermodynamic and electrical forces are equal. Any foreign invasion into the neuron's life (chemical, mechanical, or electrical, including clamping) changes the left-side reference value to an externally set value. The neuron must adjust its Na^+ and K^+ concentrations to reach a new balance while the invasion persists. That is, any invasion ("testing in the physical laboratory") changes the concentrations and voltages inside the cell. In addition, the internal biological processes can trigger internal processes that contribute (from the measurement's point of view) foreign currents or processes, see also section 3.4.1. The feedback from the clamping methods introduces a compensating current. Even a simple conductance-measuring device is active: it applies a test voltage that generates a test current, which can sensitively affect the ionic composition of the neuron (recall that concentrations may differ by up to six orders of magnitude). Although modern electronic devices attempt to conceal this effect, they subtly influence biological operation.

From our result, it follows that for more chemical elements, a per-element coupled set of equations exists, plus the sum concentration agreed electrical fields that are valid simultaneously and define the process variable (aka mem-

brane potential) of the neural regulatory circuit. In a more symmetrical form

$$0 = \frac{U_{electric}^{Total}(c, \Delta z, d)}{2} + \sum_k U_{thermal}^{C_k}(d) + invasion \quad \text{for all } k; \text{ inside and outside} \quad (3.36)$$

$$c = \sum_k C_k^{ext} = \sum_k C_k^{int} \quad (3.37)$$

In the classical measurement by HH, the sum of the concentrations of K^+ and Na^+ in the cytoplasm and extracellular fluid (in mM) is 380 and 390, respectively, while the concentration of the Cl^- are 450 and 460; practically all equal, as our analysis has derived. The deviations are well within the uncertainty of the measured values.

That is, we have an unidirectional electrical potential difference and oppositely directed thermal voltages. In the balanced state, the two potentials must be equal. When changing, say, one concentration, the other concentration on the same side, and the surface charge density of the condenser, changes accordingly. The changed electrical voltage re-adjusts the thermodynamic voltage and, consequently, the concentrations in the other segment. There is no simple way to express, say, $[Na^+]_i$ in the function of $[K^+]_i$ or vice versa, and similarly for the other ions. Everything moves. The size matters: the amount of ions in the external segment is many orders of magnitude higher, and the gradients are inversely proportional to that amount.

As the figure suggests, an external invasion, typically an electrical voltage on the intracellular side, changes the balanced state, and due to the parameters being linked, changes all concentrations. When moving the system out of its balanced state, in any way, a driving force appears that moves the system towards finding a new balanced state. However, recall that the ions are slow, so *the changes are not instant*.

Table 3.1 shows our results for three different famous published cases (Table 2.1 in [23]). It looks like the calibration of the concentration of negative and positive ions needs scaling, and the measurement of Ca^{2+} comprises issues (the Ca^{2+} concentration in all cases is inaccurate and had to be corrected).

When calculating the values in Table 3.1, we used $\Delta z = 1.05$ and employed the membrane thickness data from various publications to demonstrate that our theoretical approach is correct and that using the correct data can yield even the absolute values of the membrane potential. The higher-than-expected value of Δz may suggest that the ions and the membrane's double layers in the electrolyte may also play a role. Dedicated, complete investigations can reveal the details.

Transient state

In a special type of invasion, when suddenly a large amount of Na^+ rushes in (i.e., the trigger is external, but the ions derive from a biological process), the

Table 3.1: The summary data table for equilibrium (Data from Table 2.1 in[23])

Ion	Inside [mM]	Outside [mM]	$U_{therm} \frac{RT}{zF} \log\left(\frac{[C]_o}{[C]_i}\right)$ [mV]	U_{electr}^{theor} [mV]	U_{electr}^{exper} [mV]	U_{membr} [mV]	Width [nm]	E_{membr} [MV/m]
Squid (HH) T=293 K ^o							3	14.0
K^+	400	20	-75.5 $58\log\left(\frac{20}{400}\right)$	=	41.9	41.4	-62.0	
Na^+	50	440	+54.8 $58\log\left(\frac{440}{50}\right)$	=				
Cl^-	40	560	-66.5 $58\log\left(\frac{560}{40}\right)$	=	41.9	42.0	63.0	
Ca^{2+}	0.27*	10	+45.5 $29\log\left(\frac{10}{0.27}\right)$	=				
* used for the published value 0.4								
Frog muscle (Conway) T=293 K ^o							8	9.61
K^+	124	2.25	-101 = $58\log\left(\frac{2.25}{124}\right)$		88.9	83.6	-125.4	
Na^+	10.4	109	+59.2 $58\log\left(\frac{109}{10.4}\right)$	=				
Cl^-	1.5	77.5	-99.4 $-58\log\left(\frac{77.5}{1.5}\right)$	=	88.9	82.7	124.1	
Ca^{2+}	0.021*	2.1	+58.0 $29\log\left(\frac{2.1}{0.021}\right)$	=				
* used for the published value 4.9								
Typical mammalian cell T=310 K ^o							6	11.1
K^+	140	5	-89.7 $62\log\left(\frac{5}{140}\right)$	=	57.7	57.3	-85.8	
Na^+	15	145	+61.1 $62\log\left(\frac{145}{15}\right)$	=				
Cl^-	4	110	-89.2 $-62\log\left(\frac{110}{4}\right)$	=	57.7	56.9	85.9	
Ca^{2+}	0.04*	5	+60.8 $31\log\left(\frac{5}{0.04}\right)$	=				
* used for the published value 0.0001								
Na^+ ions only [40] T=310 K ^o							6	10.0
Na^+	12	145	+66.6 $62\log\left(\frac{145}{12}\right)$	=	60.7	66.6		

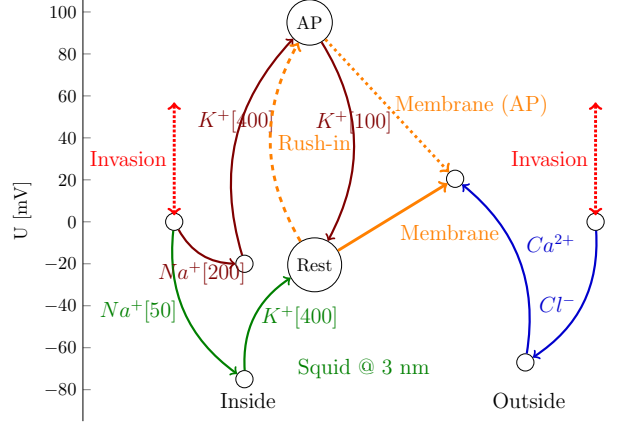


Figure 3.8: The mechanism of producing AP (using numbers referring to the case of squid shown in Table 3.1). In the resting state, the inside thermal potentials follow the green paths. When the rush-in of Na^+ ions suddenly increases the internal concentration to [200], the potential on the internal side of the membrane increases to +90 mV. The neuron must decrease its Na^+ concentrations to restore the resting potential, mainly by releasing an action potential through the AIS. If the clamping fixed the mebrane's voltage at the peak voltage of the AP, the neuron must release K^+ ions until its reaches concentration [Na50,K7].

current throughput of the "resting ion channels" is not sufficient anymore: the internal Na^+ concentration in the proximal layer drastically increases, and so does the membrane's potential. At that point, the neuron is out of balance: although most of the Na^+ ions can flow out through the AIS, the system also changes its K^+ concentration through the limited capacity "resting ion channels" until the balanced state is regained. This latter effect must not be confused with the capacitive current, which (owing to its reversed direction) causes the phenomenon known as "hyperpolarization".

Compare the figure to Fig. 7.2 in [2]. Notice the two-sided negative feedback effect: a change in the value of the "command potential" or "command concentration" (the "setpoint" in the language of control theory) changes the other value. In the case of voltage invasion, the ion concentration ratio also changes. In this case, the feedback amplifier is represented by the delicate balance of the electrical and thermodynamic force fields. In the case of clamping, another (electrical) feedback amplifier is introduced into the system, and they work against each other. The neuron attempts to restore its balance without external potential (it cannot distinguish the foreign potential from its membrane potential), and changes its concentrations accordingly. A vital difference is the *speed* of the feedback amplifier. The electrical one operates at electrical speed (not considering the electrolyte electrodes), while the thermodynamic one operates at a speed several orders of magnitude lower. When switching (or

changing) the external ("command") voltage, the neuron must first get into its "steady state". It takes time. "The command potential, which is selected by the experimenter and can be of any desired amplitude and waveform" [2]. However, if the measured data readings are not from a steady state, the measurement is wrong: the different amplifier speeds (different current speeds) surely distort the measured value.

Figure 3.8 shows the schematic course of potentials and concentrations during issuing an AP. Follow the green path: the concentration ratio of Na^+ and K^+ ions define a thermodynamic contribution in the intracellular segment. Follow the blue path: the Cl^- and Ca^{2+} ions define a thermodynamic contribution in the extracellular segment. On the left side, two positive ions are in the solution, and the membrane potential (due to charge separation) rises toward the outside segment; the ions cannot go "uphill". Similarly, the Cl^- ions cannot go uphill (due to their negative charge, the same potential is also rising for them). This single membrane potential keeps all ions in their segments. However, the Ca^{2+} ions can travel "downhill". Calcium pumps are needed: although in minimal concentration, Ca^{2+} ions flow into the intracellular segment, and they must be pumped out (BTW: this is likely also the reason why measuring Ca^{2+} concentration is inaccurate).

As discussed above, the concentrations and voltages in the extracellular segment, with their vast amount of ions, remain unchanged in resting and transient states. Follow the red path: Na^+ ions rush into the intracellular segment. They increase the concentration from [50 mM] to (say) [200 mM] (from point (Rest) to point (AP)). This sudden change increases the thermal contribution of Na^+ potential to [-20 mV], but because the total charge on the membrane significantly increases, the resulting voltage on the intracellular side of the membrane increases to [90 mV]. The neuron could decrease its potential to the resting value if it could decrease the K^+ concentration to [100 mM]. However, the ion-delivery capacity of the "resting ion channels" is insufficient: they are sized to maintain the resting state. (Furthermore, the electrical contribution appears instantly, while the thermodynamic one, due to the finite speed of ions, only with some delay [6].) Follow the dashed orange path: the large amount of rush-in Na^+ ions drastically increases the surface concentration on the membrane, so the potential suddenly changes to [90 mV]. Follow the dotted orange path: the electrical field across the membrane suddenly changes its direction from positive to negative (see the slopes of the solid and dotted orange lines) so that the excess positive ions can move "downhill" out of the intracellular space through the high-conductance ion channel array (AIS) at the end of the neuron. Its internal end is at the high membrane potential, while the external end is at the low potential of the extracellular segment.

The setpoint on the extracellular side remains fixed. On the intracellular side, at the beginning of the AP, the actual voltage is above that of the setpoint on the extracellular side, so at the two ends of the AIS, a positive driving force moves the ions toward the downstream neuron. Given that positive Na^+ ions leave the intracellular segment, its potential continuously decreases. At some point, *the voltage drops below the setpoint of the extracellular side, and the elec-*

trical field across the membrane reverses. The driving force also decreases, and a low-intensity current restores the balanced state to the previous setpoint. When measuring at the AIS, one observes that the potential rises suddenly, then decreases below the setpoint (hyperpolarizes the membrane), then a decreasing current (flowing in the opposite direction due to the changed slope of the resulting potential, the changed electrical field across the membrane, but comprising the originally rushed-in Na^+ ions) slowly restores the resting potential. Here comes the finite speed of ions again. The ions can follow the potential changes with a delay due to their finite speed, causing an apparent delay in the time course of the AP's current.

Models in neuroscience (as reviewed in [41]) almost entirely ignore these aspects. In our physical model, we see that the measurable membrane potential and current change in the function of the ions' speed, the concentration, and its time derivative. Furthermore, all mentioned quantities depend on the effective potential. As Eq. (5.1) shows, the "thermodynamic electrical field" increases as the temperature increases, and raises the setpoint. As Eq.(3.11) shows that the current decreases as the temperature increases, in line with the experimental evidence [7, 42]. The amplitude of the action potential is decreased, given that the setpoint has increased, and its duration is reduced, given that the current decreases.

3.4.2 Controlling neuronal stages

From control theory, it is known that the goal of a controlled system is to govern the application of system inputs to drive the system to a desired state while minimizing delay and overshoot, steady-state error, and ensuring control stability. The neuron implements a controller that monitors the controlled process variable (membrane voltage) and compares it with the reference or setpoint (resting potential). Recall that the geometry and the composition and concentration of electrolytes define the setpoint. The difference between the process variable's actual and desired values, known as the error signal, represents the actual offset potential. It is applied as feedback to generate a control action that brings the controlled process variable to its setpoint.

Neuron as a PID controller

Biology uses a simple Proportional-Integral-Derivative (PID) controller with proportional, integral, and derivative components. In simple words, it means that the output of the system (the output voltage measurable on the AIS) depends on the input voltage (the present: input current if the resistance is constant), the integrated input current(s) (the past), and their derivatives (the future). Notice that calculating the derivatives of the interdependent parameters needs care, as discussed in [43]. The sum of those summands defines the resulting output voltage. The biological case is more complicated than the technical one because biology works with slow ions, and the components have their temporal dependence (applying the laws created for fast currents requires

emulation, as discussed in section 3.4). Furthermore, the summands have several constituents, and the neurons handle those current-related constituents autonomously. That is, unlike what is assumed in classical physiology, the output voltage is not necessarily a simple function of the mentioned input variables. Furthermore, it is not sufficient to consider the "present" of the system. The classical models all omit the derivatives' contributions (a direct consequence of using 'clamping': in that "frozen" state, the "future" cannot be interpreted) except that of the capacitive current, which is the passive consequence of the input currents. Furthermore, the 'Integrate and Fire'-type models consider the integrated currents only in the 'charge-up' ('Computing') period of operation, and they entirely omit that different physical processes are going on in different stages of operation. That is, *even the best classical models can not describe neuronal operation completely and correctly.*

The gradients are used to adjust the process variable through their positive and negative contributions (corresponding to rising and falling edges), and the different speeds of the thermodynamic and electrical interactions minimize the delay (i.e., provide the maximum operational speed, vital for survival). The steady-state error is minimized by setting the process variable to the reference point using long-term stable parameters (geometry and overall concentration). The low-intensity current through the always-open resting ion channels provides dynamic stability in the steady state.

We considered that the neuron has a stable base state. On the one side, this resting state must be dynamically stabilized for little perturbations using as little energy as possible (and to provide a mechanism when the cell grows, divides, or ages). On the other side, it must be able to restore the state after rough perturbations as quickly as possible, causing short-time transients (when restoring the membrane's potential after issuing a spike). In both the resting and transient states, the system attempts to return to its balanced state, though the mechanisms required differ.

In its excited state, the system aims to provide an intense output that informs the downstream neurons. This overshoot is initiated by a large number of voltage-gated ion channels (distinct from the non-gated ion channels used to maintain the resting potential) distributed in the neuron's membrane. The overshoot current flows through those persistently open ion channels, which are concentrated in AIS (which has about two orders of magnitude higher channel density than the wall). The intense slow current produces a condenser-like behavior (capacitive current); the phenomena called "polarization" and "hyperpolarization" (not polarization; instead, a movement of completely separated charge in the surface layer of the electrolyte) of the membrane provide the necessary positive and negative error signals for the controller in the transient state by moving the actual potential value of the membrane above and below the resting potential value. The potentials and the electrical field's magnitudes depend on the concentration and the geometry (the finite size of the membrane). The ions' chemical nature comes into play only if the polarizability can differ for different molecules.

The time course of the overall control function $u(t)$ of the general PID con-

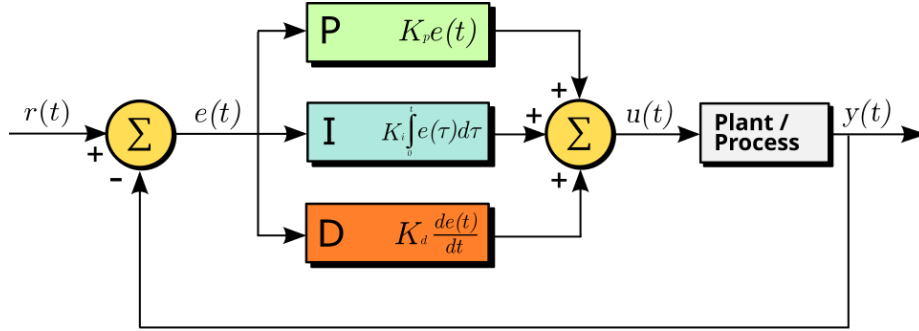


Figure 3.9: A block diagram of a PID controller in a feedback loop. $r(t)$ is the desired process variable (PV) or setpoint (SP), and $y(t)$ is the measured PV. (Wikipedia)

troller in function of the error variable $e(t)$ in Fig. 3.9 is described by

$$u(t) = K_p \left(e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt} \right) \quad (3.38)$$

where $T_i = \frac{K_p}{K_i}$ is the time integration constant, $T_d = \frac{K_d}{K_p}$ is the derivative time constant (notice the independent time constants). That is, in the world of a neuron, the "theory of everything" is

$$V_M^{OUT}(t) = K_p \left(\underbrace{\overbrace{V_M^{IN}(t)}^{\text{proportional}}}_{\text{clamping, synaptic}} + \underbrace{\overbrace{\frac{1}{T_i} \int_0^t V_M^{IN}(\tau) d\tau}^{\text{integral (parallel RC)}}}_{\text{membrane's wall}} + \underbrace{\overbrace{T_d \frac{dV_M^{IN}(t)}{dt}}^{\text{derivative (serial RC)}}}_{\text{AIS, synaptic}} \right) \quad (3.39)$$

The first term represents the constant effect of the external world (including synaptic inputs, clamping, and excitatory effects on brain tissue). The second term represents the time-averaged contribution, such as charging up the membrane, including clamping current. The voltage is mainly due to current through ion channels in the membrane's wall; it represents a parallel RC circuit. The third term represents the sum of the gradients due to all effects (the external world, the internal processes, and the outflow through the AIS) and constitutes a serial RC circuit. The underbraces of the equation identify which components of the neuron contribute the terms of the PID equation. The presence of some terms depends on the neuron's actual stage; furthermore, the R values in the parallel and serial RC circuit differ (resulting in different time constants).

One can reformulate the HH's famous Eq. (1) in [5] to the form of the PID

equation

$$\underbrace{I \times R_{AIS}}_{V_M^{OUT}} = \underbrace{I_i \times R_{AIS}}_{\text{proportional, ionic}} + \underbrace{\frac{1}{C_M \times R_{AIS}} \int_0^t \underbrace{(I_{bio} - I_{clamp})(\tau)}_{\text{integral, clamping cancels this}} d\tau}_{T_i} + \underbrace{C_M \times R_M \frac{dV_M}{dt}}_{\text{derivative, capacitive, } T_d} \quad (3.40)$$

The proportional term comprises the ionic (and external) currents. At the time when Eq. (3.40) was set up, AIS was not yet known, so HH assumed that in the proportional term, the resistance is identical to the membrane's resistance implemented by the ion channels in the wall (leading to the ideas of "leakage current" and "resting potential", misleading physiological research.) The experimental conditions canceled the integral term entirely: the negative feedback from clamping precisely counterbalances the neuron's internally generated current, so the current integrates to zero. (BTW: the "integrate and fire" type models artificially revive the forgotten integral term.) The derivative term assumes that all currents are constants, so the voltage changes only due to the capacitive current. It entirely forgets that gradients are needed to operate an electrical system (leading to the ideas of constant-voltage batteries and magically operated resistors in the equivalent circuits). Although it was known from the beginning that the APs comprise relatively steep rising and falling edges, it was forgotten that a non-constant current contributes a gradient to the derivative term. HH's formalism describes only the 'present' (as can be expected when freezing the state), has no predictive power, nor can it give an account of neuronal memories.

HH considered only the proportional term, plus in the derivative term, the voltage gradient due to the condenser (but not the capacitive current itself, directly leading to the wrong ad-hoc hypothesis of the presence of K^+ for explaining hyperpolarization). Given that clamping obscures the presence of gradient-like changes in the system, they missed that *the input currents also produce a voltage gradient and that the output current through the AIS also produces a gradient*. Furthermore, the equation shows that external currents (e.g., the rising and falling edges of step functions) also cause significant changes in the output voltage. Similarly, a "foreign" invasion (changing mechanically the positions of charges by pressure, ultrasound, magnetic pulse, changing the concentration in one of the segments, and so on) generates a change in the position of charges on the neuronal membrane. This way, it generates a voltage gradient (along with pressure and other gradients), which may trigger neuronal spikes. As discussed in section 6.1, subthreshold excitations provide direct experimental evidence that the description of the resting state includes both serial and parallel contributions.

From a control theoretical point of view, the condenser geometry and the ion concentrations set the 'always the same' value (the setpoint; see also Figure 5.1) and the currents serve to keep or restore that value. After introducing a finite-width membrane into an electrolyte, a potential difference between the two membrane surfaces is created; see Eq. (3.30). When adjusting the membrane's

potential, we must consider its ground and excited states separately, given that the perturbations in these two cases are vastly different. Nature employs various mechanisms to maintain the ground state and recover it after generating a spike. Although nature's tools are remarkably similar (channels and pumps; here, for discussing 'downhill' channels, we must consider the charge-up of different components), the significant difference in current intensity warrants a separate discussion.

When comparing Eq. (6.1) describing the operation of a simple serial RC oscillator taken from the theory of electricity, Eq. (3.40) describing HH's differential equation (having PID in mind), and the complete Eq. (3.38) describing PID in general, one can see that they describe the same process in different (experimental and mathematical) approximations. As experience shows, Eq. (6.1) describing the 'net electric' model provides a sufficiently accurate description of the AP, proving that the dominating effect comes from the derivative term and that the thermodynamic term is linearly proportional to the electrical one. Furthermore, as we emphasized by discussing the physical processes and the mathematical terms, to some measure, the "parallel RC circuit" is also present in the process, although its current amplitude is by two orders of magnitude lower than that of the "serial RC circuit". Furthermore, the time constants of the two circuits are also largely different, making the effect of the parallel circuit unnoticeable. The slow current also plays a role here: the ion currents are 'local'. Given that the channels in the membrane's wall are always open, the current on its way towards the AIS may flow out through a nearby ion channel, so only a fragment of the 'resting current' reaches the AIS. Eq. (3.38) provides a high-accuracy description of the process, but for most practical applications using Eq. (6.1) provides sufficiently good results, while Eq. (3.40) is based on an incorrect 'physical model' and requires a series of incorrect ad-hoc hypotheses to describe observations. Note that the forces have both electrical and thermodynamic components, but they are proportional to each other. That is, although the absolute values of the coefficients in the equations are not accurate, their effects are. Surely, the description of the processes is complicated, but appropriate approximations give sufficiently precise and computationally reasonable results.

Robustness of the control circle

An exciting thought experiment is discussed in [40]. "First, we consider a hypothetical membrane that is permeable only to Na^+ ions. Suppose that $[Na^+]_o$, the outside or extracellular sodium concentration, is 145 mM, and $[Na^+]_i$ is 12 mM. Moreover, suppose that initially, there is no membrane potential. The diffusion gradient for Na^+ favors Na^+ entry into the cell, and the initial Na^+ influx carries a charge that builds up on the inside of the cell. This produces a potential (recall that separation of charge produces a potential) across the membrane that impedes further Na^+ ion movement because the positive charges repel the positively charged Na^+ ion."

The paper correctly explains that in the case of Na^+ only, the potential is

set up to 66.6 mV . (We quietly add that plus a layer of the corresponding Cl^- ions on the other segment, in a similar concentration ratio, builds up; so also, the attraction between positive and negative charges contributes to the effect.) In the absence of our argumentation, it cannot explain *why* that potential builds up; only why that concentration ratio establishes: the Nernst law alone does not explain the effect. Furthermore, the statement is valid only for a well-defined semipermeable membrane. We can add that the same concentrations develop if initially identical concentrations are present on both sides. The finite width of the membrane creates an electrical potential, a consequence of charge separation, as stated. If there are no ions initially inside, then no charge separation occurs, and therefore, no ion transport begins due to the membrane's electrical potential. However, the thermal driving force exists, and the started ion transport quickly builds the electrical layer, which effectively helps to establish the resting potential.

This robustness also explains why, during replication (cell division), cells also inherit their operating ability and resting potential; the existing membrane cell continues to grow in the same way. Its voltage does not depend on its size, so it remains constant, and the voltage defined in this way adjusts the concentrations accordingly. Moreover, it explains why, during evolution, cells first formed: a couple of lipids came together to form an imperfect membrane with holes, establishing cellular electricity. As Fig. 3.7 depicts, the concentrations may adapt to the living conditions (whether living in seawater), but the general operating principles remain the same. The seawater environment enables the use of thinner membranes and lower membrane voltages; however, the same electrical field is used in all cases.

Given that neurons share the external (global) concentrations (defined by the vast amount of ions in the bulk), they provide a firmly fixed offset value to the always-the-same electrical potential. This way, the operations of the individual neurons do not interfere. Furthermore, the internal (local) concentrations can adjust themselves even following a very rough charge perturbation while issuing an AP; moreover, when the living organism is growing.

3.4.3 Goldman-Hodgkin-Katz potential

It has already been stated, based on experimental evidence, that "the membrane permeability to the ions has nothing to do with the potential generation and the ions' adsorption on the membrane surface generates the membrane potential"; for a review, see [44]. The idea itself is nonsense: in a balanced state, no ion transport happens, so even without permeability, the balanced state persists; the resting potential has nothing to do with either ions' mobility or permeability or with ion absorption [45]. Furthermore, there is no idea in Goldman-Hodgkin-Katz (GHK) about whether the 'setpoint' (why that specific concentration or potential difference) is present and why the same potential is reset after rough perturbations such as issuing an AP or replicating a cell. The causality is reversed: the potential is a static concept and is set electrically, and the two gradients form the experienced concentrations from the available solvent

molecules. For the time course of gradient formation, mobility and permeability play a role, but not in determining concentrations or the resting potential.

In our clear physical picture, the thermodynamic forces on one side and on the other side of the membrane are summed. They counterbalance each other; furthermore, jointly the effect of the neuronal condenser, see Fig. 3.7. As we emphasized, the concentration gradients are ion-specific. Furthermore, as discussed in connection with Eq.(1.2), the Nernst equation comprises a per-ion indefinite constant (a potential difference). To calculate a linear combination of terms comprising an arbitrary constant is nonsense, and so is adding absolute concentrations on the different sides of the membrane, or changing the base of the logarithm used in a calculation to match the experimental value. It is not more than number magic. The potential is described by coupled equations as discussed in connection with Eq. (3.37).

The Ca^{2+} ions do not fit into the GHK picture. One of the reasons why GHK cannot be good is that $[Ca^{2+}]$ is omitted. Biologically, it is hard to believe that Ca^{2+} does not participate in the game of life (especially since biology sees the need for Ca^{2+} pumps). Physically, a single concentration on one side of the membrane cannot maintain balance; as the different ion-exchange processes shown in Table 3.1 and Fig. 3.7 demonstrate. When adding a new ion to the solution, the sum concentration increases, and so the electrical force increases, forcing the previous elements to find new concentrations on both sides. The appearance of a new chemical element indirectly changes the concentrations of the others (a good example is the role of the negligible amount of Ca^{2+}).

Contents

3.1	Miért más	10
3.2	Concepts	12
3.2.1	Töltés	12
3.2.2	Erőtér	12
3.2.3	Feszültség	12
3.2.4	Áram	12
3.2.5	Condenser	18
3.2.6	Resistor	19
3.3	A membrán elektromosságtana	21
3.3.1	Elektrolitok	21
3.3.2	Membrán elektrolitban	23
3.3.3	Inhomogeneities	23
3.3.4	Force analysis	24
3.3.5	One segment	25
3.3.6	Two segments	26
3.3.7	Charge layers in segments	28
3.3.8	Bridging the micro- and macro views	34

Physics panel 3.4.1:**The effect of Nernst-Plack law on the GHK equation**

Warning The potential derived in **Physics panel 1.1.1** inherently comprises an additive term (a potential difference), so they must not be compared directly if they use a different reference C_k^{int} (furthermore, it is nonsense to combine quantities from the intracellular and extracellular sides additively, whether concentrations, mobilities, or Nernst voltages, as it happens in the GHK equation (see section 3.4.3).

The derivatives are spatial derivatives; the temporal derivatives (needed for describing the time course) of the concentration(s) and voltage are derived in [6]. In 'ordinary' physics, where the charge and mass are independent, changing one side changes the other in the opposite direction. In 'non-ordinary' physics [16], valid for electrolytes, the two differentials must change in the same direction, given that ions' charge and mass cannot be separated (BTW: this is why Eq. (1.2) results in a negative potential difference). (The derivation in 'ordinary physics', such as [1], Eq.(11.30), is wrong, at least because of two reasons: the same speed is assumed for mass and charge transport; furthermore, the used partial derivatives are not independent; the details are discussed throughout the paper.)

Numerics panel 3.4.1:**A rugalmas erő**

Amint a **Numerics panel 2.1.1** bemutatta, a rush-in utáni pillanatban egy ion $F_{max} = 1.6 * 10^{-12}$ [N] erőt fejt ki a membránra. A membrán ugyanekkor kényszererőt fejt ki az ionra. A membrán mozgását leíró egyenlet által meghatározva, az ionra ható rugalmas erőt a

$$F(t) = F_{max} \times e^{-\zeta * t} \times \cos(\omega \times t)$$

egyenlet adja meg. A **Numerics panel 2.1.7** szerint a rush-in után az AIS-en legfeljebb

$$F_{Na^+} = 2 * 10^3 [N/C] \times 1.602 * 10^{-19} [C] = 3.2 * 10^{-16} [N] \quad (3.41)$$

elektromos erő mozgatja az ionokat. A rush-in által generált gradiens és erő kb. ugyanilyen nagyságú, hasonlóan a termodinamikai gradiens is. Eszerint a maximális rugalmas mozgató erő több mint két nagyságrenddel nagyobb mint a maximális elektromos erő, így a csillapított rugalmas rezgés szinte tökéletesen leírja az AP-t. A rugalmas erő oscillál, ezért annak domináns hatása nem állandó.

3.4	Forming membrane's potential	35
3.4.1	States	37
3.4.2	Controlling states	46
3.4.3	Goldman-Hodgkin-Katz potential	51

Chapter 4

Tévhitek a neuronokról

4.1 A kifelé irányuló K^+ áram az AP folyamán

A V1.0 megalkotása során az egyik nagy rejtély volt, hogy az AP miért vált negatívra, amikor az AP létrejöttkor csupán Na^+ beáramlás látható. Mivel csak pozitív töltéseket találtak a környéken, feltételezték, hogy azt pozitív töltések, nevezetesen K^+ ionok, kiáramlása okozta. A folyamat azonban túl gyors, ezért HH egy feszültség rögzítő eljárást (clamping) használt [46] a fellelvezett K^+ áram kimutatására. Ennek lényege, hogy imitálták a membrán feszültségkülönböztétét (az "AP" állapot elérését, lásd a 4.2 és 4.1 ábrákat): "(after completion of the quick pulse through the membrane capacity)", majd egy negatív visszacsatolást használó áramkörrel a membrán intracelluláris tartományát a nyugalmi potenciálnál magasabb potenciálra emelte: "the potential difference across the membrane is suddenly changed from its resting value, and held at the new level by a feed-back circuit ('voltage clamp' procedure)", és ilyen állapotban mérte az áram időfüggését.

4.1.1 A folyamat formális leírása

A 4.1 ábrán láthatók azok a változások, amelyek az egyensúlyi állapotából ki-mozdított neuronban végbemennek. Ezek a folyamatok természetesen lassúak, azok szerepét a tranziens folyamatban a 3.4.1 szakaszban tárgyaljuk, itt csupán a K^+ áramra vonatkozó állításainkat illusztráljuk. Állításunkat lényegében a Kirchhoff's Voltage Law-ra alapozzuk, ami szerint egy zárt áramkörben a feszültségek összege nulla; esetünkben az áramkör a neuron belsejétől a külső környezetig tart. Ha bármilyen okból a külső és a belső potenciálszint nem egyforma, akkor mindaddig áram folyik, amíg az ki nem egyenlítődik (az áram töltést szállít, és potenciálváltozást okoz; nem ideális feszültséggenerátor). Természetesen a kiegyenlítődés nem pillanatszerű, az áramnak időfüggése van.

Az ábra bal és jobb oldalán egyaránt a függőleges skála nulla értékénél

van a nyugalmi potenciálszint. Nyugalmi állapotban (invázió nélkül) az intracelluláris szegmensben a $[Na50, K400]$ koncentrációk vannak jelen a sejten belül. Ezeket használva a zöld nyilak mentén (a Nernst feszültségek alapján, lásd a 3.1 Táblázatot) a "Rest" állapotba jutunk, ami a -21 mV magasságban található. Jobb oldalt az extracelluláris szegmensben a szintén 0 magasságban levő pontból a kék nyilak mentén ugyanilyen módon egy, a 21 mV magasságban levő pontba érkezünk. A két említett pont közötti potenciál különbség 42.5 mV, amit HH ekkorának mért. Ami ugyan korrekt érték, de elektrosztatikus feltöltődés eredménye és nem pedig az általuk feltételezett "leakage current" következtében fellépő feszültségesés. A membrán két oldala közötti feszültség (amit a folyamatos narancssárga nyíl jelképez) felfelé mutat, ami az intracelluláris térben levő pozitív ionokat megakadályozza abban hogy az extracelluláris tér felé elinduljanak, és az extracelluláris térben levő negatív ionokat abban, hogy az intracelluláris tér felé induljanak. Az extracelluláris szegmensben levő Ca^{2+} ionok számára viszont a membrán átjárható, azaz folyamatos pumpálásra van szükség. A külső és a belső oldal között nincs potenciálkülönbség, ezért áram nem folyik.

Ha bármilyen okból a membrán feszültség a küszöbérték fölé emelkedik, akkor a neuron a rendkívül gyors Na^+ beáramlás miatt a membrán potenciál kb. 100 mV értékkel megnő (egy $[Na200, K400]$ állapot áll elő, lásd a barna nyilakat), azaz a neuron átugrik az AP buborékba. Az eddig felfelé mutató narancs nyíl most szaggatva lefelé mutat: a pozitív ionok (K^+ és Na^+ is) kiáramlása szabad utat kap, a Ca^{2+} ionok számára viszont megszűnik a lejtős pálya. A membrán felületi rétegében azonban szabad ionként túlnyomó többségben a most érkezett Na^+ ionok vannak, így az ionáram az AIS-en keresztül főként ilyen ionokból áll. A membrán feszültség jelentős emelkedése maga után húzza a a neuron belső tere potenciáljának jelentős emelkedését. Áram indul, ami igyekszik a kezdeti feszültség és koncentráció viszonyokat visszaállítani. A membrán extracelluláris környezete rendkívül stabil (a koncentrációk nem változnak), a sejt belsejében viszont a feszültség ugyanilyen mértékben megemelkedik, az "Invasion" irányába. A membrán potenciál a Na^+ koncentráció változása miatt változott meg lényegesen, ezért a sejt majd azokat mozgatja az egyensúly irányába.

A természetes AP esetén nincs akadálya, hogy a neuron a korábbi állapotát bármelyik paraméter változtatásával helyreállítsa, ezért a rendszer koncentrációi elindulnak a $[Na50, K400]$ állapot felé. Ezt legegyszerűbben úgy tudja elérni, hogy a $Na[200]$ -ból igyekszik $Na[50]$ -et csinálni, a K^+ még/már megfelelő koncentrációban van. Azaz, (főként) Na^+ ionokat bocsát ki az AIS-en keresztül, amíg a potenciálkülönbség nullára nem csökken. A Na^+ koncentráció csökkentése egyúttal a membrán feszültséget is csökkenti, azaz a felesleges (a Na^+ rush-in során beengedett) ionok kibocsátása után a neuron alapállapotba áll vissza; egy körfolyamat zajlik le.

A "clamping" rögzíti a membrán feszültségét, azaz a neuron a kiindulási nyugalmi állapotot nem tudja helyreállítani; ehelyett a koncentrációk változtatásával kell új egyensúlyt találnia, miközben a mesterséges AP után a clamping (negative feedback alkalmazásával) a membrán potenciált rögzített

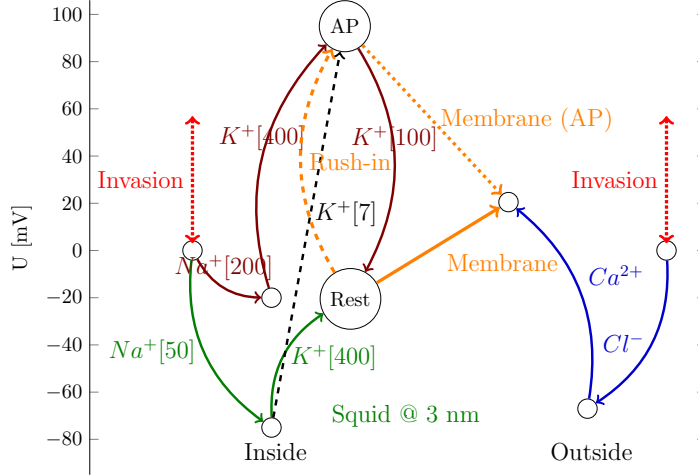


Figure 4.1: The figure is a modified version of Fig. 3.8. It is supplemented with the unbended black line showing that the clamping condition forces the neuron to find a new balanced state by compensating the clamping voltage with setting a concentration $[Na50, K7]$. For comparison, the case of natural AP is also shown. The mechanism of producing AP (using numbers referring to the case of squid shown in Table 3.1). In the resting state, the inside thermal potentials follow the green paths. When the rush-in of Na^+ ions suddenly increases the internal concentration to [200], the potential on the internal side of the membrane increases to +90 mV. The neuron must decrease its concentrations to restore the resting potential, mainly by releasing an action potential through the AIS. The clamping fixes neuron's membrane voltage at AP, so the neuron can operate only with the concentrations.

értéken tartja. Mindaddig áram folyik, amíg Na^+ és K^+ koncentrációk Nernst-feszültségei ellensúlyozni nem tudják a clamping feszültséget. Ebből a $[Na50, K400]$ állapotból kell a neuronnak úgy beállítania a potenciált, hogy a Nernst-feszültségek plusz a membránfeszültség plusz a clamping feszültség összege nulla legyen (tehát egy másik egyensúlyi helyzetet kell beállítani!), miközben a feszültség rögzítve van. Ehhez csökkentenie kell a K^+ koncentrációt, miközben a Na^+ koncentráció már lényegében jó értéken áll (az előzőhöz képest fordított eset). Még pedig nem is kis mértékben: $K^+[7]$ idéz elő csak akkora feszültségváltozást, amekkorát a clamping 100 mV rákényszerít a sejtre. Ezt a feszültséget a zöld hajlított $Na^+[50]$ és a fekete szaggatott egyenes $K^+[7]$ útvonala adja meg. A neuron ezt a $[Na50, K7]$ koncentrációt úgy tudja elérni, hogy intenzív K^+ kibocsátásba kezd.

A Na^+ esethez képest további fontos eltérés, hogy most nincs a membránon elektrosztatikus gyorsítás (a Na^+ beáramlás is ezért állt le), csupán a clamping miatt előállt termodinamikus szívóhatás kifelé. A K^+ ionok érzik a hívást, de csak diffundálni tudnak, és ettől a fiziológus "delayed current"-et lát. Valóban

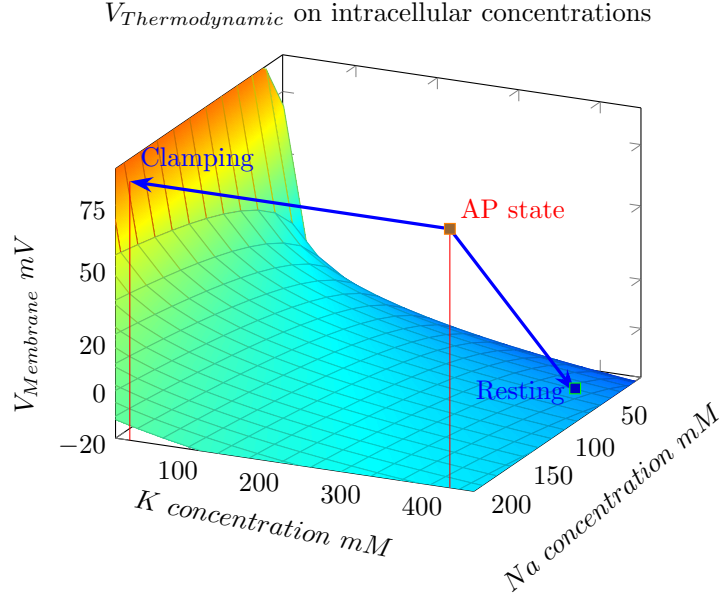


Figure 4.2: Az AP állapot mesterséges és természetes lebomlása. A természetes úton történő lebomlás esetén a neuronnak a (számára felesleges) Na^+ ionoktól kell megszabadulnia; azok kibocsátásával a kiindulási állapotba tér vissza; körfolyamat megy végbe. A clamping rögzítve tartja a membrán feszültségét, így a neuron csak az intracelluláris koncentrációk változtatásával tud új egyensúlyi helyzetet kialakítani, ami a nagyon lecsökkent K^+ koncentráció miatt csak intenzív K^+ kibocsátással lehetséges.

K^+ ionok áramlanak ki a membránon keresztül, de addig nem tudnak kilépni, amíg az elektromos potenciál nem csökken a koncentráció ekvivalens termodinamikusan hajtóereje alá. A clamping mérés alkalmával először Na^+ kezd távozni. A kibocsátás rövid szakaszon (egyidejűleg) párhuzamosan történik: a Na^+ az AIS-en kezd távozni, miközben a membránon keresztül a "felesleges" K^+ ionok távoznak. A természet furcsa játéka, hogy a termodinamikai és elektromos potenciál eredője a Na^+ ionokat a membrán felülettel párhuzamosan, az AIS felé, a K^+ ionokat pedig a felületre merőlegesen a membrán felé mozgatja, kb. az akciós potenciál feszültség csúcsának eléréséig, egyidejűleg. A két áram nem ugyanott folyik, így nem is képesek ellentétes irányú áramként interferálni és hiperpolarizációt előidézni. A hiperpolarizációt a Na^+ áram irányváltása idézi elő, amit az elektromos szemléletben a kondenzátor kapacitív áramaként, a mechanikus szemléletben a rugalmas membrán rezgésének negatív periódusaként értelmezhetünk. Azaz, a K^+ áram a clamping által előidézett artifact, amit a 'clamping' mérési feltételei okoznak, természetes körülmények között a neuronban nem fordul elő.

Szóval, innét származik a " K^+ áramlik ki a membránon keresztül az AP

második fázisában, a Na^+ beáramláshoz képest jelentős késéssel” városi legenda. Tényleg valós mérés, tényleg K^+ ion, csak éppen nem azt mérték, amit gondoltak. HH is meg kései utódaik is. Tehát nincs szó specifikus Na^+ és K^+ (egymáshoz szinkronizált) csatornákról: mindkét esetben a koncentráció és potenciálviszonyok döntenek arról, hogy milyen ionok áramlanak és merrefelé. Nem árt, ha jó modellt alkalmaz az ember arra, amit tanulmányozni akar. HH tisztában volt azzal, hogy az övék nem jó, de nem tudtak jobbat. Sajnos, a tévhitet terjesztették.

A 4.2 ábra a folyamat képszerű összefoglalását mutatja. A felület pontjai az ionokra ható Nernst-feszültséget adják meg, különböző kombinációk esetén. A felületen levő pontok egyensúlyi helyzetben vannak. A AP kezdete az addig egyensúlyi helyzetben levő pontot jóval a felületen kívülre mozgatja. Az újbóli egyensúlyi helyzetet a clamping kényszerítő hatása nélkül a neuron a megelőző állapot paramétereinek visszaállításával tudja megtalálni. Clamping esetén olyan új koncentrációkat kell beállítania, amelyek mellett a Nernst-feszültségek összege ellensúlyozni tudja a clamping feszültséget. Mindkét esetben relaxációs folyamat zajlik le, ezt a natív folyamat esetén AP-ként méri a fiziológia.

A mechanikus kép szerint a Na^+ ”robbanás” után a rugalmas membrán relaxál, azaz egy darabig nyomja, aztán meg húzza a Na^+ (és kevesebb K^+) keveréket az AIS-en át, ami oda és vissza folyó áramot eredményez. Ezt méri a fiziológia AP-ként. (Ha netán jelzett K^+ atomokat tesznek a citoplazmába, akkor tényleg lesz ellenirányban áramló K^+ is, de kevés.)

HH klasszikus kísérletében a clamping azt jelenti, hogy a bal oldali ”Invasion” úgy emeli hirtelen a membránfeszültséget az AP-nál mért csúcértékre, hogy a koncentrációk nem változnak. Az áram csak akkor szűnik meg, ha a belső szegmensek Nernst-feszültségeinek különbsége kiegyenlíti a clamping feszültséget. A töltéshordozók lassúsága miatt a folyamat nem azonnali.

4.1.2 A folyamat fizikája

A sejt lényegi működése nagyrészt a sejtmembránhoz közeli néhány nanométeres rétegekben történik, ahol néhány tucat mikroszekundumnyi időre akár több nagyságrendnyi koncentráció, feszültség és elasztikus gradiens (és ebből következően anyagáramlási sebesség) eltérések fordulhatnak elő. Továbbá, a lassú anyagáramlás potenciál- és koncentráció változásokat hoz létre, amik pillanatról pillanatra változnak, és amelyek hatása az ionok kémiai minősége miatt az egyes rétegekben jelentősen eltérő is lehet. Emiatt a történések leírása minden esetben, minden réteg és minden időszakasz esetén egyedi; nem ritkán tudományterületet is kell váltani az egyes szakaszok leírásakor, vagy azokat másféleképpen kombinálni.

A membránba ágyazott ioncsatornák két végén eltérő potenciál és koncentráció van, így az áthaladó ionok jelentős energiára és sebességre tesznek szert. Ez a gradiens azonban megszűnik, amikor kilépnek a csatornából, így az ionok csalódottan ”megállnak”. Az áthaladás mindaddig folytatódik, amíg az áthaladó ionok annyira megnövelik a másik oldal elektrokémiai potenciálját,

hogy a gradiens megszűnik. Az elektrolit belsejében a többi, nagyságrendekkel kisebb gradiens természetesen tovább működik, de csak igen lassú anyagtranszportot tud létrehozni. Minden folyamathoz lassú diffúziós hatás is társul, ezért a változások az elektrolit "bulk" részében a lassú kiegyenlítődés felé tartanak (a határoló membrán közelében más szabályok maradnak érvényben).

Amikor a Na^+ -ionok behatolnak az intracelluláris szegmensbe, azt jelentős elektrokémiai gradiens hatására teszik. Ilyen módon a közvetlenül az (néhány nanoszekundum ideig tartó) áthaladás után az ionok egy Na^+ -gazdag réteget hoznak létre, amelyből az ionok töltése (részben közvetlen ütközéssel) kiüti az ott levő, főként K^+ ionokat. Alapvetően tehát egy Na^+ réteg jön létre közvetlenül a membrán intracelluláris része felett, aminek hatására megnő a kondenzátorlemezekon levő feszültség, és mivel a membrán végében az AIS az ionok számára nyelőként viselkedik, így egy azonnali Na^+ áram indul, amint azt az elektromos modell tárgyalja. A Na^+ ionok a membránfelület mentén a nagy elektromos gradiens hatására mozognak a nagy vezetőképességű AIS-en keresztül, de kevés mennyiségű Na^+ és K^+ ion nyilván kölcsönösen bediffundál a szomszédos rétegbe. A réteg ionkoncentrációja és potenciálja a kifolyó árammal csökken. Az ionáram lassúsága miatt az ionok csak több tucat mikroszekundumnyi idő alatt érik el az AIS-t.

A pozitív ionok számára kezdetben (a Na^+ ionok áthaladás után) a membrán is le van zárva, a Na^+ réteg pedig egy további potenciálgátat jelent. Az idő múlásával azonban a membránon az utándiffúziós folyamatok és a rétegben az AP-t előidéző áram hatására csökkenő Na^+ koncentráció egyre inkább lehetővé teszi hogy a K^+ ionok is elinduljanak a membrán felé és a tapasztalt K^+ áram elinduljon, ami csak néhány tucat mikroszekundum után lehetséges. A "késleltetett K^+ csatornák" tehát olyan ioncsatornákat jelentenek, amelyek strukturálisan nem tesznek különbséget ionok között, és a késleltetést az ionrétegekben zajló tranzienst folyamatok okozzák. Ezeken a csatornákon valóban K^+ ionok folynak ki, amelyek csak akkor indulnak el, amikor a clamping voltage-t a neuronra kapcsolják (azaz, létrehozzák a neuron szegmens nem-egyensúlyi állapotát), és azért csak K^+ ionok, mert a termodinamikai gradiens csak számukra kedvező.

HH úgy találta, hogy az AP-t keltő áram két áramra bontható: "the ionic current during a depolarization consists of two more or less independent components in parallel, an early transient phase of current carried by sodium ions, and a delayed long-lasting phase of current carried by potassium ions". Meglepő az a következtetésük, hogy bár a Na^+ ionokból származó áram komponens gyorsan megjelenik, a K^+ ionokból álló komponens késéssel és tartósan megmarad. Ha magyarázatuk helytálló, egyetlen AP kibocsátása után a neuron rövid ideig tartó Na^+ és (5. és 6. ábrájuk szerint) végtelen hosszú ideig tartó K^+ áramot bocsát ki. Más szavakkal, amelyik neuron egyszer AP-t bocsátott ki, az mindörökké K^+ áramot bocsát ki. Ez neurononként 100 pA (azaz 1×10^{-10} A) telítési áram és 1×10^{11} neuron esetén 10 amper állandóan folyó áramot jelentene az agyban, továbbá folyamatosan a nyugalmi potenciál értéke alatt tartaná a membrán potenciált. A különbség megegyezne a clamping feszültség értékével; azaz eltűnne nulla clamping feszültség esetén. Más szavakkal, az extra nyugalmi feszültség

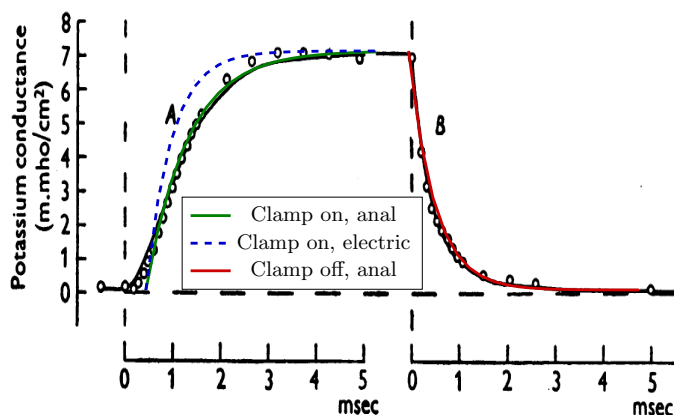


Figure 4.3: The differing time constants of the rising and falling edge of a neuronal square wave

és a K^+ áram egyaránt eltűnik clamping nélkül, a tapasztalattal megegyezően.

A mi olvasatunkban, a helyes fizika alapján, az áram mindaddig tart, amíg a clamping által okozott feszültséginvázió fennáll, vagy a neuron intracelluláris szegmensének K tartalma el nem fogy.

4.2 A neuronális rezgőkör időállandói

HH ugyanazt a hibát követték el, mint amikor a membrán axonon át történő feltöltődését mérték (lásd section 4.2). Jól észrevették, hogy a felfutó és lefutó él időállandója nem egyforma; ami önmagában bizonyítja, hogy a (párhuzamos) RC kör működésével nem lehet leírni az axonon át történő membrán feltöltést. Meg is mérték, csak nem értették, hogy a kettő nem ugyanazt a folyamatot mutatja (lásd Fig. 4.3). Továbbá, mindkét esetben "magyarázatot" adtak, ahelyett, hogy elgondolkodtak volna azon, mi az ellentmondás, és hogyan is állhat elő, amit látnak.

4.2.1 A GHK egyenlet

Contents

4.1	A K^+ áram az AP folyamán	55
4.1.1	A folyamat formális leírása	55
4.1.2	A folyamat fizikája	59

4.2	A neuronális rezgőkör időállandói	61
4.2.1	GHK egyenlet	61

Chapter 5

Nyugalmi potenciál

Concerning the origin of the membrane's resting potential, we disagree that "the resting membrane potential is the result of the [low intensity] passive flux of individual ion species through several classes of resting channels" [2]. This claim cannot be true: the flux is not passive (to move the ions, energy required), furthermore, no setpoint is defined, so some (low) current flows due to the interplay of the electrostatically set setpoint and the reversal potential of the ions inside and outside. The claim was already question-marked whether the membrane permeability is a primary property for the membrane potential generation, for a review, see [44]. That interpretation leads to causality issues [47].

It was already guessed that the membrane's resting potential is of electrostatic origin [48, 20, 45, 44, 49]. When ions are confined in a closed volume, they exist in a state of thermodynamic and electrical equilibrium. However, those equilibrium states cannot be discussed independently. We must discuss them in a non-disciplinary way because the participating ions belong simultaneously to the disciplines of electricity and thermodynamics; furthermore, we must cross the borderline between discrete and continuous points of view of nature [16]. Furthermore, the features of living matter may change when measured in physics laboratories as usual. When measuring them, we must consider that "the construction is different from anything we have yet tested in the physical laboratory" [19].

5.0.1 Deriving mebrane's "thermodynamic electrical field"

In section 1.1, we introduced the idea of the "equivalent thermodynamic electrical field"; now we calculate it for the case of neuron membrane as shown in

Physics panel 5.0.1:**The membrane's equivalent electrical thermodynamic field**

We assume that, in a balanced state, the concentrations on the two sides of the membrane are C_k^{ext} and C_k^{in} , respectively; furthermore we assume that

$$\frac{dC_k}{dz} \approx \frac{C_k^{ext} - C_k^{in}}{d}$$

To simplify the math writing, we neglect the change caused by considering the difference (maybe we should apply a near-unity factor); we use the larger concentration value instead. Using the constants and $T = 300$, we can derive the "equivalent thermodynamic electrical field" across the membrane for the case of a permeable ion channel in the membrane

$$E_{thermal}^{C_k} = -\frac{RT}{q * F} \frac{1}{C_k^{ext}} \frac{C_k^{ext} - C_k^{in}}{d} \approx -\frac{RT}{q * F} \frac{1}{d} \quad (5.1)$$

The equivalent thermodynamic electric field does not depend on concentration, but rather is linear in temperature and inversely proportional to membrane thickness (given in nm). We provide the numerical values in **Numerics panel 5.0.1** for the case of ion channels in the membrane's wall.

Numerics panel 5.0.1:**Neuron's equivalent thermodynamic electrical field**

It is valid separately for all C_k

$$E_{thermal}^{C_k}(d) = -2.585 * 10^{-2} \frac{1}{d} \left[\frac{V}{m} \right] \quad (5.2)$$

Figure 5.1 depicts the field's dependence on the membrane's width and how it compares to the charge-generated electrical field. The figure illustrates the setpoint(see section 3.4.2), the boundaries of the changes, and the interesting phenomena that occur within the shaded cross-section. The figure is an illustration, but its figures are not far from the true ones. For a 5 nm membrane thickness, it results in $5.17 * 10^6 \frac{V}{m}$, which compares well to the value provided by Eq. (3.29). (Notice that the membrane's thickness may change during the action potential, see [50]: the rush-in extra charge compresses the membrane. Similarly, the concentration of the electrolyte in proximity to the membrane also changes.) Correspondingly, by using that $U = E * d$,

$$U_{thermal}^{C_k}(5) = -25.85 \left[mV \right] \quad (5.3)$$

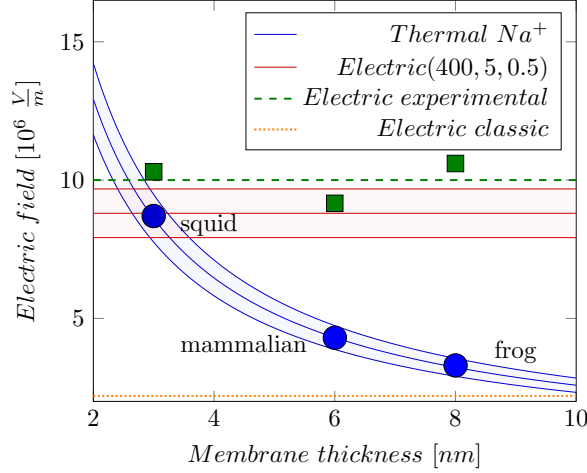


Figure 5.1: The thermodynamic "electrical field" depends on the membrane's thickness, see Eq.(5.2). The electrical gradient depends on the charged layer's concentration and thickness; see Eq.(3.29). The dots mark measured data shown in Table 3.1.

In simple words, the Nernst-Planck equation states that the changes in concentrations of ions create changes in the electrical field (and vice versa), and in a stationary state, they remain unchanged. Notice that in the case of an ion mixture, a joint charge-generated electrical field exists, and they equal, per ion, the concentration-generated "electrical fields", see section ??.

In the steady state, in contrast with the case of an ion in the infinite space, some other forces also contribute to the mentioned ones. To discuss how nature restores the steady state when a microscopic change occurs in a balanced state of a biological solution, we write the well-known Nernst-Planck equation (see Eq.(1.1)) in a slightly extended form:

$$\underbrace{(-F_{constr}(z))}_{\text{Constraint}} = \underbrace{e_{el} * E_{Gap}^{Total}(c, \Delta z)}_{\text{Electrical}} + \underbrace{e_{el} * E_{thermal}^{C_k}(d)}_{\text{Thermodynamic}} \quad (5.4)$$

$$\underbrace{(+F_{Transp}(z))}_{\text{Transport}} \underbrace{(+F_{Invasion}(z))}_{\text{External}} \quad (5.5)$$

We multiplied the usual two terms by the elementary charge, so its terms are expressed as forces, plus we added an external force (its role is discussed below). Furthermore, changing one force triggers a corresponding counterforce and/or causes the ion to move in a viscous fluid. That is, the speed of the material transport gradually changes as the steady state approached; furthermore, it

is by orders of magnitude smaller than the speed of the interactions. When neuroscience teaches that "pumps that maintain ion gradients . . . transport ions against their electrical and chemical gradients" [2], page 101, one shall ask *what kind of force acts on the ion*, keeping in mind E. Schrödinger's opinion that not "any 'new force' or what not" [19] affects ions' motion.

An interesting option is when the counterforce combines two disciplines. When Na^+ ions rush-in at the beginning into the membrane, they produce simultaneously a huge electrical and thermodynamic gradient. The repulsion among the ions creates a huge mechanical pressure that presses the elastic membrane. The counterforce starts a mechanical shock wave (soliton) [51], that is simultaneously an electrical potential wave [52] that is measured as AP. Only using cross-disciplinary discussion (non-ordinary laws) enables understanding neuronal operation. That means when describing an ionic transfer process, *we must not separate the electrical current from the mass transfer*: they happen simultaneously and mutually trigger each other. Notice that the thermodynamic term is ion-specific while the electrical term is not. To be entirely balanced, the system must be balanced to all elements. In this way, changing one concentration implicitly changes all other concentrations and the electrical field.

It was a colossal mistake to introduce equivalent circuits with their fixed-value voltage generators. It forces one to assume that the conductances of the players (membrane, synapses, AIS) change without any reason and prevents understanding how the competition between thermodynamic and electrical processes governs neuronal operation. It leads, among others, to attributing conductance change to membranes, which are simple isolators with no charge carriers that can implement charge transfer, see section ??; this way attributing the change of an electrical entity to the biological material. This assumption neglects the driving force the mentioned forces provide. Instead, it attributes the magic ability to the ion channels that they can change their transmission ability as the actual situation requires.

When discussing balanced states, no transport occurs, so the transport force cancels, and the mobility has no role. The counterforce adapts to the situation. That force may be a mechanical one: the ions sitting on the surface of the membrane press the surface due to the attractive force of ions, and the membrane mechanically provides the needed counterforce. If the boundary of the segments is not freely penetrable, the counterforce equals the difference between those two forces. In this way, no force acts on the ions; the two gradients persist. When the ions can move freely between the segments, they will move until they produce a concentration gradient (a thermodynamic force) for the given ion. When neuroscience teaches that "pumps that maintain ion gradients . . . transport ions against their electrical and chemical gradients" [2], page 101, one shall ask what kind of force acts on the ion, keeping in mind E. Schrödinger's opinion that not "any 'new force' or what not" [19] affects ions' motion. that counterbalances the electrical gradient (the electrical force) and then the transport stops. A transport force is needed to reach the Stokes-Einstein speed (see Eq.(3.11)) in a viscous fluid.

We must not forget that initially, the Nernst-Planck equation described a

transfer process (i.e., a dynamic equation), assuming the same speed for mass and charge transfers. We used it only to describe the balanced state (i.e., a static equation at zero speed) and added the term describing external (such as mechanical constraint) force that may be involved; it affects the process, but it does not belong to either of the respective fields. Commonly known mechanical constraints appear in a non-conducting layer called the membrane, where that counterforce prevents ions from penetrating the membrane layers. A membrane is a perfect isolator, i.e., no charge carriers exist between its two surfaces, so it is senseless to introduce the notion of its conductance. (There may be ion channels that deliver ions built into the membrane, but it is a different subject.) The separation results from *charge separation*, a mechanism distinct from *polarization*, as typically described in most biology textbooks. In the form we derived, the equation correctly describes the ion transfer, a current conveying ions with the speed controlled by the actual gradients, instead of assuming an ion current traveling with the apparent speed of electrical current in solids.

We must not forget that originally the Nernst-Planck equation described a transfer process (i.e., a dynamic equation), assuming the same speed for mass and charge transfers. We used it only to describe the balanced state (i.e., a static equation, at zero speed), and added the term describing external (such as mechanical constraint) force may be involved; it affects the process, but it does not belong to either of the respective fields. Commonly known mechanical constraints are a non-conducting layer, called membrane, where that counterforce prevents ions from penetrating the membrane layers. A membrane is a perfect isolator, i.e., no charge carriers exist between its two surfaces. (There may be ion channels, that deliver ions, built into the membrane, as we discuss in section ??, but it is a different subject.) The current clearly results from charge separation; a mechanism distinct from polarization most biology textbook uses. In the form we derived, the equation correctly describes the ion transfer, a current conveying ions with the speed controlled by the actual gradients, instead of assuming an ion current traveling with the apparent speed of electrical current in solids.

5.0.2 Charge layers in segment

Figure the ionic basis of a membrane potential shows and explains the case introduced by separating the cell into two segments by a finite-width membrane; although does not explain even qualitatively, *why* the ions behave apparently against the laws of thermodynamics: the left and right sides of the figure are apparently identical. On the left side (the case of infinitely thin membrane), at an exact balance of charges on each side, the potential across the membrane is zero. We explain that the finite thickness will result in a lack of balance (introduces inhomogeneity and creates a voltage and concentration gradient) proximal to the surfaces of the membrane, even if the concentrations on the two sides are the same. Changing the bulk concentration or potential in one of the segments creates a corresponding gradient across the separating membrane that increases the inhomogeneity proximal to the membrane. The ions will experi-

ence an extra force due to the gradient, but the mechanical counterforce of the membrane will keep them back in the segment. The *concentration and potential, inseparably and having the same time course*, will change across the two sides of the membrane just because of the gap's physical features the membrane represents.

We consider that the segment is composed of electrically conducting discs (the ions are free to move on the surface) and the charged discs' contribution to the electrical field at point P (see Fig. 3.3) can be calculated as known from the theory of electricity. Due to the symmetry in direction of z , in a homogenous solution, the resulting electrical field in the plane perpendicular to direction z is zero: we have contributions of the same size with opposite signs.

Here we used the abstraction of an infinitely thin "charged sheet" and that there is a step-like gradient in the electrical field in the gap. The physical reality is that the charge is represented by ions, which simultaneously represent a mass; furthermore, the thickness of the surface layer they form is by orders or magnitude thicker than the layer that the electrons form. In an equilibrium state, the forces due to the voltage and concentration gradients (see Eq.(1.1)) must be counterbalanced by an external force. An interference between science disciplines can also manifest here. We know at the same time that at the boundaries of electrolytes, different interfaces, including electrically neutral electrical double layers, can be formed by only partly known processes [53]. The presence of those structures makes drawing quantitative conclusions hard.

Simple conducting sheet (a classical condenser)

When discussing the internal electrical operation of a neuron, first we consider a condenser that obeys laws of classical electricity and has the geometrical size of a biological neuron. We apply the physics terms to that pseudo-biological object and test whether the values we derive are consistent with physiological phenomena. A tiny (according to [23], page 12, about $2 \cdot 10^{-3}$) portion of the positive ions leave their negative counterions behind and form a thin positively charged ion layer on the surface of the membrane (of course, the same happens on the opposite side with negative ions). Experience shows, that – also in biology – two very thin (but not infinitely thin) layers of ions are formed on the two sides of the membrane, see the caption of figure (Fig. 11.22 in [24]): "The ions that give rise to the membrane potential lie in a thin ($< 1 \text{ nm}$) surface layer close to the membrane". The two surfaces get covered by a sub-nanometer-thick conducting layer, on the top of a 5 nm nanometer thick insulator. So, we model the cell with two finite-width "conductive sheet" layers. These layers represent an electrical condenser (so we can calculate the internal electrical field between the plates) and they counterbalance each other's electrical field outside the condenser. We may assume that the solid surface represents the needed counterforce to keep the charges in rest.

One cannot expect geometrically plane surfaces: there are some biological objects of up to 1 nm size on the surface. So, we assume a physical 'uniformly charged sheet' with an average thickness $\Delta z = 0.5 \text{ nm}$ on the surface. For the

sake of simplicity, we assume a step-like function for the electrical field. It is zero in the segment outside the "conducting sheets" (the 'bulk' portions) as well as inside the sheets. Between the sheets, it jumps to the value of the electrical field of a charged sheet with surface density σ_A (see Eq.(3.20)). (In this picture we see that there is a jump in the value of the electrical field (and so: in the force acting on a charge) in the two sides of the physical layer. Here we still assume that a mechanical counterforce due to surface's roughness keeps the charges in place inside the charged layer.)

We assume that all unbalanced (dissociated) ions are in the layer, and the rest contains no dissociated ions, furthermore, that the ions' concentration is unchanged in the segment. By assuming a 0.5 nm thick layer on the surfaces of the membrane, our model delivers an electrical field (see Eq.(3.21))

$$E_{Classic}^{Gap}(c, \Delta z) = E_z(c, \Delta z) = 4.45 * 10^3 * c * \Delta z \quad \left[\frac{V}{m \text{ nm} * mM} \right] \quad (5.6)$$

In the case of a 400 mM solution and a 0.5 nm 'thin physical layer' it evaluates to an electrical field

$$E_{Classic}^{Gap}(400, 0.5) = 0.11 * 10^7 \left[\frac{V}{m} \right]$$

and at a 5 nm membrane thickness the voltage across the plates becomes

$$U_{Classic}^{Gap}(400, 0.5, 5) = 2 * E_{Classic}^{Gap}(400, 0.5) * 10^{-3} * d = 10.84 \text{ [mV]} \quad (5.7)$$

The number of uncompensated ions needed for the cell, using the method in [23], is $0.88 * 10^7$. Remarkably, the calculated values are about by a factor of 5 lower than the experimentally derived values. "An electrical potential difference about $50 - 100 \text{ mV}$... exists across a plasma membrane only about 5 nm thick, so that the resulting voltage gradient is about $100,000 \text{ V/cm}$ " [24]. "The number of uncompensated ions needed for the cell is $4.7 * 10^7$ " [23]. We are in the right order of magnitude but we arbitrarily assumed a layer thickness, a non-dielectrical medium. Furthermore, given that the charges are distributed uniformly in the layer, the electrical field changes linearly between the two sides of the conducting sheet. The deviation from the experienced values suggests that the dipoles' presence in the electrolyte bulk significantly changes the achievable electrical field and potential across the plates. Our simple model seems to be a strong oversimplification.

If the distance between the plates is finite, the resulting electrical field will differ from zero. With such a model, a usual parallel plate condenser can be derived, as shown in Fig. 5.2. We have two charged disks (infinite conducting sheets), and an insulator layer between them. In the classical picture, the opposite charges' amount on the two plates must be the same. As shown, a constant electrical field is present inside the membrane (across the plates of the condenser) and zero electrical field inside the parallel conducting plates, as well as outside the condenser. In this ideal picture, the charges are aligned on the border of two (infinitely thin) conducting layers and cannot move. Anyhow, the

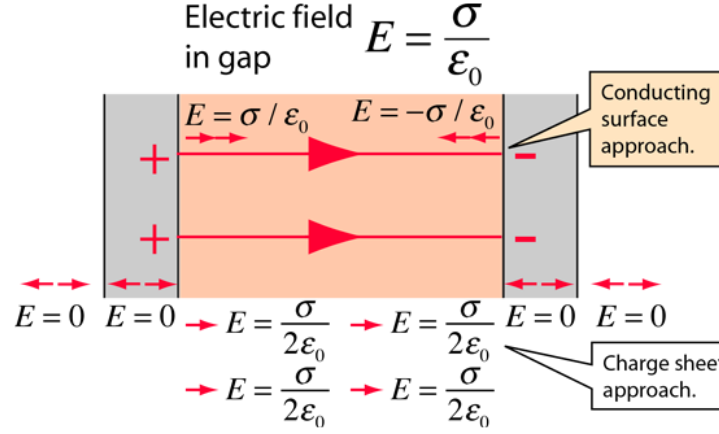


Figure 5.2: The oppositely charged parallel surfaces of the membrane are treated like conducting plates (infinite planes, neglecting fringing), then Gauss' law can be used to calculate the electrical field between the plates. Presuming the plates to be at equilibrium with zero electrical field inside the conductors, then the result from a charged conducting surface can be used.

attractive force between the opposite charges on the plates keeps them fixed in direction of z . The repulsive force between the charges with the same sign keeps their surface density in the (x, y) plane uniform: the infinitely thin plates are equipotential. The strong electrostatic force can produce an enormous acceleration for the individual ions, but the thermodynamic gradient can only change with a several orders of magnitude lower speed, allowing measurable changes in the current intensity on the surface). This picture is valid for the equilibrium state of charges, infinitely small non-dissociating charge carriers and perfectly smooth surfaces.

Figure 5.3 (the blue diagram lines) show the electrical field around the neuronal membrane. In this picture we see that there is a jump in the value of the electrical field (and so: in the force acting on a charge) in the two sides of the physical layer and we assume that the physical layer somehow emulates the well-defined boundary on the side opposite to that proximal to the membrane.

Electrical field in a two-segment electrolyte

Fig. 5.3 displays how the function shapes of electrical field change in the function of the distance from the membrane. As discussed, the physical distance between two two segments with electrical charge causes a charge condensation on the surface of the separating membrane. In the case of classic condenser, this primary charge density is abstracted as an infinitely thin layer of charges (electrons) on an ideally plain layer, and the rest of the volume is abstracted that no charge carrier is included. This case is perfectly described by the classic electricity theory. In the case of electrolytes, the separating membrane's surface

is not ideal, furthermore the electrostatic force can press some biological objects to the surface, so we must assume that the primary charge forms a uniformly charged layer with finite thickness on the surface. In the uniformly charged layer the electrical field is a linear function of the distance from the surface. The presence of dielectrical molecules in the rest of the volume creates secondary (virtual) dipole charges, that contribute to the electrical field, as shown. The two fields join to each other at the boundary of the charged layer. The electrical field on the surface of the membrane is the sum of those two fields. The opposite charges on the other plate generate another electrical field in the opposite direction.

As shown in the figure, the field steeply rises near to the membrane. At the membrane surface, it reaches its maximum value; on both sides of the membrane. Inside the membrane, there are no charges, so the field remains constant, for both types of condensers. Notice that the classic condenser can be considered as the abstraction of the real condenser to the case of zero dielectricity.

The electrical field acts on the ions, but there may be present a mechanical counterforce, plus a thermodynamic force arising from the difference of ions' concentrations. (We just notice here that *the electrical driving force does not depend on the chemical type of the ions, but the thermodynamic force does*. The mechanical counterforce is adaptive, per ion, so it can keep different types of ions in rest.) If there are 'holes' (ion channels) in the membrane, there is no mechanical counterforce, so the ions could move under the effect of the difference between the electrical and thermodynamic forces. The latter depends on the chemical nature of the ions, so it natively provides selectivity: the resulting forces are different for ions of different types. The usual "downhill" method of explanation must be corrected: the *resulting potential* instead of the *electrical potential* defines how the ions move.

The figure shows two ions (green and red) at the axis of the ion channel in two positions (the empty circles along the axis of the ion channel), at the beginning and end of a movement, when the ion channel is open. On the diagram lines, filled circles show the electrical fields on the diagram lines at the position of the ions.

In the charged layer the ion *density* is constant, that is the thermodynamic force is zero while the electrical force decreases with the distance from the membrane's surface

If the channel opened, the mechanical counterforce cancels, the ions will move under the effect of the resulting force, that comprises not only the thermodynamic and electrical forces, but also the force to move the ion with the Stokes-Einstein speed. In the position at the entrance of the ion channel (see the red ions), if the channel is open, the large thermodynamic and electrical forces 'push' the ions into the channel. Here the concentration gradient quickly cancels, but the huge potential gradient will accelerate it toward the other segment. After passing the membrane with a high speed, the ion faces a quickly decreasing field, and its speed quickly decreases. The leave of ions changes both concentration and charge density in the vicinity

In the position farther from the membrane's surface (see the green ions)

the ions must travel 'uphill'; that is, they cannot travel instantly toward the membrane. They must wait until the gradient change caused by the ions pushed into the gap reaches their position. As the figure shows, the ions quickly move out from the region with high charge density and very slowly refill that region with ions from the region farther from the membrane. This speed difference results in a quick change in the local electrical field. In the region of the other plate, the ions decrease the (opposite) local potential. As the result, there are less ions ready to travel on the one side, and the potential difference continuously decreases on the other, so the flow of ions stops: the ion channel shuts itself down (and, on both sides, starts to regenerate the potential as it was before the channel's cap was removed). That is, the ion channel is opened by a slow mechanical cap, but it is closed by a fast electrical shutter. While the potential self-closes the channel, the electrical fields on the two sides of the membrane have to relax: the ions and dipoles have time to change their position and even their polarization and ionization state. The local gradients will rearrange concentration and electrical field in the segment.

As depicted, in the two regions, there is a significant difference in the value of the effective force acting on the ions (they are using $E_{accelerate}$ and E_{assist} , so they produce different Stokes-Einstein speeds (see Eq. (3.9)). Based on those speeds, one can introduce "fast" and "slow" ions and correspondingly, speak about '*slow*' and '*fast*' currents that the ions represent at a macroscopic level. The corresponding speeds also differ by orders of magnitudes. For this study, we assume the diffusion, potential-assisted and potential-accelerated speeds, in m/s to be 10^{-4} , 10^{-1} (also inside neurons [27]), 10^{+3} , respectively (used only to estimate the order of magnitude of some respective operating times). When staging, we assume the greater of the mixing speeds as 'infinitely large' and omit the time that the process needs, while discussing how the slower process proceeds.

We are in line with the estimation given in [24] that in the case of resting potential, the scale of the gradient that accelerates the ions across the ion channel is calibrated approximately as kV/cm . Recall that we are still speaking about the resting state and only about the extra gradient evoked by the finite-width membrane. We are at the boundaries of the macroscopic and microscopic worlds. We derived our integrand from the picture of discrete charges but integrated it into the picture of continuous charge distribution. We assume an atomic layer (a skin) on the surface. However, the layer itself can also be modeled as having just a few ions under their mutual repulsion on the surface or a few atomic layers on top of each other, depending on the concentration and voltage in the bulk on the two sides. (The diagram line is valid in the plane crossing the membrane and the ion channel.)

We assumed that the membrane's width is 5 nm . An ion channel is depicted in the middle of the figure with a diameter of about 1 nm . Furthermore, we assume that the ion's size and, correspondingly, the thickness of the atomic layer in the electrolyte on the surface of the membrane is about 0.1 nm . For comparison, recall that the size of the tip of the clamp pipette is in the range of $1,000\text{ nm}$ and the size of the soma in the range of $10,000\text{ nm}$.

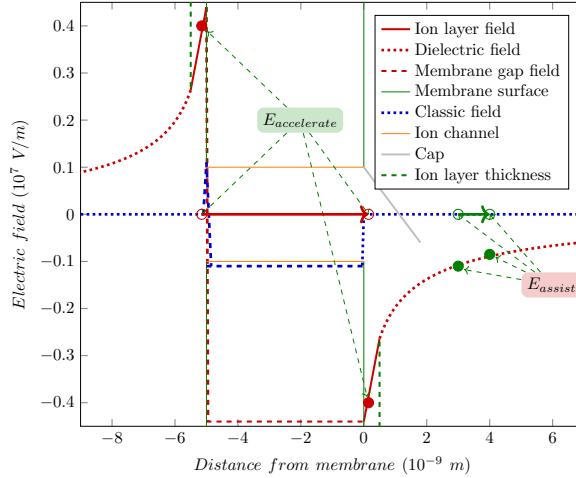


Figure 5.3: The neuronal membrane's electrical field in the function of the distance from the membrane's surface and the bulk potential. The thickness of the atomic layers proximal to membrane's surfaces are also shown.

Our results align with the observation (see caption of 11.22 in [24]: "A small flow of ions carries sufficient charge to cause a large change in the membrane potential. The ions that give rise to the membrane potential lie in a thin ($< 1 \text{ nm}$) surface layer close to the membrane". See the dotted line in Fig. 5.3; notice that the x-scale on the figure spans 1 nm only. The amount of unbalanced ions is in the range of 10^7 , and so is the amount of rush-in ions. In addition, those ions on the high-concentration side rush-in to the low-concentration side and cause the large change in the membrane potential. Their absolute amount is small compared to the total number of ions in the cell, but it is significant compared to the number of unbalanced ions.

Membranes with ions channels

The attraction between the ions in the two skin layers prevents the ions in the layers on the two sides from diffusing into/from the bulk without a current drain in the layer for an extended period. This steady state results from the interplay of the concentration and the potential described by Eq. (1.1). The gradients change gradually within the segments and drop suddenly across the membrane. No current can flow through the membrane; there is no leakage current.

Here, we assume that no ion channels are in the excellently isolating wall (ion channels would mean a current drain and, therefore, a voltage drop).

5.0.3 Goldman-Hodgkin-Katz potential

It was already stated, based on experimental evidence, that "the membrane permeability to the ions has nothing to do with the potential generation and the ions adsorption on the membrane surface generates the membrane potential", for a review see [44]. The idea itself is nonsense: in a balanced state, no ion transport happens, so even without permeability, the balanced state persists; the resting potential has nothing to do with either ions' mobility or permeability nor with ion absorption [45]. Furthermore, there is no idea in GHK about whether the 'setpoint' (why that specific concentration or potential difference) is present and why the same potential is reset after rough perturbations such as issuing an AP or replicating a cell. The causality is reversed: the potential is a static concept which is set electrically, and the two gradients form the experienced concentrations and potentials from the available solvent molecules. For the time course of forming the gradients, mobility and permeability play a role, but not in determining the concentrations or the resting potential.

In our clear physical picture, the thermodynamic forces on the one side and on the other side of the membrane are summed. They counterbalance each other; furthermore, jointly the effect of the neuronal condenser, see Fig. 3.7. As we emphasized, the concentration gradients are ion-specific. Furthermore, as we discussed in connection with Eq.(1.2), the Nernst equation comprises a per-ion indefinite constant (a potential difference). To calculate a linear combination of terms comprising an arbitrary constant is nonsense, and so is adding absolute concentrations on the different sides of the membrane, or changing the base of the logarithm used in a calculation to match the experimental value. It is not more than number magic. The potential is described by coupled equations as discussed in connection with Eq. (3.37).

The Ca^{2+} ions do not fit into the GHK picture. One of the reasons why GHK cannot be good is that $[Ca^{2+}]$ is omitted. Biologically, it is hard to believe that Ca^{2+} does not participate in the game of life (especially since biology sees the need for Ca^{2+} pumps). Physically, one concentration alone on one side of the membrane cannot keep balance; as the different ion exchange processes, Table 3.1 and Fig. 3.7 show. When adding a new ion to the solution, the sum concentration increases, and so the electrical force increases, forcing the previous elements to find new concentrations on both sides. The appearance of a new chemical element indirectly changes the concentrations of the others (a good example is the role of the negligible amount of Ca^{2+}).

In the light of our findings, we must reconsider a few key conceptions. We derived (see Eq.(3.30)) that the gap voltage depends on the thickness of the membrane and the overall concentration. Furthermore, we demonstrated that the gap voltage is equivalent to the resting potential (correctly: potential difference) and is independent of ion mobility or ion composition in the solution. Derived values, such as the Goldman-Hodgkin-Katz potential, should be rethought. The idea is nonsense: in a balanced state, no ion transport happens, so even without permeability, the balanced state persists; as experiments proved, the resting potential has nothing to do with either ions' mobility or permeability nor with ion

absorption [45]. The causality is reversed: the potential is a static notion and is set electrically, and the two gradients form the experienced concentrations from the available solvent molecules. For the time course of forming the gradients and maintaining the balance, mobility and permeability play a role but not in determining the concentrations or the resting potential.

The Ca^{2+} ions do not fit into the GHK picture. However (see our Table 3.1), it provides a thermodynamic contribution in the same order of magnitude as the charge separation by the membrane. If we add the potential due to Ca^{2+} , we arrive at the correct result that the total balance of the thermodynamic contributions equals zero, that is, the resting potential is zero. The electrical potential contribution is an additional term, resulting in the resting potential.

When adding a new ion to the solution, the sum concentration increases, and so the electrical force increases, forcing the previous elements to find new concentrations on both sides. The appearance of a new chemical element indirectly changes the concentrations of the others (a good example is the role of the negligible amount of Ca^{2+}). No linear summing or similar assumptions can be made. The potential is described by coupled equations as discussed in connection with Eq.(3.37). As we discussed in connection with Eq.(1.2), the integral form of the Nernst equation implies an undetermined additive constant. To calculate a linear combination of terms comprising an arbitrary constant is nonsense, and so is changing the base of the logarithm used in a calculation to match the experimental value.

5.0.4 Moving in the membrane's potential

In Fig. 5.3, an ion channel is depicted in the middle of the figure with a diameter of about 3 nm for visibility. Furthermore, we assume that the ion's size and, correspondingly, the thickness of the atomic layer in the electrolyte on the surface of the membrane is about 0.1 nm (although with their electrical double layers, they can also reach the size of 1 nm). For comparison, recall that the size of the tip of the clamp pipette is in the range of 1,000 nm and the size of the soma in the range of 10,000 nm.

Passing through ion channels

Figure 5.3 also hints at how ions can move in the proximity of the membrane. On the left side, the electrical field increases toward the membrane, so the ions cannot move in that direction without an initial force being applied. The case is the same on the right side, because the charge is the opposite. An ion must gain energy to get closer to the membrane, which means, at the same time, higher potential energy. This situation explains why the ions do not diffuse into the bulk region. At their present positions, the counterforces by the membrane (and the dipole layer on it) balance the attraction from the opposite side. The ion channels can work in continuous mode (the "resting ion channels") and impulse mode (gated by "caps" on the channels, in transient mode).

Notice that the resulting electrical field cannot accelerate an ion except when the ion is at the beginning of the open ion channel across the membrane. Here, the mechanical counterforce is missing, and the field drops to the value generated by the condenser inside the plates. Here, a vast electrical field accelerates the ions to a high speed (although a force needed to move against the viscosity decreases the electrostatic acceleration). The empty red circles show an ion that traverses through the channel. The figure shows their electrical fields on the two sides, at the beginning and end of the channel. (On the right side, the electrical field is negative, but so is the charge of the ions.)

Moving in the polarized electrolyte

The electrical acceleration sharply decreases after leaving the ion channel (see the right side). As shown by the difference of the vertical positions of the filled red and green balls, the difference in the electrical field values is huge across the membrane ($E_{accelerate}$) and considerable inside the dielectric electrolyte (E_{assist}) as shown by the difference in the electrical field values of the green balls. The field still accelerates, but significantly less intensely than the one in the gap. Suppose we assume that the ion quickly accelerates to its Stokes-Einstein speed. In that case, the travel speeds in those space regions are proportional to the difference of the fields in the shown positions and the reciprocals of their travel times.

The passed ions decrease the concentration, and the electrical field on the high-concentration side increases on the low-concentration side. Similarly, the gradients decrease and increase, changing locally the gradients. At the exit, the field continues to accelerate the ion. However, the increasing potential and concentration increasingly decelerate the passing ions, so they quickly brake them to the assisted speed. Then, the effect of the electrical field cancels, and the ions will move with their diffusion speed ("the ion stops").

Ion selectivity

As Eq.(5.5) states, the electrolyte separated by a permeable membrane is balanced when the electrical and thermodynamic forces are equal. The $E_{thermal}^{C_k}(d)$ "electrical field" depends on the chemical quality of the ion; the electrical field does not. That means the resulting driving forces acting on the different ions are different. While one ion experiences a resulting force and moves to the other segment, the other remains resting. Of course, the ions moving across the segments change the overall concentrations in both segments (and so the electrical field contribution), indirectly affecting the balanced state of the other ions. When a mixture of ions is present in the volume, all ions feel a different driving force, and the solution as a whole will be balanced when the resulting forces for all chemical ions are balanced. The permanently open ion channels only passively participate in the process. The driving forces of the ions are different, and they produce, depending on the actual concentrations, the selectivity of the channels.

When Na^+ ions rush into the intracellular layer, they roughly increase the overall concentration and the potential. All other ions, including K^+ , feel a driving force. The targeted membrane potential is set electrically, and the driving gradients may behave controversially during the transient period. The actual voltage gradient may temporarily direct the chemical gradient in opposite directions. Phenomena such as 'over-relaxation', in this sense, similar to 'hyperpolarization', may happen.

However, the layer itself can also be modeled as having just a few ions under their mutual repulsion on the surface or a few atomic layers on top of each other, depending on the concentration and voltage in the bulk on the two sides. (The voltage gradient diagram line validates the plane crossing the membrane and the ion channel.)

Our results align with the observation (see caption of Fig. 11.22 in [24]). The amount of unbalanced ions is in the range of 10^7 , and so is the amount of rush-in ions. In addition, those ions on the high-concentration side rush into the low-concentration side and cause a large change in the membrane potential. Their absolute amount is small compared to the total number of ions in the cell, but it is significant compared to the number of unbalanced ions.

Chapter 6

Action Potential

As discussed, a smaller number of ion channels is distributed over the surface of the membrane, and the overwhelming majority of ion channels is concentrated in the AIS.

6.1 The physical process

In the resting state, we have a condenser that has some low-conductance resistors (the resting ion channels) and a very low intensity (nearly constant) distributed current flows, under the effect of a slightly varying (nearly constant) membrane voltage. Due to the membrane potential, a current through the always-open channels flows on the dendritic surface and the resting channels' conductivity represents a drain with sufficient transmission. The resting current practically does not reach the AIS: the ion channels along the current's path practically "shunt" the current. This is a static state that classical neuroscience assumes: some static current in and some static (leakage) current out; no significant gradient. *In this state*, an almost correct model is a condenser with a *parallelly* switched resistor; an integrator-type RC circuit, see Table 6.1 (although when approaching the threshold voltage, the AIS plays some role. The smaller gradient changes, such as subthreshold excitations, produce "mini-AP waves" [32], providing direct experimental proof that in this state the parallel RC circuit is not correct). Some perturbations must be counterbalanced, but their amount does not exceed a predefined threshold.

In the transient (perturbed) state, the process variable (the membrane's potential) exceeds the threshold. That event acts as a fast trigger signal. The amount of charged particles (and so: the membrane's voltage) suddenly increases, the created gradient drives a current. When Na^+ ions rush into the intracellular layer, they roughly increase the overall concentration and the potential in that thin layer. All nearby ions feel a driving force. The targeted membrane potential is set electrically, and the driving gradients may behave

in unexpectedly during the transient period. The actual voltage gradient may temporarily reverse the direction of the chemical gradients.

Our results align with the observation (see caption of 11.22 in [24]), that *the significant processes occur in a thin layer of the electrolyte proximal to the membrane surface*. The amount of unbalanced ions is in the range of 10^7 , and so is the amount of rush-in ions. In addition, those ions on the high-concentration side rush into the low-concentration side and cause a significant change in the membrane potential (and concentration). Their absolute amount is small compared to the total number of ions in the cell, but it is significant compared to the number of unbalanced ions in that layer. However, the layer itself can also be modeled as having just a few ions under their mutual repulsion on the surface or in a few atomic layers on top of each other, depending on the concentration.

Suppose that at the beginning of an AP a large amount of Na^+ ions are transferred from the extracellular to the intracellular side. In that case, the K^+/Na^+ concentrations change from 145/15 to 45/115 and the corresponding thermodynamic voltage contribution changes from +61 [mV] to -25 [mV] (i.e., altogether a 86 [mV] sudden increase in membrane potential). For the Na^+ ions, the resultant potential changes from +4 [mV] to -83 [mV]. That means when the AP begins, the "Na-K pump" stops (if the resulting potential provides the driving force for the exchange pump). The only way for the neuron to remove the excess Na^+ ions is to generate a current toward the AIS (where the other end of the ion channels remained at the extracellular potential), until the Na^+ -specific driving force disappears. The K^+ -specific driving force, changes from -90 [mV] to -176 [mV], i.e., the K^+ intake gets more intensive, misleading researchers into believing that it causes the observed hyperpolarization. However, as Fig. 3 in [32] demonstrates, it occurs instantly, not with a delay; furthermore, the "leakage current" and the stimulus current are negligible. The low extracellular K^+ concentration plus the low number of channels in the membrane's wall do not enable a significant increase in the intracellular K^+ . (Given that complex changes occur, including changes in the electrical potential that also change the concentrations, different waves start; the statement is not strictly valid.)

As the current flows, the thermodynamic force decreases, and at some point the resultant force changes its sign; so does the current direction. This is seen as the "condenser current" (and mis-identified as an intense K^+ current). The ions move slowly, and due to the enormous electrostatic repulsion, their "electrostatic fluid" must be continuous. A special "damped wave" is formed (with all secondary mechanical, optical, etc. effects) and the condenser current reverses. Electricity interprets the phenomenon as "capacitive current" in the equivalent circuit, and thermodynamics interprets it as changed chemical concentration (or the negative amplitude of the damped elastic oscillation). Our discussion adds that electrical repulsion generates a pressure wave, or shock wave, within the neuron's closed volume.

The neuronal AP generation can be entirely described by the 3-D line in Fig. 6.1. The three dashed diagram lines are drawn in three perpendicular planes; see the figure's caption and the discussion below for the details.

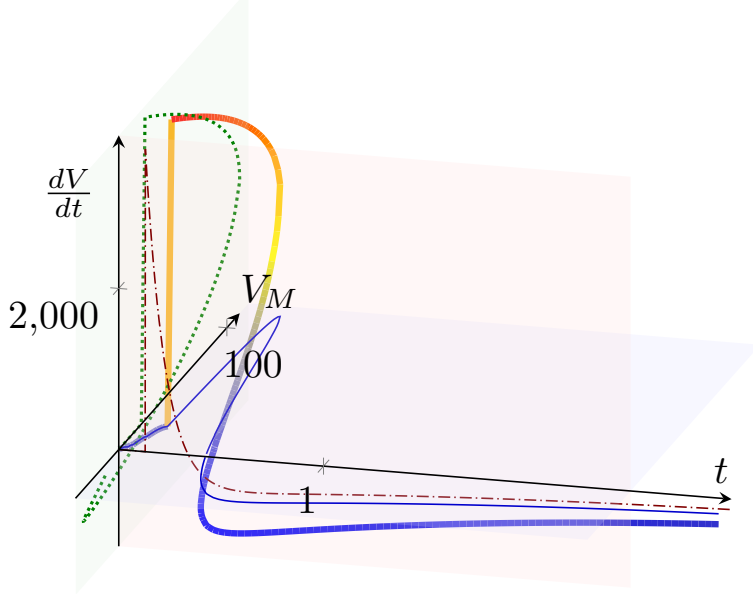
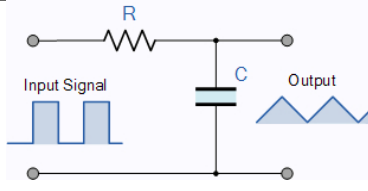
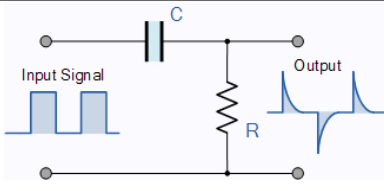


Figure 6.1: A summary of producing neuronal AP as a 3-D figure. The thick, solid 3-D line shows how the color-coded voltage gradient drives the membrane potential over time. The three dashed diagram lines are the projections of the spatial curve to the boundary planes of the cube. The physiology delivers them as 2-D plots; the figure explains how those parameters are interrelated. The red line shows the voltage gradient produced by the rush-in of the Na^+ ions, see also the middle inset in Fig. 6.2. The blue diagram line shows the AP, see also the bottom inset in Fig. 6.2. The interdependence of the gradient and the voltage is called a phase plot; see, for example [55].

6.2 The oscillator model

Despite the early warning that ‘it was not possible to separate the change into resistance and capacity components’ [10], a commonly accepted truism was that neurons, in some sense, behave as electrical oscillators. HH introduced the idea explicitly that the electrically equivalent circuit of a neuron is an RC oscillator. We must also add the invention from about three decades later: “Neurons ensure the directional propagation of signals throughout the nervous system. The functional asymmetry of neurons is supported by cellular compartmentation: the cell body and dendrites (somatodendritic compartment) receive synaptic inputs, and the axon propagates the action potentials that trigger synaptic release toward target cells. *Between the cell body and the axon sits a unique compartment called the axon initial segment.* The AIS was first described 50 years ago [i.e., nearly two decades after HH published their study], and its molecular composition and organization have been progressively elucidated during the

Table 6.1: RC circuit types

The RC Integrator	The RC differentiator
$V_{out}^{Integrator} = \frac{1}{RC} \int_0^t V_{in} dt$	$V_{out}^{Differentiator} = RC \frac{dV_{in}}{dt}$
Low Pass Filter	High Pass Filter
Resting state	Transient state
	
(Mostly) resting state	(Mostly) transient state
Ion channels in the membrane	Ion channels in the AIS

following decades. Recent years have also brought crucial insights into the functions of the AIS: how *ion channels at its surface generate and shape the action potential*.” [29] We provide the physics and mathematics of how AIS shapes the action potential (or more precisely, we show what an important role it plays in forming AP).

Inventing AIS changed the viewpoint of neuroscience [29], a half century later, after setting up the commonly accepted model by HH. ”The AIS is located at the proximal axon and is the site of action potential initiation. This reflects the high density of ion channels found at the AIS. ... The summation of synaptic inputs gives rise to action potentials at the AIS, a 20–60 μm long domain located at the proximal axon/soma interface that has a high density of voltage-gated ion channels.” As discussed in [30], see also their Figure 1, the structure of the AIS is known to the most minor details. As the illuminating investigations in 2008 [27] revealed, the AIS has very dense ion channels. That is, from an electrical point of view, those parallelized channels can be abstracted as a *discrete conductance* (or resistance) between the membrane and the axon. The membrane itself can be abstracted as a *distributed condenser* with no resistance. This model is in contrast with the viewpoint of biophysics, that the membrane plus AIS is considered a distributed element, where the capacitor and condenser cannot be separated. The current model is the result of projecting the physical processes in the resting state to the transient state. Notice the important point: ”Neurons are also anatomically polarized, as they can be subdivided into a somatodendritic input domain and an axonal output domain” [26]; providing direct evidence that (unlike in HH’s model) *the input and output currents (and voltage time derivatives) are independent*.

In our model, the AIS gets part of the membrane, and this leads to crucial changes. (BTW, the name is slightly misleading: the AIS is part of the neuronal oscillator, and it forwards a traveling potential wave to the axon instead of

belonging to it.) 'Although by definition a neuron must have an axon to assemble an AIS, the relationship between AIS assembly and axon specification in vivo has not been determined yet' [26]. Anyhow, a neuron can be abstracted (in the disciplinary view of electricity) as comprising a resistor-like and a capacitor-like component. However, in the resting and the transient states, they are combined in different ways.

In the resting state, we can model neurons as a *parallel RC* circuit (see the "integrator" in Table 6.1), where "random" currents flow in and out. Practically, the output voltage is constant (the resting potential): the ion channels and pumps are distributed over the membrane's surface, and due to the low speed of currents, at low local voltage gradients, the AIS is not involved. Only the synaptic inputs cause a measurable gradient on the AIS. In this native state, there is no gradient; in this sense, it is similar to the case of clamped operation, where the gradient is (apparently) suppressed by the feedback. HH did not see any structural elements on the membrane, so logically, they modeled it as a distributed resistor and capacitor, which really has resemblance with a *parallel RC oscillator*. However, they made a wrong choice of the circuit type, and their choice (likely due to inertia) was repeated in good textbooks such as ([23] Figure 3.1 or [1] Figure 1.1), and it is a commonly accepted fallacy even today [56].

In the transient state, the high-transmission ion channels in the AIS (high conductance R) replace the low-transmission ion channels (low conductance R) in the membrane's wall, and the rush-in current flows out entirely on the AIS (for the model, in this state, we omit the low-transmission ion drain). Those two elements form a *serial RC* circuit (see the "differentiator" in Table 6.1) that acts as a damped oscillator and produces a current pulse, known as AP. Neuroscience mistakenly claims it is a parallel *RC* circuit; that statement is valid only in the resting state. However, the number and intensity of the 'resting' ion channels in the membrane's wall are insignificant for producing AP, so we use the approximation that only the 'transient ion channels' in the AIS are active when generating an AP.

In the electrical view, the process can be modeled as follows: a sudden voltage gradient (see Figure 6.2 middle inset) appears on the membrane (as a discrete capacitor), and a current flows out (see Figure 6.2 bottom inset) through the serially connected ion channel array (as a discrete resistor). The neuron forms a simple *serially* (*not parallelly*, as assumed by Hodgkin and Huxley [5] and mistakenly claimed by neurophysiology) connected *RC* oscillator. For the physical and mathematical description, see section 6.3, furthermore, section 4 in [34]; for its algorithmic specifics [35], for further details [33]. Notice that the difference between the rush-in and the resultant gradients emphasizes the role of the finite resources: without the AIS, no turn-back of the resultant gradient would occur.

The fundamental difference between the parallel and serial *RC* circuits is that *the parallel circuit cannot produce output voltage with opposite sign*. As discussed, the capacitive current, which by definition changes its direction and so generates an opposite voltage on the resistor represented by the AIS, perfectly describes the so-called "hyperpolarization". It is not the effect of a K^+ current:

the resting ion channels do not have sufficient conductance. Not knowing about the AIS leads to assuming that the resting ion channels work also as transient channels (in other words, assuming a parallel RC circuit instead of the serial one, furthermore, that the output current flows through the membrane instead of the axon). Some K^+ current certainly exists; for the magnitude of currents during AP, see [32].

The shape of the output waveform depends on the ratio of the pulse width to the RC time constant. When RC is much larger than the pulse width, the output waveform resembles the input signal, even with a square wave input. (In the case of the neuronal oscillator, the shape of the front side of the spike is similar to the one derived for the rush-in current, while the back side is very prolonged.)

6.3 Mathematics of the ActionPotential

Although the implementation of a neuronal circuit works with ionic currents and has a charge transmission mechanism drastically different from the one based on free electrons used in metals, the operation can align (to some measure) with the instant currents of physical electronics (as we suggest, the effect of the slow current and finite size can be reasonably imitated with current generators generating a special current shape). The input currents in a biological circuit are a large rush-in current through the membrane in an extremely short time, and much smaller gated synaptic currents arriving at different times through the synapses. Those currents generate voltage gradients, and those gradients direct the operation of the RC oscillator (Notice that due to the multi-physical effects, the parameters R and C may change in the function of time and the actual voltage). As it is well known from the theory of electrical circuits, the output voltage measured on the output resistor is as follows:

$$V_{out}^{Differentiator} = RC \frac{dV_{in}}{dt} \quad (6.1)$$

where the input voltage is as follows:

$$\frac{d}{dt} V_{in} = \sum \frac{d}{dt} V_{IN}^{Component} - \frac{d}{dt} V_{OUT}^{AIS} \quad (6.2)$$

that is, the (temporally gated) sum of the input gradient that the currents generate, plus the gradient of the output current through the AIS [30, 29]. The latter term can be described as follows (see the discussion around also Eq.(3.38):

$$\frac{d}{dt} V_{OUT}^{AIS} = \frac{1}{C} \frac{V_{internal} - V_{external}}{R_{AIS}} \quad (6.3)$$

The input currents are slow (we typically imitate them using a current generator with a special form). In the simplest case, the resulting voltage derivative comprises only the rush-in contribution and the AIS contribution described by

Equation (6.3). The gating implements a dynamically changing temporal window and dynamically changing synaptic weights. The appearance of the first arriving synaptic gradient starts the “Computing” stage. The rush-in gradient starts the “Delivering” stage [35]. The charge collected in that temporal window is as follows:

$$Q(t) = \int_{t_{0,i}}^{t_{thr}} dt \sum_{\substack{i \\ t_{0,i} \leq t \leq t_{thr}}} I_{syn,i}(t) \quad (6.4)$$

The total charge is integrated in a mathematically entirely unusual way: this is *NOT* a simple “integrate and fire”: the collected charge defines the upper bound of the integration time. Through adjusting the integration time window, the neuron selects which one(s) out of its upstream neurons can contribute to the result; the individual contributions are the result of a bargain between the sender and the receiver. The upstream neuron decides when the time window begins, but the neuron decides when a contributing current terminates. *The beginning of the time window is set by the arrival of a spike from one of the upstream neurons, and it is closed by the neuron when the charge integration results in a voltage exceeding the threshold.* The speculations [43] about the spatial length of the spike are correct, but its “payload length” is defined by the receiving neuron. The computation (a weighted summation) is performed on the fly, excluding the obsolete operands (the idea of Analog In-Memory Computing (AIMC) [57] almost precisely reproduces the charge summation except for timing). The only operation that a neuron can perform is that unique integration. Its result is the reciprocal of the weighted sum, and it is represented by the time when a spike is sent out (*when* that charge on the membrane’s fixed capacity leads to reaching the threshold). Notice that the time is defined on the neuron’s local time scale and is interpreted similarly by the downstream neurons: the data transmission time matters.

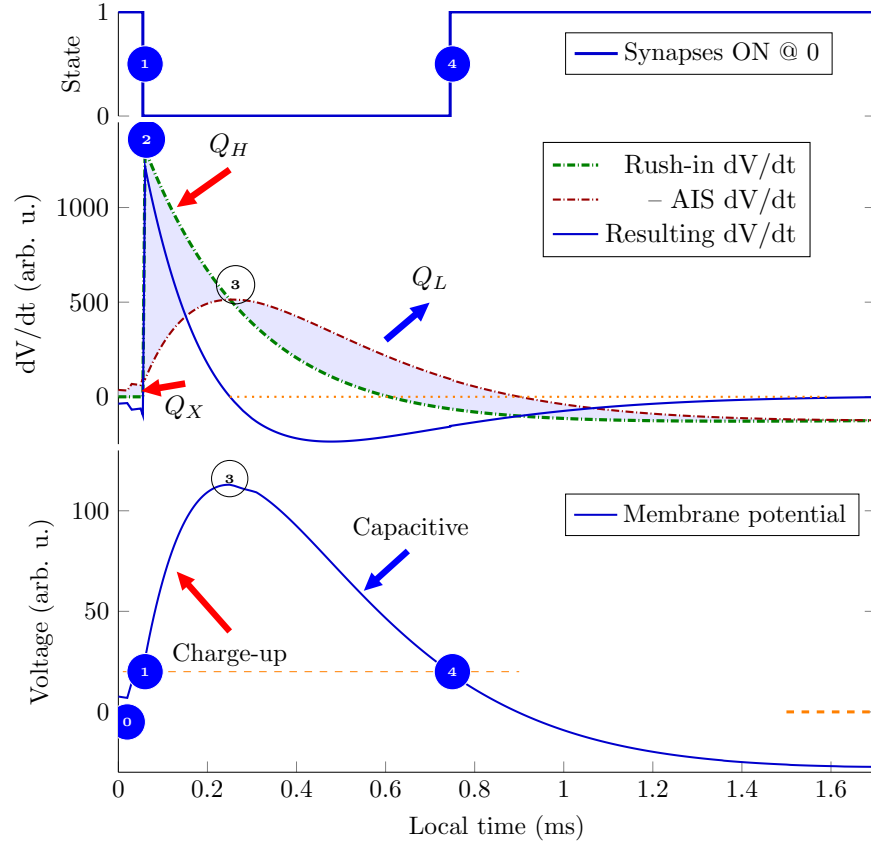


Figure 6.2: The physical processes describing the membrane's operation. The rush-in Na^+ ions instantly increase the membrane's charge, and the membrane's capacity discharges (producing an exponentially decaying voltage derivative). The ions are created at different positions on the membrane, so they take different times to reach the AIS, where the current produces a peak in the voltage derivative. The resulting voltage derivative, the sum of the two derivatives (the AIS current is outward), drives the oscillator. Its integration produces the membrane potential. When the membrane potential crosses the threshold, it switches the synaptic currents on or off.

Appendices

Dichotomies

List of Tables

3.1	The summary data table for equilibrium (Data from Table 2.1 in[23])	43
6.1	RC circuit types	82

List of Figures

3.1	The measured current, instead of conductance, (Fig. 2 from [5]), demonstrating that two different physical effects result in the rising and falling edge of the measured current.	17
3.2	The PSP shape measured by [11] and fitted by function forms given by Eq.(3.15).	18
3.3	The electrical field of a disc of charge can be found by superposing the point charge fields of infinitesimal charge elements. This can be facilitated by summing the fields of charged rings.	26
3.4	The oppositely charged parallel surfaces of the membrane are treated like conducting plates (infinite planes, neglecting fringing), then one can use Gauss's law to calculate the electrical field between the plates. Presuming the plates are at equilibrium with zero electrical fields inside the conductors, the result from a charged conducting surface can be used. "from Hyperphysics by Rod Nave, Georgia State University". © http://hyperphysics.phy-astr.gsu.edu/hbase/	31
3.5	The neuronal membrane's <i>electrical field</i> in the function of the distance from the membrane's surface. The thickness of the 'atomic layers' proximal to the membrane's surfaces are also shown. The electrical field is linear inside the uniformly charged atomic ion layer on the surface and is described by Eq.(3.25) in the electrolyte. For visibility, a layer thickness 0.5 nm is assumed. For illustration, a simple capped ion channel in the membrane's wall is also displayed.	36
3.6	The electrical and the thermal "electrical fields" at different concentrations, see Eq.(3.29). The thermal electrical field varies with the thickness parameter and depends on the concentration; see Eq.(5.2). Balanced states are where the sloping lines cross the horizontal ones. The dot marks where HH provided measured data.	38

- 3.7 The mechanism for controlling neuronal operation is the same in all cases shown in Table 3.1. It appears that the thicker membrane produces a higher membrane electrical potential, requiring a higher thermal potential to compensate. However, it can solve the task using a lower salt concentration, since the electrical membrane potential accelerates ion flow through the membrane channels, thereby speeding up the operation. The electrical fields for the different species have similar values, as shown in Fig. 5.1. Furthermore, the total concentration decreases as the membrane thickness and potential increase. 40
- 3.8 The mechanism of producing AP (using numbers referring to the case of squid shown in Table 3.1). In the resting state, the inside thermal potentials follow the green paths. When the rush-in of Na^+ ions suddenly increases the internal concentration to [200], the potential on the internal side of the membrane increases to +90 mV. The neuron must decrease its Na^+ concentrations to restore the resting potential, mainly by releasing an action potential through the AIS. If the clamping fixed the membrane's voltage at the peak voltage of the AP, the neuron must release K^+ ions until it reaches concentration [Na50,K7]. 44
- 3.9 A block diagram of a PID controller in a feedback loop. $r(t)$ is the desired process variable (PV) or setpoint (SP), and $y(t)$ is the measured PV. (Wikipedia) 48
- 4.1 The figure is a modified version of Fig. 3.8. It is supplemented with the unbended black line showing that the clamping condition forces the neuron to find a new balanced state by compensating the clamping voltage with setting a concentration [Na50,K7]. For comparison, the case of natural AP is also shown. The mechanism of producing AP (using numbers referring to the case of squid shown in Table 3.1). In the resting state, the inside thermal potentials follow the green paths. When the rush-in of Na^+ ions suddenly increases the internal concentration to [200], the potential on the internal side of the membrane increases to +90 mV. The neuron must decrease its concentrations to restore the resting potential, mainly by releasing an action potential through the AIS. The clamping fixes neuron's membrane voltage at AP, so the neuron can operate only with the concentrations. 57
- 4.2 Az AP állapot mesterséges és természetes lebomlása. A természetes úton történő lebomlás esetén a neuronnak a (számára felesleges) Na^+ ionoktól kell megszabadulnia; azok kibocsátásával a kiindulási állapotba tér vissza; körfolyamat megy végbe. A clamping rögzítve tartja a membrán feszültségét, így a neuron csak az intracelluláris koncentrációk változtatásával tud új egyensúlyi helyzetet kialakítani, ami a nagyon lecsökkent K^+ koncentráció miatt csak intenzív K^+ kibocsátással lehetséges. 58

4.3	The differing time constants of the rising and falling edge of a neuronal square wave	61
5.1	The thermodynamic "electrical field" depends on the membrane's thickness, see Eq.(5.2). The electrical gradient depends on the charged layer's concentration and thickness; see Eq.(3.29). The dots mark measured data shown in Table 3.1.	65
5.2	The oppositely charged parallel surfaces of the membrane are treated like conducting plates (infinite planes, neglecting fringing), then Gauss' law can be used to calculate the electrical field between the plates. Presuming the plates to be at equilibrium with zero electrical field inside the conductors, then the result from a charged conducting surface can be used.	70
5.3	The neuronal membrane's <i>electrical field</i> in the function of the distance from the membrane's surface and the bulk potential. The thickness of the atomic layers proximal to membrane's surfaces are also shown.	73
6.1	A summary of producing neuronal AP as a 3-D figure. The thick, solid 3-D line shows how the color-coded voltage gradient drives the membrane potential over time. The three dashed diagram lines are the projections of the spatial curve to the boundary planes of the cube. The physiology delivers them as 2-D plots; the figure explains how those parameters are interrelated. The red line shows the voltage gradient produced by the rush-in of the Na^+ ions, see also the middle inset in Fig. 6.2. The blue diagram line shows the AP, see also the bottom inset in Fig. 6.2. The interdependence of the gradient and the voltage is called a phase plot; see, for example [55].	81
6.2	The physical processes describing the membrane's operation. The rush-in Na^+ ions instantly increase the membrane's charge, and the membrane's capacity discharges (producing an exponentially decaying voltage derivative). The ions are created at different positions on the membrane, so they take different times to reach the AIS, where the current produces a peak in the voltage derivative. The resulting voltage derivative, the sum of the two derivatives (the AIS current is outward), drives the oscillator. Its integration produces the membrane potential. When the membrane potential crosses the threshold, it switches the synaptic currents on or off.	86

Index

- Action Potential, 41
- ATP hydrolysis, 59
- Brain Initiative, 2
- clock reset, 40
- complete measurement, 35
- Coulomb's law, 34, 43, 45, 50
- delayed current, 42
- equivalent circuit, 50
- equivalent electrical field, 60
- event, 26
- Feynman R. P, 31
- force
 - stochastic, 33
- force field, 33
- Galilei, Galileo, 45
- Galillei's relativity principle, 48
- heat absorption by neuron, 59
- instant interaction, 40
- interaction
 - attributes, 49
- ion channel, 42
- Johnston Daniel, 8
- Kepler's Law, 45
- Kirchhoff's Laws, 44
- lassú áram, 10
- law of motion, 13
- laws of motion, 34, 39, 57
- Lotka-Volterra differenciálegyenlet, 24
- mathematics of finite speeds [58], 45
- Maxwell equations, 42
- melting
 - lipids, 10
- Minkowski, Herman, 45
- Mozgástörvény, 2
- mozgástörvények, 2
- neuron model
 - McCulloch-Pitts, 41
 - multi-compartment, 42
- Newton törvények, 2
- non-ordinary law, 48
- Onsager's relations, 35
- ordinary laws, 25, 27
- partial derivative, 47
- rendetlen törvények, 8
- retarded time, 43
- Schrödinger, Erwin, 25
- science
 - disciplinarity, 31
- slow current, 27
- spatiotemporal, 26
- stochastic force, 33
- Stokes-Einstein relation, 76
- telegrapher's equations, 40

- teljes mérék, 24
- time-space, 45
- time-space coordinate, 26

- von Neumann
 - 'zigzag' way, 3

- Wu Samuel Miao-sin, 8

Glossary

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